Mega Architecture for Extreme Weather Energy Harvesting

A modular resilience campus with stress-limited kinetic storage and storm-tolerant energy capture

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Proposal and engineering feasibility assessment

Abstract

Destructive EF5 tornadoes in the United States motivate a dedicated energy-harvesting architecture intended to intercept and convert a usable portion of locally incident storm energy. Conceived as industrial infrastructure with residential use and storm refuge excluded, the proposed campus couples protected wind collection to a flywheel buffer and a separately sized energy reserve, assigning transient exchange and extended outage support to equipment suited to each duty. For eight conceptual steel rings totaling 100,000 tonnes, a 30 m mean radius and 340 RPM give approximately 15.85 GWh of gross kinetic energy, accompanied by an idealized hoop stress of 8.96 GPa. Restricting the same bank to an illustrative 500 MPa allowable stress reduces gross storage to approximately 0.885 GWh and estimated electrical delivery to 0.597 GWh under the assumed discharge conditions, exposing the gap against an 8.4 GWh seasonal service requirement. An adaptive GPU implementation supports investigation of that constrained design, with computational benefit determined by complete solves at matched accuracy. Progress toward a pilot requires measured component limits and independent protection, keeping the proposed resilience service grounded in achievable energy delivery.

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1 Prerequisites

Harvesting energy from extreme weather should strengthen community electrical resilience without transferring unacceptable risk into the infrastructure intended to provide that service. At the delivery boundary, the required load profile sets the energy obligation, with site hazards and qualified material limits constraining the equipment available to meet it. Numerical acceleration supports the resulting engineering calculations only within declared error limits; authority over protective action belongs to independent hardware.

1.1 Engineering status and qualification boundaries

At the proposed scale, calculated capacity describes a research case whose equipment limits still require qualification. Certified material properties constrain the stresses permitted in service, and measured losses determine how much stored energy reaches the load over the required interval. Testing against the site's environmental loads connects those separate component limits to the envelope of the assembled plant.

Construction readiness requires the selected components to satisfy the civil and electrical criteria adopted for the site, including the loads imposed during faults. Within that qualified envelope, independent protection initiates isolation or controlled rundown in response to hazardous equipment states. Supervisory predictions provide additional information without acquiring authority to suppress an overspeed trip or extend an approved limit.

1.2 Required project data

Across the local weather record, the frequency and duration of hazardous winds determine both collection opportunities and exposure to damaging loads. Measurements of equipment performance connect that resource to net energy delivery, with supplier quotations and utility tariffs providing the corresponding cost basis. Until these project inputs are available, annual yield and lifecycle cost require scenario ranges, as does computational performance pending matched-accuracy benchmarks.

For every assumed input, a recorded measurement boundary and uncertainty range should identify the design decision affected by its value. Carrying that information across component interfaces prevents an isolated favorable calculation from acquiring the status of a campus rating without the mechanical and electrical checks required for integration.

2 Conceptual motivation and scope

During a storm, high instantaneous wind power may coincide with equipment shutdown and a prolonged interruption of utility service. Designing for resilience requires a reserve that covers the load across that interruption, including intervals in which collection is unavailable. The resulting service obligation gives captured power its engineering context, with recharge opportunity and survivability governing its contribution to recovery.

2.1 EF5 motivation and infrastructure purpose

The occurrence of destructive EF5 tornadoes in the United States is a central motivation for this mega-architecture proposal, giving rise to the pursuit of infrastructure that intercepts a usable portion

of extreme-weather energy. The 20 May 2013 Newcastle–Moore tornado in Oklahoma provides a documented example of the catastrophic damage associated with this intensity class. [1] Across the central United States, the project gives particular attention to candidate regions commonly described as Tornado Alley, with site selection governed by local hazard histories and the energy crossing the installation's capture boundary.

For the proposed facility, the intended function is energy interception and conversion, with controlled dissipation assigned to protective components that absorb loads or arrest selected debris. Housing, living accommodation, and storm refuge are excluded from the architecture's intended use; community benefit is delivered by electrical service to separately located loads. During hazardous weather, remote supervision and exclusion of people from exposed collection areas preserve this industrial purpose, with maintenance access confined to an accepted equipment state.

At the capture boundary, a usable energy contribution means net electricity or process heat delivered after conversion and auxiliary losses. Mechanical work dissipated in a protective barrier belongs to a separate impact-management balance, preventing structural resistance from being counted automatically as harvested energy. Whether the installation reduces damaging wind or debris exposure outside its footprint is an additional research question requiring coupled flow and impact measurements, leaving tornado weakening and regional damage reduction outside the demonstrated results.

Within the Enhanced Fujita scale, EF5 corresponds to damage-based estimates of three-second gusts exceeding 200 mph, with no upper wind-speed bound assigned to that category. [2] A selected 320 mph numerical load case and a bounded statistical wind distribution describe research assumptions, carrying neither a universal EF5 design wind nor a site-specific return period. The architectural objective is to absorb and recover part of the locally accessible storm-energy input within qualified equipment limits, keeping the intercepted fraction and its effect on surrounding loads as quantities to be measured.

2.2 Motivation

Across successive outages, essential service requires enough stored energy to bridge each interruption and enough recharge opportunity to restore the reserve. Ordinary-weather generation provides recurring input to this cycle, reducing reliance on rare storm encounters whose timing and recoverable energy are uncertain. Any proposed increase in storm collection must improve delivered service within the accepted risk and lifecycle cost.

In the conceptual arrangement, omnidirectional wind intakes supply electricity to eight large flywheels, with superconducting conversion and support retained as optional equipment choices. Underground siting offers protection against some external hazards, although deformation of a compliant exterior must stay mechanically separated from the precision machinery. Surplus electricity may support a materials-processing branch after the service reserve has been allocated, linking its permitted duty to the campus energy balance. Coupled numerical models assess these interfaces using the relationship between rotor radius, speed, and stress as the starting constraint on storage capacity.

For a flywheel, the intended discharge duty governs the rotor geometry and the supporting machinery needed to sustain rotation. Bearing losses deplete the stored inventory between exchanges, with conversion losses reducing the electricity delivered during discharge. Containment must address the mechanical energy left in the rotor during credible failures, connecting its design directly to the speed interval and energy rating of the complete installation. [3]

2.3 Scope and research questions

Inside the collector's tested wind envelope, measured net output sets the electrical input available for storage. Accepting that input requires adequate charging capacity and headroom, followed by discharge within the storage equipment's qualified limits. A simpler installation delivering the same essential service supplies the comparison needed to judge whether the additional campus branches justify their cost and complexity.

At the site boundary, resource assessment measures accessible energy, with the collector model resolving the fraction recoverable under the imposed loads. Dispatch allocates the accepted electricity under independent protection, preserving the reserve assigned to essential service. Facility wind measurements and certified material allowables are currently unavailable, leaving yield and stress limits as illustrative research inputs. Measured loss curves and manufacturer quotations are also needed to connect the equipment studies to efficiency and cost; civil certification and procurement require their own professional review. Full atmospheric forecasting lies outside the proposed calculations, and any computational advantage awaits executed GPU benchmarks.

Table 1: Scope and research questions

Item	Engineering treatment	Required decision outcome
Eight rotors, each 12,500 tonnes	Retained as a conceptual comparison case	A benchmark for quantifying the energy and stress limits, without construction approval
30 m mean radius and 340 RPM	Retained for the first calculation, then screened by stress	A dimensionally complete case that exposes the main mechanical constraint
Storm capture	Retained as a research branch	Revenue and emergency availability require site-specific assessment
Ten-week peak support	Retained as a service requirement	Storage must be sized from net delivered energy and recharge opportunities
HTS bearings and machines	Retained as optional technology branches	Conventional equipment supplies the initial comparison basis
Underground storage	Retained as a siting option	Containment, groundwater, access, and common-cause hazards determine suitability
Debris-to-materials processing	Retained as an optional downstream phase	Feedstock quality and process economics require independent proof
CuPy hybrid computation	Developed into a numerical architecture	Accuracy and performance must be demonstrated for selected workloads
Quantum simulation	Retained as exploratory research	The baseline project functions without a quantum processor

3 Step-by-step implementation

An energy-harvesting installation must satisfy mechanical limits and the required electrical duty at the same component interfaces. Screening the energy and hazard boundaries at the outset exposes infeasible combinations of rotor size and service duration without committing resources to detailed optimization. For candidates that pass this screen, uncertainty assessment and service-equivalent economics determine whether further integration is justified.

3.1 System primer and architectural organization

A short burst of generation places a power demand on the converter, although sustaining an outage places an energy demand on the reserve. Separating these duties permits the collection equipment and storage system to operate at independently controlled speeds. Electrical sectionalization then limits the paths by which a fault in a storm-facing module reaches stored-energy assets.

3.1.1 From moving air to useful community service

During charging, motor torque increases a rotor's angular speed, accumulating mechanical work as kinetic energy; during discharge, generator torque extracts work over the permitted speed interval. At the electrical interface, power specifies the rate of exchange, with usable energy setting its possible duration. A 400 MW converter paired with 400 MWh of usable storage would sustain rated discharge for one hour under constant-power, loss-free assumptions, illustrating why either rating alone leaves part of the duty unspecified.

Between collection and storage, a protected electrical bus removes the need for a shared shaft that would transmit unpredictable intake forces to the flywheel. Independent generator control accommodates each wind module's speed, with storage converters accepting electricity within the available charging headroom. Properly coordinated isolation limits the propagation of collector faults, allowing the flywheel's transient duty to stay separate from the longer-duration reserve.

3.1.2 Proposed campus organization

Across the campus, shared failure paths threaten continuity if a collector disturbance disables the stored-energy system needed during recovery. Assigning separate electrical zones and protective functions limits that coupling, with acceptance tests tied to the physical duty of each zone. The reserve then supports essential service within its available energy margin, independent of whether storm collection continues.

Table 2: Proposed campus organization

Campus layer	Main function	Proposed physical organization	Design condition that governs acceptance
Resource observation	Estimate available wind and approaching hazards	Independent weather feeds, local sensors, protected communications	Missing or conflicting data produces a conservative state transition
Wind capture	Generate electricity inside a tested load envelope	Distributed, isolatable collector modules with a bypass path	Measured net output and demonstrated shutdown response
Debris interception	Protect machinery from selected projectile classes	Accessible barriers and separated collection bays	Defined impact spectrum, clogging limits, and safe maintenance access
Power conversion	Control voltage, current, and energy routing	Sectionalized AC or DC buses with local converters	Fault isolation and thermal ratings verified for each mode
Fast storage	Buffer gusts and rapid changes in demand	Contained flywheel modules with independent monitoring	Rotor integrity, bearing response, and safe rundown demonstrated

Campus layer	Main function	Proposed physical organization	Design condition that governs acceptance
Energy reserve	Support hours, days, or seasonal service targets	Technology selected separately from the flywheel buffer	Net duration demonstrated using site load and recharge sequences
Process-energy branch	Use suitable surplus electricity or heat	Physically separated thermal storage and optional process equipment	Product quality, emissions, and interruption tolerance established
Community support	Supply essential loads and recovery services	Dedicated feeders to separately located community loads	Service priorities agreed with the community and utility
Numerical environment	Support design and bounded supervisory decisions	Offline HPC plus a restricted advisory interface	Traceable model state, uncertainty, and fallback behavior

3.1.3 Modes of use

Ordinary weather provides the recurring energy needed to replenish the reserve and leaves unreserved flywheel capacity available for selected balancing services. Approaching hazards shift the plant toward protected states according to charging headroom and the qualified envelope of each module. Following an event, inspection findings govern restart as external appearance alone leaves hidden damage and obstructed flow paths unresolved.

Table 3: Modes of use

State	Entry condition	Permitted activity	Response of independent protection
Normal generation	Wind, grid, and equipment health inside approved limits	Economic dispatch and routine charging	Local trips retain authority
Elevated readiness	Hazard forecast or local threshold crossed	Reserve allocation, controlled pre-spin, selective intake closure	Block commands that exceed tested limits
Restricted generation	Wind or debris approaches the qualified envelope	Derated generation, increased bypass, restricted maintenance	Isolate a module on adverse sensor readings
Protective shutdown	Overspeed, vibration, flood, pressure, or electrical trip	Controlled isolation and engineered rundown	Protection operates without the numerical adviser
Islanded service	Utility supply absent and islanding authorized	Supply prioritized loads using available reserve	Shed lower-priority demand prior to reserve exhaustion
Recovery	Hazard has passed and inspections are accepted	Sequential restart and condition assessment	Prevent automatic restart of damaged equipment

3.2 Design basis and mathematical notation

A radius interpreted as a diameter changes calculated kinetic energy by a factor of four at fixed mass and RPM. Declaring the geometry and unit system at the outset prevents this ambiguity from propagating into the stress calculation or service estimate. Definitions beside each symbol connect the numerical inputs to their physical interpretation.

3.2.1 Units and conceptual assumptions

For the conceptual bank, metric tonnes convert to kilograms using 1,000 kg per tonne, ensuring consistency with the SI equations for force and energy. The primary geometry uses a 30 m mean radius, with a separate 30 m diameter case exposing the sensitivity to that interpretation. Allowable stresses serve as illustrative design scenarios and carry no certification for a particular steel grade.

Table 4: Units and conceptual assumptions

Quantity	Symbol	Baseline or treatment	Unit
Rotor count	N_r	8	Dimensionless
Mass of one rotor	m_r	1.25×10^7	kg
Total rotor mass	M	1.00×10^8	kg
Mean ring radius	R	30	m
Baseline speed	n	340	RPM
Illustrative steel density	ρ_s	7,850	kg m^{-3}
Illustrative Poisson ratio for disk screening	ν_s	0.30	Dimensionless
Standard gravity for load examples	g	9.80665	m s^{-2}
Example peak electrical support	P_{peak}	20	MW
Daily support period	t_d	6	h day^{-1}
Service period	N_d	70	days
Minimum-to-maximum speed ratio	r_ω	0.50 in worked examples	Dimensionless
Example efficiency of discharge conversion	η_{dis}	0.90, assumed	Dimensionless
Illustrative air density	ρ_a	1.225	kg m^{-3}
Diameter used for collector comparison	D_c	100	m
Uniform wind speed used for comparison	U	60	m s^{-1}

3.2.2 Symbols used across the physical and numerical models

Across the coupled calculations, a familiar symbol may represent a physical field in one model and a numerical object in another. Distinguishing magnetic flux density from graph incidence, for example, prevents a matrix operation from inheriting an unintended physical interpretation. Each definition needs its domain and units wherever it first enters an equation.

Table 5: Symbols used across the physical and numerical models

Symbol	Definition	Unit or mathematical type
I	Mass moment of inertia about the spin axis	kg m^2
ω	Angular speed	rad s^{-1}
v	Rim speed	m s^{-1}
E_{rot}	Gross rotational kinetic energy	J

Symbol	Definition	Unit or mathematical type
E_{use}	Mechanical energy available over the permitted speed interval	J
E_{del}	Electricity delivered under stated loss assumptions	J or MWh
σ_θ	Circumferential stress in the ring approximation	Pa
σ_{allow}	Allowable stress after the chosen design reductions	Pa
τ	Shaft torque	N m
A_c	Consistently defined upstream capture area	m^2
C_P	Power coefficient using the declared reference area and wind speed	Dimensionless
η	Efficiency for a specified conversion boundary	Dimensionless
p	Fluid pressure	Pa
$\boldsymbol{u}$	Fluid velocity vector	$\mathrm{m\ s}^{-1}$
$\boldsymbol{d}$	Structural displacement vector	m
$\boldsymbol{B}$	Magnetic flux density	T
B	Oriented incidence matrix of the graph; distinct from magnetic flux density	Matrix
$\boldsymbol{J}$	Electrical current density	$\mathrm{A\ m}^{-2}$
T	Temperature	K
$\mathcal{G}$	Mesh or component connectivity graph	Graph
$\boldsymbol{q}$	Discrete physical state	Vector with field-dependent units
$\boldsymbol{\theta}$	Uncertain model parameters	Vector with parameter-dependent units
π	Target probability density	Density with respect to the specified measure
N	Number of scalar algebraic unknowns in a selected numerical system	Integer
n_{nz}	Number of stored nonzero matrix entries	Integer
V_f	Free memory on the GPU measured at the declared planning point	Bytes
B_{run}	Memory allowed for the planned workload	Bytes
B_{sparse}	Memory allowed for sparse operators and their assigned workspace	Bytes
ε_Q	Allowed error in a selected quantity of interest	Quantity-dependent

3.3 Flywheel energy, stress, and duration calculations

Increasing angular speed raises kinetic energy quadratically, with the accompanying centrifugal stress imposing a limit on the speed attainable by a given material. The permitted operating interval fixes the mechanical energy available for extraction, and conversion losses reduce its electrical delivery. Comparing that delivery with the load profile determines whether the bank meets the proposed service duration.

3.3.1 Rotational energy and geometry

For rotation about a fixed axis, cylindrical distance r weights each mass element by its squared distance from that axis. Expressing speed n in revolutions per minute gives angular speed ω in radians per second, which enters the rigid-body relation for kinetic energy.

$$\omega = \frac{2\pi n}{60}, \qquad I = \int r^2 \, \mathrm{d}m, \qquad E_{\mathrm{rot}} = \frac{1}{2} I \omega^2.$$

A thin ring of mass m concentrated near radius R has the largest inertia among the simple same-mass geometries considered here at a fixed outer radius. For a uniform solid disk or uniform annular disk, the inner and outer radii are R_i and R_o.

$$I_{\mathrm{ring}} \simeq mR^2, \qquad I_{\mathrm{disk}} = \frac{1}{2} m R_o^2, \qquad I_{\mathrm{annulus}} = \frac{1}{2} m \left(R_o^2 + R_i^2 \right).$$

For the conceptual ring geometry, substituting the stated mass and speed quantifies the gross energy associated with rotation, excluding material limits and losses. Across the eight identical rotors, E_{bank} denotes the sum of those individual kinetic-energy contributions.

$$\omega = 35.6047 \ \mathrm{rad\,s^{-1}}, \qquad v = R\omega = 1{,}068.14 \ \mathrm{m\,s^{-1}}.$$

$$E_{\mathrm{bank}} = \frac{1}{2} M R^2 \omega^2 = 5.70463 \times 10^{13} \ \mathrm{J} = 15.8462 \ \mathrm{GWh}.$$

Each rotor carries approximately 1.981 GWh in this idealized case, with the bank total obtained by summing eight identical contributions. Converting the calculated joules into electrical-energy units uses the exact joule-to-gigawatt-hour conversion.

$$1 \ \mathrm{GWh} = 3.6 \times 10^{12} \ \mathrm{J}.$$

For comparison with larger energy estimates, the exact conversion places an interval of 1 to 2×10^{14} J at approximately 27.8 to 55.6 GWh, well above the 15.85 GWh calculated for this bank.

3.3.2 Why rim speed controls the steel feasibility screen

A free, thin, uniform ring develops circumferential tension as each mass element follows its curved path. For a ring with cross-sectional area S, balancing the required centripetal force on a short arc against that tension yields the screening relation between material density and rim speed.

$$\sigma_\theta S \, \mathrm{d}\phi \simeq \rho_s S R \, \mathrm{d}\phi \, R\omega^2, \qquad \sigma_\theta \simeq \rho_s R^2 \omega^2 = \rho_s v^2.$$

Table 6: Design sensitivities of the thin ring at fixed density. Each row changes one input independently.

Change	Rim speed	Hoop stress	Gross energy
Radius multiplied by a; mass and RPM fixed	$\times a$	$\times a^2$	$\times a^2$
RPM multiplied by b; mass and radius fixed	$\times b$	$\times b^2$	$\times b^2$
Mass multiplied by c; radius and RPM fixed	Unchanged	Unchanged	$\times c$

At the stated radius and speed, the thin-ring relation gives approximately 8.96 GPa of circumferential stress under a centrifugal load case that excludes gravity and bearing reactions. Fabrication defects and joints disturb the local stress field, with thermal gradients adding strain that the uniform-ring approximation omits. Fatigue and motion around the supports require separate dynamic assessment, limiting this result to an initial mechanical screen.

$$\sigma_\theta \simeq 8.9563 \times 10^9 \text{ Pa} = 8.9563 \text{ GPa}.$$

Reducing the maximum speed until the ring reaches the assumed allowable stress yields a material-limited upper bound within this idealized geometry. The corresponding rim speed is $v_{\max}$, with angular speed $\omega_{\max}$ and gross stored energy $E_{\max}$.

$$v_{\max} \simeq \sqrt{\frac{\sigma_{\text{allow}}}{\rho_s}}, \qquad \omega_{\max} \simeq \frac{1}{R}\sqrt{\frac{\sigma_{\text{allow}}}{\rho_s}}, \qquad \frac{E_{\max}}{M} \simeq \frac{\sigma_{\text{allow}}}{2\rho_s}.$$

At a fixed allowable stress and density, increasing ring radius lowers permissible RPM without raising ideal specific energy. Improving the ratio of allowable strength to density raises that energy bound, subject to fabrication and the discharge-speed interval required by the machinery. Additional mass increases total storage linearly, accompanied by higher foundation loads and more energy for containment to manage during a failure.

A bore-free, thin, isotropic solid disk offers a second analytic screen. Under linear elastic plane-stress conditions with a traction-free outer edge, its maximum central stress follows the stated disk relation; a bore or thick three-dimensional geometry requires another solution.

$$\sigma_{\max,\text{disk}} = \frac{3+\nu_s}{8}\rho_s\omega^2 R_o^2.$$

For the same 30 m outer radius and 340 RPM, that expression gives approximately 3.69 GPa, alongside half the thin-ring energy at the same mass. Changing the geometry improves one part of the stress distribution, although the corresponding energy must be recalculated.

3.3.3 Comparison of rotor geometries

Rotor geometry redistributes inertia and stress, which prevents equal mass and outer size from guaranteeing equal energy capacity or mechanical margin. Comparing idealized rings and disks on one mass-and-speed basis exposes how geometry trades stored energy against the first analytic screen for stress.

Table 7: Comparison of rotor geometries

Calculated geometry	Total mass	Radius interpretation	Speed	Rim speed	Gross energy of the bank	Analytic screen for stress	Interpretation
Thin-ring baseline	100,000 t	30 m mean radius	340 RPM	1,068.1 m/s	15.846 GWh	8.956 GPa hoop stress	Exceeds a justified baseline steel screen
Thin ring with the 15 m-radius interpretation	100,000 t	15 m mean radius	340 RPM	534.1 m/s	3.962 GWh	2.239 GPa hoop stress	Still requires major speed reduction and full design assessment
Uniform solid disk	100,000 t	30 m outer radius	340 RPM	1,068.1 m/s	7.923 GWh	3.694 GPa central stress for $\nu_s = 0.30$	Separate geometry with lower energy and different stress distribution

At a 30 m radius, the conceptual 340 RPM corresponds to a rim speed above 1 km/s, explaining the large centrifugal stress despite the modest revolution count. Assessing rotational duty requires the speed specification to retain its geometric context.

3.3.4 Usable energy and an illustrative stress-limited capacity table

Toward the lower end of discharge, falling rotor speed increases the torque required for a given output power and may reduce the converter's available voltage. Stable rotor motion and the reserved energy margin impose additional constraints on the permitted minimum. Defining the minimum-to-maximum angular-speed ratio as r_ω connects that lower limit to the extractable mechanical energy.

$$E_{\text{use}} = \frac{1}{2} I \left(\omega_{\max}^2 - \omega_{\min}^2 \right) = E_{\max} \left(1 - r_\omega^2 \right), \qquad r_\omega = \frac{\omega_{\min}}{\omega_{\max}}.$$

For an assumed constant efficiency for discharge conversion, with standby losses and reserve deductions excluded, the delivered electricity is approximated by the stated conversion relation.

$$E_{\text{del}} \simeq \eta_{\text{dis}} E_{\max} \left(1 - r_\omega^2 \right).$$

These scenarios retain the original total mass and 30 m mean radius. Values of 250, 500, and 1,000 MPa are illustrative *allowable* stresses after all required design reductions; they are neither material certifications nor recommendations for a particular steel grade. With $r_\omega = 0.50$ and $\eta_{\text{dis}} = 0.90$, the assumed delivery factor is 0.675.

Table 8: Usable energy and an illustrative stress-limited capacity table

Assumed allowable ring stress	Maximum rim speed	Maximum speed at 30 m	Gross energy of the bank	Delivered energy under the stated assumptions	Continuous delivery at 20 MW	Equivalent 6 h peak-support days
250 MPa	178.46 m/s	56.80 RPM	0.4423 GWh	0.2986 GWh	14.93 h	2.49 days
500 MPa	252.38 m/s	80.33 RPM	0.8846 GWh	0.5971 GWh	29.86 h	4.98 days
1,000 MPa	356.92 m/s	113.61 RPM	1.7693 GWh	1.1943 GWh	59.71 h	9.95 days

For a 12,500 tonne ring at a 30 m mean radius, the steel section has an area of approximately 8.45 m^2 under the uniform-ring assumption. Maintaining the assumed material quality requires a three-dimensional stress assessment as well as a fabrication plan matched to that geometry. Finite-section corrections then refine the thin-ring screen, providing the allowable speed used for hardware design.

3.3.5 Ten-week service and charging-rate consistency

Across a repeated service cycle, the stored inventory must cover discharge and then recover within the available charging interval. Comparing cumulative load with charging power exposes a configuration that holds enough nominal energy yet fails to replenish it on schedule. Both constraints must be satisfied under the same conversion and loss assumptions.

Over seventy days, supplying the conceptual 20 MW cooling-related peak for six hours per day requires 8.4 GWh at the electrical delivery boundary, excluding any additional energy consumed in storage or conversion.

$$E_{\mathrm{service}} = P_{\mathrm{peak}} t_d N_d = 20\ \mathrm{MW} \times 6\ \mathrm{h\,day^{-1}} \times 70\ \mathrm{days} = 8.4\ \mathrm{GWh}.$$

At the illustrative 500 MPa limit, the original 100,000 tonne bank delivers approximately 0.597 GWh under the assumed speed interval and conversion efficiency, excluding standby losses. Reaching 8.4 GWh by scaling the same idealized design would require approximately 1.407 million tonnes of rotating material, shifting the design comparison toward other storage technologies with more suitable duration and material requirements.

For the charger's electrical power P_{ch}, a constant conversion efficiency η_{ch} sets the maximum rate at which rotor energy increases. The requested increase ΔE_{rot} then defines a lower bound on charging time t_{ch}, with standby consumption and interruptions extending the actual interval.

$$t_{\mathrm{ch}} \geq \frac{\Delta E_{\mathrm{rot}}}{\eta_{\mathrm{ch}} P_{\mathrm{ch}}}.$$

Table 9: Ten-week service and charging-rate consistency

Charging or service scenario	Energy calculation	Design consequence
1 kW of machine rating per 25 kWh of energy permits a rapid refill	25 kWh/1 kW = 25 h excluding losses	That ratio describes a 25 h ideal charge interval
A 480 MW machine bank would fill 12 GWh during a 30 min storm pulse	It accepts at most 0.24 GWh in 30 min excluding losses	Approximately 2% of 12 GWh is added in that interval
A complete 12 GWh refill in 30 min	Requires at least 24 GW of average accepted power	Converter, collector, grid, and heat-rejection sizing change completely
Four events, each 500 MW for 1 h	Supplies 2 GWh excluding losses	Below the 8.4 GWh service requirement
A 10 GWh store supports 70 days continuously	Average output is 5.95 MW excluding losses	Peak shaving and continuous town supply are different services

3.3.6 Seasonal retention and the loss budget

Eliminating contact at a bearing removes a friction pathway without eliminating the system's other energy demands. Magnetic interactions and residual gas continue to drain rotor energy, with conversion

introducing further losses during charging and discharge. Continuous electricity for vacuum and refrigeration must enter the same storage accounting boundary, particularly over long holding intervals.

Seasonal retention must account for auxiliary demand at the same energy boundary used for stored energy. Measurements across the speed range connect the loss budget to rotor state; temperature and pressure sweeps then quantify how thermal and vacuum conditions alter retained energy. Application selection in flywheel research incorporates these system-level losses. [3]

For a constant equivalent-loss screen, let E_0 denote the initial energy at the selected accounting boundary and f_L the fraction that may be lost during holding time t_h. Dividing that permitted loss by the holding interval bounds the average power associated with equivalent loss $\overline{P}_{L,\mathrm{equiv}}$.

$$\overline{P}_{L,\mathrm{equiv}} \leq \frac{f_L E_0}{t_h}.$$

Under a 10% loss allowance over seventy days, an initial 10 GWh inventory permits approximately 595 kW of average equivalent loss, falling to 59.5 kW for a 1 GWh inventory. Preserving the same fractional retention at lower capacity requires a correspondingly smaller auxiliary demand. For the illustrative 0.8846 GWh gross bank capacity, the simplified allowance is approximately 52.7 kW.

For a separate illustrative model with a fixed daily fractional energy loss ℓ_d, retention after N_d days follows a geometric decay relation. Its daily loss fraction is a scenario input that would need replacement by measured flywheel data.

$$\frac{E(N_d)}{E(0)} = (1 - \ell_d)^{N_d}.$$

Daily losses of 1% and 0.5% leave approximately 49.5% and 70.4% of the initial energy after 70 days, respectively. Any assertion of weeks-to-months storage should show comparable retention curves using measured auxiliary and rotor-loss data.

3.4 Wind resource, capture limits, and storm survival

Across a collector boundary, extreme wind provides a transient kinetic-energy flux whose recoverable fraction is constrained by equipment loading and storage headroom. A short power peak contributes to seasonal service only for the interval in which the plant accepts it safely. Time-resolved flow and protection states connect this encounter energy to annual yield, including the effects of debris and subsequent downtime.

3.4.1 Tornado terminology and the resource boundary

In the United States, the Enhanced Fujita scale assigns tornado intensity from damage indicators and estimated winds. [2] Hurricane categories use a different averaging convention, requiring tornado loading to retain a separately specified wind-field scenario. Any associated EF classification must follow the observations supporting that scenario.

Near-ground tornado winds vary with height, making vertically resolved loading essential for structures spanning that variation. Measurements by Kosiba and Wurman support this treatment of the hazard input, with recoverable electricity requiring an additional collector model that resolves geometry and conversion losses. [4]

In resource planning, the relevant kinetic energy crosses the installation's local capture boundary during an encounter. Latent heat released across a hurricane belongs to a different energy reservoir, and a regional storm total includes air that may never reach the site. Restricting the calculation to the accessible stream prevents those broader atmospheric quantities from inflating the proposed collector yield.

3.4.2 An idealized wind-power calculation

Uniform steady flow normal to an upstream area A_c admits a simple kinetic-power relation that provides a first resource screen. A tornado requires separate spatial and temporal averaging prior to using that relation as a site-loading or energy estimate.

$$P_{\mathrm{kin}} = \frac{1}{2}\rho_a A_c U^3, \qquad A_c = \frac{\pi D_c^2}{4}.$$

Defining C_P as captured mechanical power divided by incident kinetic power permits a first estimate of electrical output after generator/converter efficiency η_g is applied. The upstream capture area and its reference velocity must match those used in the incident-power calculation, with the measurement boundary stated alongside the coefficient.

$$P_{\mathrm{elec}} = C_P \eta_g P_{\mathrm{kin}}.$$

Using the assumed 100 m diameter, 60 m/s uniform flow, and air density of 1.225 kg/m^3, the calculated incident power is approximately 1.039 GW. Choosing $C_P = 0.30$ and $\eta_g = 0.90$ yields approximately 280.6 MW of electricity. Those coefficients are hypothetical inputs for this example; measured performance of the proposed structure is still required.

Table 10: An idealized wind-power calculation

Duration of the assumed uniform flow	Incident kinetic energy	Electrical energy for $C_P = 0.30$, $\eta_g = 0.90$	Fraction of the 8.4 GWh service requirement
10 min	173.18 MWh	46.76 MWh	0.557%
30 min	519.54 MWh	140.28 MWh	1.670%
60 min	1,039.08 MWh	280.55 MWh	3.340%
Four separate 30 min encounters	2,078.16 MWh	561.10 MWh	6.680%

Each row assumes uninterrupted exposure for the stated duration, with storage losses reducing the amount ultimately delivered at the service boundary. Protective shutdown may shorten an encounter, and damage-related downtime may remove later encounters from the charging schedule. Roughly sixty of the assumed 30 min encounters would supply 8.4 GWh excluding storage losses, illustrating why an instantaneous power estimate alone leaves seasonal service unresolved.

For ideal actuator-disk momentum theory, steady incompressible flow must approach an unconfined disk with uniform axial loading, and the ideal wake carries no rotation. Let U_∞ denote undisturbed wind speed and U_d the disk-plane speed; their relation defines axial induction factor a_i. The disk power coefficient $C_{P,d}$ then uses mechanical disk power P_{disk} and swept area A_d, retaining a reference convention separate from the upstream capture area.

$$U_d = U_\infty(1 - a_i), \qquad a_i = 1 - \frac{U_d}{U_\infty}.$$

$$C_{P,d} = \frac{P_{\text{disk}}}{\frac{1}{2}\rho_a A_d U_\infty^3} = 4a_i(1-a_i)^2 \le \frac{16}{27}, \qquad 0 \le a_i \le \frac{1}{2}.$$

Under these actuator-disk assumptions, $a_i = 1/3$ gives the ideal Betz limit with disk area retained in the coefficient's denominator. Changing to an upstream streamtube area changes the reference convention, requiring every reported coefficient to identify its area and velocity basis. A duct also alters the pressure field beyond its throat, extending the control volume needed for a fair comparison. Across staged turbines, one energy ledger must follow the shared air stream, preventing repeated allocation of its incoming energy.

3.4.3 A time-dependent control-volume calculation

A nonuniform storm field requires a boundary flux integral; one velocity is insufficient for spatially varying inflow. The unit normal $\boldsymbol{n}_{\text{in}}$ points into capture surface A_{in}, and its dot product with velocity defines the inward normal component. Retaining only the positive part excludes outward flow from the incoming-energy flux.

$$u_n^+ = \max(\boldsymbol{u} \cdot \boldsymbol{n}_{\text{in}}, 0)\,.$$

With the inward-flow convention fixed, integration over the capture surface gives the time-dependent incoming power, preserving spatial variations that a single representative wind speed would discard.

$$P_{\text{kin,in}}(t) = \int_{A_{\text{in}}} \frac{1}{2}\rho_a \|\boldsymbol{u}\|^2 u_n^+ \,\mathrm{d}A.$$

Pressure work contributes to the complete fluid-energy balance, leaving inlet kinetic energy alone insufficient to represent the duct's work exchange. Outgoing flow carries energy beyond the turbine, and transient storage within the fluid separates instantaneous inlet power from instantaneous shaft power during nonsteady intervals. A single control-volume ledger tracks those contributions and prevents repeated allocation of the same available energy across multiple intake faces or turbine stages.

At the plant boundary, $P_h(t)$ denotes harvest power after collector-side losses, and $P_{\text{accept}}(t)$ denotes the charging power permitted by equipment ratings and available headroom. For an encounter extending from t_0 to t_1, the accepted energy is bounded by the smaller of these powers at each instant.

$$E_{\text{accepted}} \le \int_{t_0}^{t_1} \min[P_h(t), P_{\text{accept}}(t)]_+ \,\mathrm{d}t.$$

Within the accepted-energy integral, the positive-part convention excludes negative charging contributions. Forecast errors alter reserve allocation and available headroom, with equipment downtime removing collection over the affected intervals. Debris-induced pressure loss reduces recoverable power, and a restricted grid connection limits the destination for excess generation. Retaining their joint time histories connects these effects to the energy actually accepted by storage.

3.4.4 Collector modules and the role of existing demonstrations

For a turbine operating in hazardous weather, the generation envelope defines the conditions permitting power production, with a separate survival envelope governing exposure after shutdown. Challenergy's

published Magnus-turbine specification gives 10 kW at 11 m/s, a generation cut-out speed of 35 m/s, and a 70 m/s withstand value. The final figure describes survival without assigning an output at that wind speed. [5]

Aerodynamic forces on the rotating cylinders arise from the Magnus effect, supplying a device-specific precedent for the collection mechanism. Extrapolation to a campus collector requires new load and output measurements, as these specifications provide neither a tornado-capture rating nor a scaling law for the proposed building.

Field tests of a gravo-aeroelastically scaled downwind turbine connect a flexible demonstrator's measured blade response to the scaling target used in its design. That study provides a precedent for testing compliant aerodynamic components, with its experimental scope preserved in related comparisons. Extending the result to a tornado-exposed building would require a new structural model and experiments matched to that building's load envelope. [6]

Table 11: Collector modules and the role of existing demonstrations

Design branch	Potential role	Principal uncertainty	Required next result
Conventional wind generation outside the high-hazard intake	Supply recurring energy and an economic baseline	Site yield and storm downtime	Measured or justified modeled annual generation
Modular storm-tolerant turbines	Extend useful generation within a qualified envelope	Combined gust, debris, and direction-change response	Instrumented load and output tests
Magnus-effect modules	Explore reduced sensitivity to changing horizontal wind direction	Scaling, cylinder-drive consumption, and net power	Net electrical power curve including all drives
Flexible downwind modules	Investigate load alleviation	Coupled aeroelastic stability and fatigue	Scaled and full-component response correlation
Protected ducted turbines	Move machinery into an accessible enclosure	Pressure losses, recirculation, blockage, and intake loads	Consistent control-volume efficiency measurement
Direct tornado-core interception	Exploratory hazard-research case	Encounter probability and survivability	A research justification independent of the base energy business case

3.4.5 Screens, filters, and staged flow control

Within an intake, screening removes selected debris classes at the cost of a pressure drop that reduces energy available downstream. Comparing separate flow states at equal area and density, a velocity reduction to 40% leaves 6.4% of the original kinetic-energy flux. Applying that cubic ratio across a screen in one continuous streamtube would require a compatible continuity and pressure balance.

Along one steady incompressible streamtube, continuity requires compensating area changes or flow diversion. The cubic ratio alone cannot quantify the loss across a screen under that constraint.

$$\frac{P_2}{P_1} = \left(\frac{U_2}{U_1}\right)^3 = 0.4^3 = 0.064.$$

For incompressible flow encountering a resistance at volumetric flow rate Q_v, the associated hydraulic power loss is approximated by pressure drop Δp times flow rate. This quantity should appear in the collector's energy ledger.

$$P_{\text{screen,loss}} \simeq \Delta p\, Q_v.$$

Coarse impact barriers should be placed where inspection and replacement are practicable, with fine-particle treatment confined initially to a controlled side stream or a downstream material-handling stage. Across the entire storm flow, fine media would impose a pressure drop that reduces downstream power; accumulating debris could increase that drop and change the load on the barrier. Restoring the intake's tested condition then requires practical cleaning access. Any proposal to filter the full flow must demonstrate that the protection gained justifies those energy and maintenance consequences.

Under steady, inviscid, incompressible conditions along a streamline, Bernoulli's relation describes the exchange between static pressure and kinetic energy. Elevation z is measured from a declared datum, making the gravity term sensitive only to height. At fixed elevation, an ideal contraction accelerates the flow at the expense of static pressure without adding total pressure to the air stream.

$$p + \frac{1}{2}\rho_a U^2 + \rho_a g z = \text{constant along an ideal streamline.}$$

Practical ducts introduce viscous and separation losses, with adverse pressure gradients increasing the likelihood of detached flow. Bypass doors and shutters should be designed around tested pressure transients; their intermediate positions and failure state must stay compatible with the governing structural load case.

3.4.6 Site frequency and annual yield

At one facility, harvestable encounters follow the local storm tracks and wind histories, leaving regional storm counts insufficient for predicting recharge. Debris or flooding may close an intake independently of wind speed, with grid availability further constraining export. Representing these conditions jointly defines a site-specific mean annual occurrence rate λ_k and net accepted energy E_k for each event class.

$$\mathbb{E}[E_{\text{annual,storm}}] = \sum_k \lambda_k \, \mathbb{E}[E_k \mid k] \, .$$

The annual expectation follows from the event-class contributions without requiring independence, although service performance still requires the chronological sequence. Repair after one encounter may reduce collection during the next, particularly if correlated grid damage also removes export or recharge opportunities. Routine renewable charging should provide the base energy plan until site measurements support a dependable contribution from extreme winds.

3.5 Civil structure, containment, and environmental interfaces

Protecting stored mechanical energy requires civil structures to carry both ordinary loads and credible failure consequences. During the same hazard, drainage and access must preserve the response paths needed by the equipment, with neighboring occupied properties separated from rotor chambers and exposed collectors. Independent load paths limit the common-cause failures that would otherwise threaten recovery service.

3.5.1 A campus with a defined hazard rating

Around the site, the dam-like exterior redirects wind pressure and runoff, placing corresponding demands on its foundations and drainage. Underground rotor chambers face a different combination of stored-energy consequences and water ingress, requiring their own hazard criteria. Access openings interrupt both forms of protection and must be assessed as potential paths for pressure, fragments, or flooding.

Below ground, surrounding material offers possible fragment shielding, although groundwater creates a concurrent ingress hazard. Site elevation and water exclusion must address that exposure without assuming that burial alone provides complete protection.

ASCE's account of ASCE 7-22 identifies tornado provisions within the wind-load standard. [7] For separately located community safe rooms, FEMA guidance addresses a protective function excluded from the energy-harvesting campus. [8] Qualified civil review must identify the adopted editions applicable to the site, then relate the selected criteria to the building's occupancy and equipment hazards. Referring generally to dam or shelter construction leaves those project-specific requirements unresolved.

3.5.2 Functional separation of structures

A structure qualified for one hazard may lack the capacity required for another load case at the same site. Separating neighboring occupied properties from stored-energy equipment permits each civil element to follow acceptance criteria suited to its load path. The arrangement must also prevent a shared access or drainage failure from defeating both protective functions.

Table 12: Functional separation of structures

Civil element	Proposed purpose	Coupled hazard to examine	Acceptance deliverable
Primary building structure	Carry wind, gravity, and equipment-support loads	Wind pressure with debris impact and loss of openings	Site-specific structural design basis
Rotor foundation	Maintain alignment and transfer dynamic loads	Differential settlement, seismic motion, and neighboring module failure	Geotechnical and rotor-dynamic coupling assessment
Rotor containment	Limit consequences of credible rotor failure	Fragment trajectories, secondary impacts, and pressure rise	Energy-informed containment analysis and representative tests
Vacuum vessel	Maintain low gas density around the rotor	External pressure buckling and rapid air ingress	Pressure-boundary qualification and controlled venting strategy
Water barrier	Limit flood and groundwater intrusion	Hydrostatic uplift, scour, drainage failure, and saturated soil	Compound-flood design and inspection plan
Boundary to off-site community loads	Deliver electricity without residential or refuge use of the collection campus	Rotor, electrical, cryogenic, and process incidents affecting adjacent property	Physical separation and protected feeder interfaces
Materials-processing enclosure	Contain process hazards	Fire, hot material, carbon monoxide, and contaminated feedstock	Independent process-safety review

Across a vacuum enclosure, atmospheric pressure imposes a sustained external load, with rotor failure presenting a separate demand for fragment arrest and energy management. A shared wall would require qualification for both cases, as resistance to pressure alone provides no measure of containment capacity. Underground shielding must extend to credible escape paths along access shafts and neighboring rooms, preserving the complete failure boundary.

3.5.3 Tensegrity and compliant elements

Within a tensegrity assembly, prestress and geometry determine how the structure redistributes load and resists deformation. Elastic analysis of these effects supports assessment of a particular cable-and-strut arrangement, avoiding an assumed resilience advantage based solely on its structural class. [9]

For the campus exterior, an independently supported, replaceable screen offers a bounded test of compliant response and damage tolerance. Bearings would occupy a separately assessed foundation, preventing movement of that screen from closing rotor clearances or obstructing bypass paths. Maintenance access and replacement procedures would form part of the screen's acceptance conditions ahead of any permanent structural role.

Under changing load, cable slackening and connection slip alter the prestressed geometry, potentially reducing stiffness at the interval of greatest aerodynamic demand. Buckling interrupts a compression load path, with subsequent motion and repeated loading extending the fatigue assessment into damaged states. Prestress monitoring and replaceable connections support inspection only with an accompanying maintenance plan. Scaled experiments must preserve the stiffness and aerodynamic ratios controlling the selected response, limiting extrapolation to the behavior represented by those ratios.

3.5.4 Footprint and construction access

Eight circular envelopes of 30 m radius occupy approximately 22,620 m^2 without separation between the rotor envelopes. Containment walls and access passages increase that footprint, with drainage and maintenance areas requiring additional space outside the rotating envelopes. This calculated geometric minimum leaves the proposed four-block campus unresolved, as city-block dimensions vary and the eventual excavation and exclusion zones must follow the construction and hazard studies.

At the conceptual mass, fabrication and installation logistics constrain the rotor geometry ahead of service-load qualification. Segmenting the assembly introduces joints whose fatigue behavior and residual stresses require measurement; the assembled rotor must also permit inspection and balancing. With a composite overwrap, differential thermal strain loads the interface. The radial-stress solution and interface durability must be checked against a qualified manufacturing process. Adding the overwrap alone leaves the 340 RPM target unsupported.

3.6 Bearings, motor-generators, and cryogenic systems

Under continuous gravity loading, a large flywheel also subjects its supports to speed-dependent motion and electromagnetic forces. Stable clearance requires the bearing response to accommodate these loads throughout the permitted speed range and during rundown. At each rotor state, the conversion and cryogenic systems must satisfy the corresponding torque and heat budgets.

3.6.1 Gravity load and magnetic support

For one conceptual rotor, approximately 122.6 MN of static weight sets the initial support demand. Imbalance superimposes a rotating load, with electromagnetic transients changing the reaction during mode transitions. Dynamic load cases must capture these excursions around the gravity-supported equilibrium.

$$W_r = m_r g \simeq 1.226 \times 10^8 \text{ N}.$$

Under an ideal field arrangement, the magnetic-pressure relation supplies an optimistic force-per-area scale. Here μ_0 is vacuum permeability and B is the magnitude of the relevant field in the idealized gap.

$$p_B \simeq \frac{B^2}{2\mu_0}, \qquad A_{\text{ideal}} \simeq \frac{W_r}{p_B}.$$

At 1 T, using $\mu_0 \simeq 4\pi \times 10^{-7}$ H/m gives approximately 398 kPa of ideal magnetic pressure, corresponding to about 308 m^2 of support area per rotor. Raising the field to 2 T reduces that ideal area to roughly 77 m^2, subject to the field geometry and sustainable superconducting current. Stable support additionally requires restoring stiffness and a reaction structure capable of carrying the load with the assigned margins, leaving these ideal areas as scales for analysis without specifying bearing dimensions.

For a 300 kW flywheel, published superconducting-bearing development provides a component-scale demonstration of magnetic support. [10] Extending that duty to 12,500 tonnes requires a force-capacity assessment coupled to measurements of restoring stiffness and damping over the permitted displacement range. Thermal stability must sustain the support across the intended service cycle, with measured losses entering the storage budget at that same duty.

3.6.2 Bearing architecture choices

Across the permitted speed interval, a bearing must preserve clearance and support a qualified response to loss of control or refrigeration. Low friction improves retention only as part of an arrangement that also addresses vacuum conditions and touchdown loads. Comparing complete support systems connects energy loss to the stability and fault behavior required of the rotor.

Table 13: Bearing architecture choices

Option	Potential benefit	Limitation requiring measurement	Suggested research role
Mechanical bearings	Established machine interfaces and direct inspection	Contact loss, wear, lubrication, and extreme load rating	Baseline for smaller demonstrators
Active magnetic bearings	Controllable force and monitoring	Continuous control power, sensor integrity, and backup behavior	Candidate for modular rotors
HTS magnetic bearings	Contact-free support using superconducting behavior	Cryogenic load, flux motion, stiffness, and scale-up	Separate component experiment
Hybrid support	Assign different load components to suitable devices	Coupled stability and load-transfer behavior	Evaluate after single-support models are checked
Distributed rim support	Spread reactions across multiple locations	Alignment, varying support loads, and ring deformation	Exploratory structural and control study

A touchdown bearing must be assessed at the speed and stored energy associated with the fault it is intended to handle. Catching a rotor near rest gives no basis for assuming that the same device tolerates full-speed support loss. An engineered rundown should reduce energy along a qualified path, with an independent electrical sink available for the duty assigned to it. Passive containment must address the residual consequences after the limits of those active responses are reached.

3.6.3 Machine rating and torque envelope

During full-power testing, a 36.5 MW HTS propulsion motor demonstrated conversion at a specific mechanical and thermal duty. [11] Applying a machine of that class to reversible flywheel service requires confirmation of its torque-speed envelope and suitability for the proposed charging cycle.

Mechanical power P_{mech} equals the work rate associated with shaft torque τ and angular speed ω for the declared sign convention. This relation supplies the first consistency check between a machine's power rating and its rotor duty.

$$P_{\text{mech}} = \tau\omega.$$

At 36.5 MW and 120 RPM, the calculated torque is approximately 2.90 MN m. The same power at 340 RPM would imply about 1.03 MN m, although an actual machine requires an approved torque-speed curve. A useful simplified envelope combines converter power and maximum permissible torque.

$$|P_{\text{mech}}(\omega)| \leq \min(P_{\text{converter,max}}, \tau_{\max}|\omega|)\,.$$

Near rest, finite torque limits acceleration, invalidating a startup estimate based on constant power. With several machines coupled to one rotor, the torque distribution must avoid unacceptable torsional excitation and account for the redistribution following a drive fault. Machine count follows the qualified shaft-speed envelope and mechanical interface, with load sharing checked against each converter's thermal and fault duty. Procurement must tie these requirements to an identifiable supplier specification covering reversible storage service, including the approved torque-speed curve.

3.6.4 HTS and conventional machine comparison

Within superconducting machinery, reduced resistive loss and greater field loading may improve conversion, accompanied by refrigeration demand and a quench-recovery obligation. Assessing those competing effects requires the cryostat and cooling plant to share the same service boundary as the machine. Conventional equipment provides the reference for determining whether the superconducting branch improves net delivery or lifecycle cost.

Table 14: Comparison of HTS and conventional machines

Criterion	Branch using conventional machines	Branch using HTS machines	Method for project decisions
Electrical duty	Select a machine for the full variable-speed motoring and generating envelope	Confirm that the superconducting design supports the same envelope	Compare matched duty cycles
Mass and footprint	Allow space for machine, cooling, and service access	Include cryostats, refrigeration, and access in the comparison	Compare installed system dimensions
Losses	Measure copper, iron, converter, and mechanical losses	Add cryogenic wall power and relevant superconducting losses	Integrate losses over the actual dispatch history
Fault response	Coordinate converter and winding protection	Add quench detection and thermal recovery constraints	Test the combined electrical and thermal response

Criterion	Branch using conventional machines	Branch using HTS machines	Method for project decisions
Procurement	Obtain an application-specific supplier offer	Obtain a supplier commitment for the proposed rating and duty	Use current quotations and acceptance criteria
Maintenance	Evaluate inspection and repair access	Include cryogenic service and cooldown recovery	Estimate downtime from the proposed service plan

HTS equipment should proceed if it improves the installed system's performance or economics under the intended duty. A stationary campus may tolerate conventional machine mass more readily than a ship or aircraft, reducing the value of compactness relative to maintenance and cooling cost.

3.6.5 Refrigeration and vacuum losses

At 30 K under ordinary cryogenic pressure conditions, nitrogen is solid, excluding liquid-nitrogen cooling for that cold stage. A compatible cryocooler or other refrigeration arrangement must meet the heat load at the temperature required by the machine's field and current limits. This temperature choice links conductor performance directly to wall-power demand.

For a cold stage at temperature T_c rejecting heat to a warm stage at T_h, the Carnot coefficient of performance sets an ideal upper bound. With a measured practical coefficient of performance $\mathrm{COP}_{\mathrm{real}}$, the required wall power follows from the cold-stage heat-removal rate Q_c in watts.

$$\mathrm{COP}_{\mathrm{Carnot}} = \frac{T_c}{T_h - T_c}, \qquad P_{\mathrm{cryo,wall}} = \frac{Q_c}{\mathrm{COP}_{\mathrm{real}}}, \qquad \mathrm{COP}_{\mathrm{real}} < \mathrm{COP}_{\mathrm{Carnot}}.$$

At the cold stage, radiative exchange and conduction along supports or current leads contribute heat that refrigeration must remove continuously. Rotating interfaces and AC-field losses add duty-specific contributions, requiring a thermal budget checked against measured cooldown and steady-state behavior. Vacuum-pump electricity enters the same auxiliary accounting, with rapid air ingress assessed as a transient that may overwhelm the normal cooling arrangement.

Table 15: Ideal refrigeration limits for $T_h = 300$ K and $Q_c = 1$ kW. Practical wall power exceeds these reversible lower bounds.

Cold-stage temperature (K)	Carnot COP	Minimum wall power (kW)
30	0.111	9.00
50	0.200	5.00
77	0.345	2.90

3.6.6 Pre-spin and emergency energy release

Pre-spin places the machine inside a favorable control range, although the initial charging work occupies part of its energy capacity. Defining initial angular speed as ω_0, the fraction s_0 measures that speed relative to the permitted maximum; the quadratic kinetic model then determines stored energy and available headroom.

$$s_0 = \frac{\omega_0}{\omega_{\max}}.$$

$$\frac{E_0}{E_{\max}} = s_0^2, \qquad \frac{E_{\text{headroom}}}{E_{\max}} = 1 - s_0^2.$$

At half maximum speed, the rotor already contains one quarter of its maximum kinetic energy. This state preserves three quarters of the gross energy headroom, subject to the allowable lower speed and reserve policy. Pre-spin changes the starting state; it leaves the rotor's inertia unchanged.

During emergency rundown, the energy-release path must limit deceleration to the machine's qualified loads, with independent trips initiating the response. Deliberate contact with a sand bed and intentional HTS quenching are excluded from ordinary braking, leaving containment responsible for failure consequences beyond controlled rundown. If the utility grid rejects exported power, an independent electrical sink must accommodate both the discharge rate and the total energy assigned to it.

3.7 Storage portfolio and electrical dispatch

At the instant of a rapid change in demand, converter power and repeated cycling govern the flywheel's buffering duty. Extending support across hours or days introduces a different constraint, as standby loss consumes an increasing fraction of the energy held in reserve. Independent sizing assigns each service to equipment whose energy inventory and loss behavior match its required duration.

3.7.1 Separate fast power from extended energy service

For the same electrical load, extending discharge duration changes the relative contribution of power equipment and stored-energy capacity to cost. A fair portfolio comparison must also preserve cycling duty and recharge assumptions, as these alter lifetime delivery. Levelized storage-cost analysis provides a method for that comparison, with project prices requiring current supplier quotations. [12]

Table 16: Separate fast power from extended energy service

Architecture	Assigned role	Reason to investigate	Main limitation for this project	First comparison metric
Modular flywheel bank	Rapid buffering and frequent cycling	Mechanical state is directly related to speed	Material stress, standby losses, containment, and auxiliary power	Net cost per required power-buffering service
Electrochemical storage	Electrical reserve over the selected duration	Modular energy and converter sizing	Chemistry-specific safety, degradation, and replacement requirements	Delivered energy under the project duty cycle
Pumped or gravity-based storage	Longer-duration reserve where geography permits	Separate power equipment and stored potential energy	Site, civil works, water or lifting-system constraints	Site-specific cost and environmental impact
Thermal storage for cooling	Reduce the summer electrical peak indirectly	The original service centers on air-conditioning demand	Cooling capacity, heat leakage, and chiller performance	Avoided electrical peak at the required comfort level
Thermal storage for industrial heat	Decouple materials processing from short power pulses	Heat serves the process without reconversion to electricity	Process temperature and thermal cycling	Useful process heat delivered
Hydrogen or another fuel pathway	Explore extended reserve for a suitable site	Energy inventory is separable from converter rating	Conversion chain, storage safety, and utilization rate	Net delivered electricity and full lifecycle cost
Hybrid campus	Combine a small fast store with a separately sized reserve	Each subsystem serves a defined time scale	Integration and dispatch complexity	Reliability-adjusted cost of the entire service

Across these candidates, cost and efficiency rankings require a shared service definition and project-specific quotations. Dividing the flywheel bank into smaller modules improves access and isolation, although equivalent material and geometric assumptions give essentially the same stress-limited energy per unit mass. Modularity changes maintainability and failure consequences without removing the material constraint on specific energy.

3.7.2 Energy balances with clear boundaries

During exchange with the electrical bus, charging and discharging affect rotor energy on opposite sides of the conversion efficiencies. Let P_{ch} and P_{dis} denote those electrical powers at the flywheel converter interface, with $P_{\mathrm{rot,loss}}$ accounting for rotor-side dissipation. Their balance gives a reduced model for the changing mechanical inventory.

$$\frac{\mathrm{d}E_{\mathrm{rot}}}{\mathrm{d}t} = \eta_{\mathrm{ch}} P_{\mathrm{ch}} - \frac{P_{\mathrm{dis}}}{\eta_{\mathrm{dis}}} - P_{\mathrm{rot,loss}}.$$

At the campus bus, auxiliary electricity must enter once, either within a declared converter efficiency or as separate demand. Utility import $P_{\mathrm{grid,in}}$ and export P_{export} describe exchange beyond the site, with P_{load} denoting served community demand. Optional industrial consumption P_{process} follows a lower service priority, and P_{aux} sustains equipment within the chosen accounting boundary. Using this common electrical interface makes the balance consistent with the mechanical storage model.

$$P_h + P_{\mathrm{grid,in}} + P_{\mathrm{dis}} = P_{\mathrm{load}} + P_{\mathrm{process}} + P_{\mathrm{aux}} + P_{\mathrm{ch}} + P_{\mathrm{export}}.$$

Adding a storage technology requires its own charge and discharge terms at the same electrical boundary. During mode transitions, the solver must preserve energy and avoid counting auxiliary losses twice, rejecting simultaneous charge and discharge unless distinct devices and a justified physical duty require both flows.

3.7.3 Service priorities and reserve

Under forecast uncertainty, dispatch must preserve an agreed emergency reserve that covers the selected essential-load scenarios. Economic use draws on capacity beyond that allocation, with cycling wear included in the cost of repeated exchange. Optional process demand receives lower priority if interruption follows a qualified procedure protecting the equipment and product.

For cooling storage, thermal energy and electricity require separate units. If the chiller coefficient of performance is $\mathrm{COP}_{\mathrm{cool}}$, the electrical input required to produce a thermal cooling quantity Q_{cool} is approximated for a constant-COP comparison.

$$E_{\mathrm{elec,cooling}} \simeq \frac{Q_{\mathrm{cool}}}{\mathrm{COP}_{\mathrm{cool}}}.$$

Temperature-dependent performance and storage heat leakage should replace this simple ratio in the chronological campus model. A thermal store sized in MWh of cooling should never be presented as an equal number of MWh of electrical storage without that conversion.

3.7.4 Grid interfaces

Within a converter-fed network, the available fault current and grounding arrangement govern which protection devices detect and isolate a disturbance. Waveform requirements impose additional harmonic limits at the utility interface, with islanded service requiring authorization and controls suited to the separated network. Black start must provide an initial voltage and frequency reference without relying on an energized utility connection.

Selecting the distribution voltage requires resolution of these interconnection conditions, keeping the conceptual 13.2 kV and 24.9 kV options provisional until the utility and equipment interfaces have been assessed.

In grid-stabilizing machinery, stored rotational energy supports a service distinct from the dispatch of a variable-speed energy store over a large speed interval. The Moneypoint project description gives approximately 3,500 MWs for its added flywheel and 4,000 MWs for the combined rotating system, corresponding to about 0.972 MWh and 1.111 MWh. These energy-like inertia measures differ dimensionally from a mass moment of inertia in kg m^2, and their full values give no unrestricted multi-hour discharge rating. [13]

3.8 Debris recovery and optional materials processing

Among recovered storm debris, contamination and variable composition limit the fraction acceptable to a manufacturing process. Processing adds heat demand and product-quality requirements that must be satisfied independently of the electricity-harvesting function. Sharing infrastructure is justified only if qualified feedstock and process economics support that additional duty.

3.8.1 Recovery begins with feedstock qualification

Within a captured debris stream, damaged building materials may introduce constituents incompatible with the intended product, and mixed soil offers no assurance of silica purity. Waste characterization must identify these constituents prior to routing accepted fractions toward reuse. Treated wood and recovered metals require handling suited to their composition and destination, preventing a common intake from implying a common processing route.

Fine particulate requires controlled collection, particularly if salts or biological contamination complicate handling. Estimating manufacturing yield requires measured captured mass and composition, with the electricity used for collection entering the process economics alongside the recoverable material value.

An unidentified contaminant may damage processing equipment or introduce a hazard into its exhaust, making isolation and sampling essential at the collection boundary. Characterization determines which fractions permit reuse and which need a documented disposal route. Safe intake flow and machinery protection govern collector design; increasing collection solely to supply a furnace would require a separate environmental and economic assessment.

3.8.2 Glass and silicon are different process routes

With a suitable silica-containing mixture, melting produces glass under composition-dependent conditions, leaving the silicon chemically bound in the material. Producing elemental silicon requires reduction, followed by purification matched to solar or electronic impurity limits. The resulting process chain extends well beyond melting, linking its feasibility to feedstock quality and successive purification performance. [14]

For the reduction furnace, a simplified net carbothermal reaction specifies the overall material balance, with intermediate chemical species requiring a more detailed process model to predict yield and heat demand.

$$\mathrm{SiO_2} + 2\mathrm{C} \rightarrow \mathrm{Si} + 2\mathrm{CO}.$$

Carbon monoxide from the reduction route requires process-specific hazard controls, with reducing-agent consumption and emissions included in its lifecycle inventory. For semiconductor-grade material, the impurity specification governs purification beyond the furnace and the contamination controls surrounding each subsequent stage. Until these requirements are quantified for captured feedstock, polysilicon production from storm sand is an exploratory process branch.

3.8.3 Process options and boundaries

At the process boundary, feedstock composition determines which recovery routes are chemically compatible with the collected material. Product acceptance then constrains the required treatment and its energy consumption. Accounting for emissions along that route connects the recovered output to a defined destination and environmental comparison.

Table 17: Process options and boundaries

Branch	Candidate input	Useful product or service	Required qualification	Proposed priority
Material sorting and recycling	Captured debris after hazard screening	Recovered metal, compatible mineral fractions, or properly handled waste	Composition, contamination, and destination acceptance	First materials-related pilot
Glass or glass-ceramic research	Assayed, prepared mineral feed	A defined glass or ceramic formulation	Composition, melting behavior, durability, and emissions	Optional laboratory phase
Thermal process store	Electricity converted to heat	Stable heat supply to a selected process	Temperature range, insulation loss, and cycle durability	Compare ahead of any foundry installation
Metallurgical silicon	Qualified silica and reducing agent	Silicon meeting a declared metallurgical specification	Reduction chemistry, gas treatment, and impurity control	Separate industrial feasibility study
Solar or electronic silicon	Qualified intermediate material	High-purity silicon	Dedicated purification and contamination management	Outside the initial campus scope

For a furnace requiring continuous heat, a short storm pulse leaves an energy gap after collection ends. A thermal buffer may bridge that gap, or an independent electrical supply may sustain heating directly, with either arrangement entering the process cost and energy balance. Separate ventilation and fire protection must address the furnace atmosphere within an industrial enclosure isolated from neighboring occupied properties and flywheel-containment hazard zones.

3.9 Multiphysics model hierarchy

Across the campus, atmospheric motion and structural deformation exchange forces on scales distinct from electrical dispatch. Rotor dynamics and electromagnetic heating add faster local responses whose resolution governs stability and material limits. A hierarchy of coupled models assigns each decision to the least costly representation that preserves its controlling physics, concentrating refinement where the estimated error exceeds the permitted bound.

3.9.1 Model the questions at the scale that answers them

An inconsistent energy inventory or excessive centrifugal stress excludes a rotor configuration regardless of the atmospheric model's detail. Analytic storage balances and mechanical screens must accordingly precede expensive field calculations, resolving whether the proposed service has a physically plausible starting point. Further refinement targets uncertainty that these simpler calculations leave open, such as local stress concentration or collector pressure loss.

Table 18: Model the questions at the scale that answers them

Model level	Question answered	Suggested representation	Main interface to the next level
Service and resource model	How much energy must be delivered, and how what charging frequency is available?	Chronological load, weather, and availability model	Power histories and scenario probabilities
Campus network model	Where does energy flow during each state?	Lumped electrical, thermal, and storage balances	Converter setpoints and thermal loads
Collector model	What forces and net power arise within the tested envelope?	Finite-volume CFD with suitable turbulence treatment	Surface pressure, torque, debris trajectories, and flow rates
Structural model	Do components sustain the required loads and cycles?	Finite-element elasticity, contact, fatigue, and selected nonlinear behavior	Deflection, allowable speed, and failure indicators
Machine and bearing model	What force, torque, loss, and stability are available?	Electromagnetic and rotor-dynamic models	Force maps, torque curves, and loss maps
Thermal and cryogenic model	Do temperatures stay within the intended limits?	Heat equation or calibrated thermal network	Temperature-dependent material and machine properties
Uncertainty model	How do uncertain inputs alter performance and risk?	Bayesian inference, ensembles, and rare-event estimation	Credible intervals and risk-informed design comparisons

3.9.2 Flow equations and regime selection

Within a constant-density Newtonian fluid, pressure enforces the incompressible velocity constraint, coupling motion across the domain. The governing momentum balance uses velocity $\boldsymbol{u}$, dynamic viscosity μ, and body force per unit volume $\boldsymbol{f}$, with pressure denoted by p.

$$\nabla \cdot \boldsymbol{u} = 0,$$

$$\rho_a \left(\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} \right) = -\nabla p + \mu \nabla^2 \boldsymbol{u} + \boldsymbol{f}.$$

In a compressible regime, density evolves with momentum, requiring an energy equation and an equation of state to close the thermodynamic model. Mach number provides an initial indicator of compressibility, although the permitted approximation must also preserve density variations affecting the selected output. Pressure transients and heat release may change that assessment, preventing execution time alone from determining the governing equations.

Comparing inertial transport over characteristic length L with viscous diffusion gives Reynolds number Re for speed U and kinematic viscosity ν_a. Dividing that flow speed by sound speed a gives Mach number Ma, which measures a different aspect of the regime.

$$\mathrm{Re} = \frac{UL}{\nu_a}, \qquad \mathrm{Ma} = \frac{U}{a}.$$

For the illustrative 100 m collector length, a 60 m/s wind and assumed kinematic viscosity of 1.5×10^{-5} m^2/s give a Reynolds number near 4×10^8. These inputs specify a comparison case, with field application requiring site measurements. At that scale, a practical full-domain engineering model would represent unresolved turbulent motion using a declared treatment; reducing the grid size alone leaves that unresolved contribution outside the calculated field.

3.9.3 Conservative discrete flow and graph connectivity

Within cell i, multiplying conserved-state density $\boldsymbol{q}_i$ by volume V_i gives the inventory whose change must match the net transfer and local production. Source density $\boldsymbol{s}_i$ contributes inside the cell, with outward face-integrated flux $\boldsymbol{F}_{ij}$ transferring the quantity to neighboring cells in $\mathcal{N}(i)$. Equal and opposite fluxes across each shared face preserve the finite-volume balance.

$$\frac{\mathrm{d}(V_i \boldsymbol{q}_i)}{\mathrm{d}t} + \sum_{j \in \mathcal{N}(i)} \boldsymbol{F}_{ij} = V_i \boldsymbol{s}_i, \qquad \boldsymbol{F}_{ij} = -\boldsymbol{F}_{ji} \text{ for a shared internal face.}$$

Adjacency identifies which cells exchange flux, with face geometry and constitutive laws setting the magnitude and units of each exchange. Across a refinement interface, the summed fine-face flux must match the corresponding coarse-face transfer to preserve the integral balance. AMReX provides an established implementation route for block-structured refinement and parallel data management, making it a candidate backend once the required physical operators have been defined. [15]

3.9.4 Structural and rotor-dynamic equations

Within a rotating structure, acceleration produces inertial force via the mass matrix M_s, with damping represented by C_s. A compatible formulation adds the speed-dependent gyroscopic matrix $\mathsf{G}(\omega)$, capturing coupling introduced by rotation. The internal-force vector $\boldsymbol{f}_{\text{int}}$ evaluates structural response at the current displacement $\boldsymbol{d}$, temperature, and spin state.

Applied aerodynamic force $\boldsymbol{f}_{\text{aero}}$ and bearing force $\boldsymbol{f}_{\text{bearing}}$ enter separately; their coupling is checked directly. The term $\boldsymbol{f}_{\text{other}}$ contains the additional loads declared for that case.

$$\mathsf{M}_s\ddot{\boldsymbol{d}} + [\mathsf{C}_s + \mathsf{G}(\omega)]\,\dot{\boldsymbol{d}} + \boldsymbol{f}_{\text{int}}(\boldsymbol{d}, T, \omega) = \boldsymbol{f}_{\text{aero}} + \boldsymbol{f}_{\text{bearing}} + \boldsymbol{f}_{\text{other}}.$$

After recovery of the analytic ring and disk limits, resolving the actual rotor geometry reveals stress redistribution around joints and changes in section. Prestress fixes the initial load state, with directional material response requiring an appropriate anisotropic law where applicable. Contact and discontinuities need local constitutive treatment, and rotor-dynamic assessment must test critical-speed crossings and whirl beyond the static stress calculation. Repeated-load fatigue then uses the resulting stress history over the proposed duty.

3.9.5 Electromagnetic and thermal coupling

In a magnetoquasistatic regime, neglecting displacement current simplifies the field equations subject to the relevant temporal and spatial scales. The selected material law relates magnetic field intensity $\boldsymbol{H}$ in A/m to flux density $\boldsymbol{B}$ in tesla, with electric field $\boldsymbol{E}$ in V/m and current density $\boldsymbol{J}$ in A/m^2 describing electrical response. Applying these relations to a rotating conductor requires a consistent reference frame and treatment of its motion.

$$\nabla \times \boldsymbol{H} = \boldsymbol{J}, \qquad \nabla \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}, \qquad \nabla \cdot \boldsymbol{B} = 0.$$

Under the isotropic local HTS approximation, E_c is the criterion electric field in V/m used to define critical current density $J_c(B, T)$ in A/m^2. The dimensionless exponent n_E controls how sharply electric field rises with current. Calibration must match the material and regime under study. Current magnitude J is defined from $\boldsymbol{J}$, and the vector law aligns the local electric field with that current for this screening model.

$$J = \|\boldsymbol{J}\|.$$

$$\boldsymbol{E} = E_c \left(\frac{J}{J_c(B, T)} \right)^{n_E} \frac{\boldsymbol{J}}{J}, \qquad J > 0.$$

Within HTS tape, field orientation and mechanical strain alter the critical current, requiring more detail than the isotropic screening law for a qualified machine. Current sharing also changes where heat is deposited as the conductor approaches its limit. The electromagnetic contribution q'''_{em}, measured in W/m^3, follows the field-current work rate in the material rest frame; for a moving conductor, the comoving electric field separates this heating from mechanical work.

$$q'''_{\text{em}} = \boldsymbol{J} \cdot \boldsymbol{E}.$$

For heat transfer within the material, material density ρ_m in kg/m^3 and specific heat c_p in J/(kg K) determine the heat capacity per unit volume. The conductivity tensor k in W/(m K) maps the temperature gradient into conductive heat flux. Writing the total volumetric heat source as q''' in W/m^3 gives the heat equation for a stationary solid or material-coordinate formulation. An Eulerian model of moving material includes the corresponding advective derivative.

$$\rho_m c_p \frac{\partial T}{\partial t} = \nabla \cdot (\mathsf{k} \nabla T) + q'''.$$

At a coupled interface, equal and opposite force transfer must also be consistent with the velocities used to compute mechanical work. Thermal exchange requires the heat leaving one submodel to enter the receiving model with the same sign convention and magnitude. Recording an interface energy residual exposes discrepancies introduced by interpolation or loosely timed exchanges, preventing their appearance as a fictitious efficiency improvement.

3.10 Feasibility of the proposed GPU numerical approach

On a GPU, repeated numerical kernels offer enough parallel work to offset overhead if their data stay close to the computation. Setup and synchronization may erase that advantage in a small or irregular problem, making complete-solve timing essential to backend selection. The proposed CuPy implementation treats acceleration as a measured property at matched accuracy, without assigning a speed guarantee to the full multiphysics program.

3.10.1 Assessment of the combined approach

For selected intensive calculations, an open-source implementation combines GPU field arrays with connectivity derived from the physical discretization. Different governing equations require compatible numerical methods, giving each part of the hybrid solver a defined mathematical responsibility. Feasibility of this software organization leaves its accuracy and performance subject to measurement on the proposed cases.

Metropolis–Hastings then samples uncertainty around the computed response, leaving physical time evolution to the conservative solver. CuPy supplies an execution backend for supported array and sparse work, with full-system accuracy and performance still requiring demonstration.

Hou and colleagues demonstrate CuPy with vectorized sparse matrix-vector products in large-scale topology optimization, providing a direct structural-design example for the proposed acceleration branch. [16] That application supports testing comparable project kernels, although its measured performance cannot be transferred unchanged to tornado CFD or superconducting-bearing calculations. Each workload introduces its own solver and memory behavior, which must be measured within the campus software.

For a model that requires block-structured refinement, AMReX offers a developed route for managing adaptive meshes and parallel field data. [15] PETSc/TAO addresses the algebraic and optimization side of large GPU-based calculations, providing a complementary backend if convergence or distributed execution exceeds the custom prototype's scope. [17] These alternatives allow the project to retain its physical interfaces without committing every numerical task to one Python implementation.

Repeated numerical work justifies GPU execution only where the workload exposes enough parallelism and data reuse to offset setup and transfer cost. Batched scenarios expose independent work across designs, as persistent field arrays reduce transfers during repeated local updates. Reusing a suitable sparse operator amortizes setup across many solves; alternatively, a reduced-order model may lower the

cost of repeated uncertainty likelihoods. Efficiency must still appear as lower elapsed time, energy use, or memory demand at matched accuracy, and the combined implementation has yet to undergo that benchmark.

3.10.2 What each term contributes

Within the hybrid computation, graph connectivity identifies permitted interactions, with field arrays storing the quantities exchanged along them. Deterministic solvers advance the physical equations, and an outer statistical calculation explores uncertainty in their inputs and outputs. Adaptivity changes resolution or representation within the accepted error budget, preserving these mathematical roles across backend choices.

Table 19: What each term contributes

Term in the requested approach	Precise interpretation	Useful project role	Limitation to preserve
Adjacency-based	Store or generate local connections between mesh entities or physical components	Face fluxes, finite-element connectivity, thermal networks, and circuit topology	Connectivity alone omits geometry, constitutive laws, and boundary conditions
Hybrid	Select compatible methods for different domains and scales	CFD plus structural FEM, network dispatch, and calibrated reduced models	Interface exchanges must conserve work, heat, and mass as applicable
Numerical tensor	Store multidimensional physical fields as arrays; optionally approximate them with tested low-rank representations	GPU field updates, ensembles, and data organization	Compression and quantum computation require separate mathematical constructions
Metropolis–Hastings	Sample a declared probability distribution using an acceptance rule	Parameter inference and conditional rare-event studies	Physical time evolution still requires a conservative deterministic solver
CuPy GPU execution	Implement supported array and sparse operations on the GPU	Repeated dense batches, local kernels, and selected sparse operations	Data movement, memory bandwidth, and supported APIs limit benefit
Automatic adaptation	Adjust allowed numerical settings using diagnostics	Mesh, time step, precision, batch size, and backend selection	Safety limits and model assumptions require separately controlled authority
Automatic sparse policy	Assemble a sparse operator only if it fits the resource and accuracy plan	Repeated pressure, diffusion, elasticity, or network solves	Matrix-free execution may be preferable for a regular stencil

3.10.3 Adjacency is a physical data structure

Across a discretized domain, neighboring cells exchange physical quantities along connections defined by geometry and boundary conditions. Representing cells or components as vertices gives a graph with vertex set $\mathcal{V}$ and edge set $\mathcal{E}$, whose binary adjacency entries A_{ij} record the permitted interactions.

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}).$$

$$A_{ij} = \begin{cases} 1, & \text{if vertices } i \text{ and } j \text{ share a modeled interaction,} \\ 0, & \text{otherwise.} \end{cases}$$

For a simple diffusion network with positive edge conductances stored in diagonal matrix W, an oriented edge-by-vertex incidence matrix B gives a weighted graph Laplacian. Boundary conditions determine its null space and solvability.

$$\mathsf{L} = \mathsf{B}^{\mathsf{T}}\mathsf{W}\mathsf{B}.$$

In a thermal network, positive edge conductance relates a temperature difference to heat flow; a resistive electrical network gives the analogous current-voltage interpretation. Those physical weights make the Laplacian well-defined, and an unweighted connectivity matrix leaves the units and constitutive response unspecified. General fluid or elastic operators require their own discretizations, with incompressible flow retaining a globally coupled pressure constraint. Updating only a selected subgraph against stale surrounding values risks missing propagated disturbances or violating that pressure condition.

On regular grids, index arithmetic generates neighboring locations without allocating a separate graph object for each cell. Small campus networks use an authoring tool such as NetworkX to inspect component connections and detect topology errors ahead of execution. The field solver should instead receive compact numerical connectivity buffers or matrix-free indexing, preserving that connectivity without Python-object overhead at every degree of freedom.

3.10.4 Tensor arrays and optional compression

In a discretized velocity field, spatial axes locate each value and a component axis distinguishes the vector directions. A leading scenario axis accommodates independent realizations within the same array layout, facilitating batched computation. This classical tensor organization supplies a data container without introducing compression or a quantum algorithm.

$$\mathcal{U} \in \mathbb{R}^{N_b \times N_c \times N_z \times N_y \times N_x},$$

Here N_b is the number of scenarios in the batch, and N_c counts the stored components per grid location. Grid dimensions N_x, N_y, and N_z specify the spatial extent of the array. At a fixed byte width b_v per value, raw dense storage follows the stated product of byte width and tensor extents, excluding allocation overhead and solver workspace.

$$M_{\text{array}} = b_v N_b N_c N_z N_y N_x.$$

For a field with correlated variation across dimensions, a low-rank representation may reduce storage by retaining fewer independent modes. In the Tucker approximation, an order-d array $\mathcal{X}$ is represented by a core $\mathcal{C}$ and factor matrices $\mathsf{U}^{(j)}$, with r_j columns specifying the retained rank in mode j. The approximation error then determines whether these ranks preserve the physical quantities required from the full array. [18]

$$\mathcal{X} \approx \mathcal{C} \times_1 \mathsf{U}^{(1)} \times_2 \mathsf{U}^{(2)} \cdots \times_d \mathsf{U}^{(d)}.$$

$$N_{\text{stored,Tucker}} = \prod_{j=1}^{d} r_j + \sum_{j=1}^{d} n_j r_j, \qquad N_{\text{stored,dense}} = \prod_{j=1}^{d} n_j.$$

In the storage count, n_j denotes the original extent of dimension j, and $\times_j$ specifies multiplication along that tensor mode. Resource savings require retained ranks that reduce both storage and execution at the accepted error, including the cost of constructing or adapting the approximation.

Near a sharp front or localized contact, concentrated spatial variation may require additional modes, with intermittent vortices changing the necessary ranks over time. These structures limit the compression available at a fixed physical tolerance.

Ordinary field arrays provide the reference against which compression error is assessed at each design-driving observable. A small global error may conceal an unacceptable shift in peak stress, requiring local checks ahead of using the approximation to select rotor speed. For risk calculations, corresponding tests must preserve failure indicators in the distribution tails, where a small field perturbation may move a sample across the failure boundary.

3.10.5 Where GPU execution is likely to be worth testing

Across large repeated kernels, regular memory access helps parallel hardware amortize launch and transfer costs. Small control tasks or frequent topology changes may instead favor CPU execution, particularly if the required data already reside on the host. Measured cost across the complete workflow determines placement at the accepted accuracy.

Table 20: Where GPU execution is likely to be worth testing

Workload	Candidate GPU strategy	Comparison baseline	Reason for caution
Many rotor-design scenarios	Vectorized arrays for energy, stress screens, and constraints	Vectorized CPU calculation	Small batches may be faster on the CPU
Local finite-volume updates	GPU kernels using neighboring values and persistent field buffers	CPU reference stencil	Boundary handling and memory traffic may dominate
Repeated elliptic solves	CSR or matrix-free operator with an appropriate preconditioner	CPU sparse solver and PETSc backend	Convergence and global communication determine time to solution
Thermal networks	Batched solves or sparse time integration	CPU network model	Small graphs offer little GPU parallelism
Structural design sweeps	Repeated assembly patterns and sparse operator evaluation	Established finite-element backend	Contact and changing topology raises setup cost
Electromagnetic parameter maps	Batched material laws and selected field solves	Independent machine-model reference	Strong nonlinearity and anisotropy may demand specialized methods
Bayesian likelihood evaluation	Parallel independent chains or batched forward models	CPU sampler using the same posterior	Successive states of one chain are dependent
Reduced-order dispatch ensembles	Batched time histories and constrained optimization subproblems	CPU vectorized model	Branch-heavy mode logic may reduce GPU efficiency
Visualization preparation	On-device reductions, slicing, and decimation	CPU post-processing	Full-field transfers erase compute gains

3.10.6 Open-source scope and dependency boundaries

For a CuPy-based release, the application's MIT License applies to CuPy itself, with NVIDIA runtime components governed by their respective terms. [19] [20] A reproducible distribution must identify these dependency boundaries alongside the application code, preserving the licensing information associated with each included component.

Across repeated runs, backend and package versions determine the available interfaces, with build options selecting the kernels and solvers exposed by that configuration. Recording these settings beside test results ties each output to an executable environment whose driver and runtime compatibility has been checked. [21] Separating physics interfaces from accelerator kernels supports migration, with any replacement backend still requiring numerical comparison and performance measurement.

3.11 Automatic adaptivity and the sparse-computation control knob

Under a fixed physical tolerance, an adaptive solver may trade discretization effort against iteration cost and memory demand. A sparse representation is admissible only if its complete workspace fits the measured budget and its solution preserves the required accuracy. Automatic selection then compares total solve cost across the admissible candidates.

3.11.1 Adaptation should optimize a declared error budget

Within an approved set of numerical configurations, automatic planning seeks lower execution cost without changing the physical question. Estimated error restricts approximation, with measured memory use excluding candidates that exceed device capacity. The governing regime determines which models enter the set, keeping a resource shortage from silently redefining the simulated physics.

$$\min_{\mathcal{A}\in\mathcal{A}_{\text{approved}}} T_{\text{solve}}(\mathcal{A}) \quad \text{subject to} \quad \widehat{e}_Q(\mathcal{A}) \le \varepsilon_Q, \quad M_{\text{peak}}(\mathcal{A}) \le B_{\text{run}}, \quad \text{required conservation and stability checks pass.}$$

In the constrained planning problem, $\mathcal{A}$ denotes a numerical configuration selected from the approved set $\mathcal{A}_{\text{approved}}$. Its estimated error $\widehat{e}_Q$ in the chosen quantity of interest must stay below ε_Q, and peak allocation M_{peak} must fit the run budget B_{run}. Elapsed time T_{solve} includes setup as well as solution and required data transfers. If every available candidate violates a requirement, the planner should save a checkpoint and stop or invoke an approved alternative resource plan.

3.11.2 Proposed controls at the top of the implementation

At startup, typed control settings define the permitted numerical choices and resource limits, making the planner's authority inspectable. Proposed default values require tuning against representative measurements, with the selected configuration retained in the run record.

Table 21: Proposed controls at the top of the implementation

Control knob	Suggested initial value or choices	Purpose	Guard condition
ADAPT_MODE	auto by default after case checks; review and manual available	Select approved numerical changes automatically	Only approved methods may be selected
SPARSE_POLICY	auto, force_csr, matrix_free, or cpu; default auto	Automatically activate a budgeted sparse approach for suitable heavy tasks	Forced CSR still must satisfy memory and solver checks
GPU_MEMORY_FRACTION	0.60	Limit planned use of measured free memory on the GPU	Account for all persistent and transient allocations
GPU_RESERVE_GIB	2.0 for the development example	Preserve headroom for runtime variation	Reduce workload or use CPU if the device cannot support this reserve
SPARSE_MEMORY_FRACTION	0.20	Limit the sparse subsystem's share of free memory	Also limited by memory left after other allocations
SPARSE_WORKSPACE_FACTOR	3.0 as an initial estimate	Estimate operator plus assigned working memory	Replace the estimate with measured high-water use
SPARSE_MIN_GAIN	1.20	Require a measured or conservatively predicted time benefit ahead of automatic switching	Compare complete repeated-solve cost at matched accuracy
SPARSE_DENSITY_SCREEN	0.01 as a candidate-screen value	Avoid considering dense-like matrices for CSR by default	Density alone never determines the backend
MAX_CELLS	Derived from the workload memory model	Limit mesh size	Exceeding the cap triggers resource reassessment
MAX_AMR_LEVELS	Case-specific	Bound refinement depth	Adequacy of the finest level must still be checked
REFINE_TOL and COARSEN_TOL	Quantity-specific with distinct thresholds	Refine important regions and avoid switching chatter	Coarsening must preserve conservation and error limits
ADAPT_MIN_DWELL_STEPS	A positive case-specific interval	Avoid repeated backend or mesh oscillation	Immediate protective numerical checks override the dwell interval
CFL_TARGET	0.4 only as an initial advection example	Set a starting time-step target	Actual stability limit follows the selected scheme
PRECISION_POLICY	fp64_reference or mixed_checked	Use double precision as the initial comparison	Lower precision requires error and convergence agreement
TENSOR_POLICY	arrays by default; optional low_rank_auto	Separate ordinary field storage from approximation	Rank limits and quantity-specific error checks govern compression
MCMC_ADAPT_POLICY	warmup_only	Tune proposals ahead of retained sampling	Production sampling uses a fixed target and approved kernel
RANDOM_SEED	Recorded user-selected integer	Reproduce stochastic initialization and sampling	Separate streams for chains and independent experiments
SAFETY_PROFILE	locked_external	Keep plant protection outside numerical tuning	Software knobs cannot relax physical protection settings

3.11.3 Sparse storage is governed by bytes and operations

For an assembled sparse operator, memory use includes both numerical values and the indices locating them. In an $N \times N$ CSR matrix, n_{nz} values occupy b_v bytes each, column indices occupy b_i bytes each, and $N + 1$ row pointers identify row boundaries. The raw estimate excludes allocation overhead and the additional arrays required by the chosen solver.

$$M_{\mathrm{CSR}} \simeq n_{\mathrm{nz}}(b_v + b_i) + (N + 1)b_i, \qquad M_{\mathrm{dense}} = N^2 b_v.$$

With an average of k stored entries per row, nonzero count grows approximately as kN, giving linear CSR storage at fixed value and index widths. A fully stored dense operator instead grows quadratically, which makes representation cost diverge with problem size under the same scalar precision. Iteration count and preconditioner behavior require a separate estimate of total solution cost.

$$n_{\mathrm{nz}} \simeq kN, \qquad M_{\mathrm{CSR}} \propto N, \qquad M_{\mathrm{dense}} \propto N^2 \quad \text{for fixed } k, b_v, b_i \text{ as } N \text{ grows.}$$

Table 22: Sparse storage is governed by bytes and operations

Example operator	Unknowns	Stored entries	Raw storage	Qualification
Seven-entry-per-row CSR, FP32 values and int32 indices	1,000,000	Approximately 7,000,000	60,000,004 bytes, about 57.22 MiB	Excludes solver vectors, preconditioner, and temporary assembly data
Seven-entry-per-row CSR, FP64 values and int32 indices	1,000,000	Approximately 7,000,000	88,000,004 bytes, about 83.92 MiB	Same sparsity with larger values
Dense FP32 matrix	1,000,000	1,000,000,000,000	4 TB, about 3.64 TiB	Storage is excessive for this local operator
Matrix-free regular stencil	1,000,000	Operator entries generated as needed	Field and coefficient storage only, with problem-specific size	May outperform CSR by avoiding index and matrix-value traffic

On a multiphysics mesh, one cell may carry several scalar unknowns, making the algebraic system larger than the physical cell count suggests. Counting the actual degrees of freedom and block structure aligns the memory plan with the arrays that the solver allocates.

3.11.4 A concrete "reasonably sized" sparse-memory policy

Prior to case allocation, the planner measures free GPU memory V_f, providing the reference for its run and sparse-subsystem limits. The run fraction f_{run} corresponds to `GPU_MEMORY_FRACTION`, with reserved bytes B_{reserve} preserving uncommitted capacity. A separate fraction f_{sparse} limits the sparse allocation after persistent fields B_{fields} and other assigned storage B_{other} have been accounted for, keeping that subsystem within the total run budget.

$$B_{\text{run}} = \max[0, \min(f_{\text{run}} V_f, V_f - B_{\text{reserve}})] \,.$$

$$B_{\text{sparse}} = \max[0, \min(f_{\text{sparse}} V_f, B_{\text{run}} - B_{\text{fields}} - B_{\text{other}})] \,.$$

During initial planning, estimate assigned sparse workspace using a factor $\gamma_{\text{work}} \geq 1$, then replace that estimate with measured allocations for the selected solver and preconditioner.

$$M_{\text{sparse,planned}} = \gamma_{\text{work}} M_{\text{CSR}}, \qquad M_{\text{sparse,planned}} \leq B_{\text{sparse}}.$$

A worked memory example with 20 GiB free, a 60% run fraction, and a 2 GiB reserve gives a 12 GiB run budget. If other fields and allocations need 7 GiB, a 20% sparse fraction permits at most 4 GiB for the assigned sparse subsystem. That is a planning allowance, with measured high-water usage still required ahead of a long run.

During factorization, fill-in may expand sparse storage well beyond the assembled operator, requiring solver-specific workspace estimates. Iterative methods retain Krylov vectors according to their algorithm and restart policy, with multilevel preconditioners adding coarse operators and transfer maps. Communication buffers enter distributed runs as a further allocation, making measured high-water use essential to replacing the provisional workspace factor of three.

3.11.5 Automatic activation procedure

Boundary conditions and discrete operator properties constrain the admissible solver family, giving backend selection a mathematical basis. Expected reuse then determines how many solves share assembly and preconditioner setup, with the quantity-of-interest tolerance fixing the required accuracy. Automatic switching must compare candidates under these same conditions.

For a representative pilot, sparse and matrix-free candidates should be compared with a CPU reference, with setup timed separately from iterations. Required transfers belong in the elapsed time, and measured peak allocation determines whether the candidate fits its assigned memory budget.

After correctness and memory checks pass, the planner activates sparse execution only if matched-accuracy timing demonstrates the configured benefit. For a regular local stencil, matrix-free indexing may reproduce the discrete equations without storing the assembled operator, offering a candidate whose cost must enter the same comparison. Reuse amortizes setup, although changes in geometry or material state require replanning if they alter connectivity or convergence behavior.

Table 23: Automatic activation procedure

Planning result	Numerical action	Required record
Regular local stencil with cheap coefficient access	Prefer a tested matrix-free kernel	Timing, memory, and error against the same discrete operator
Irregular sparse operator reused many times	Consider CSR with a suitable preconditioner	Assembly cost, setup cost, iteration history, and peak memory
Small system or low batch count	Permit CPU execution	End-to-end comparison including transfers
Sparse solver exceeds its assigned budget	Try a tested lower-memory solver or backend	Reason for fallback and unchanged accuracy target
Whole domain exceeds one GPU	Use an approved domain-decomposed solver or a larger resource	Partition, halo, and global-solver cost
Every candidate fails the accuracy or memory requirement	Save a checkpoint and stop the case	Unmet requirement and recoverable state

Memory pressure gives the planner no authority to remove modeled interactions or reduce physical loading. Discarding graph edges or cells changes the represented domain, and altering material properties changes the problem itself. Mesh coarsening is permitted only after its error estimate and conservative transfer satisfy the declared acceptance conditions; otherwise, the case must use another approved resource plan or stop.

3.11.6 CuPy API compatibility and solver choice

Within a selected CuPy release, sparse classes and solver routines expose capabilities that the backend adapter must check prior to allocating the case. CSR and CSC cover common compressed representations, with coordinate and diagonal formats serving particular construction needs; support for block formats requires a separate capability check.

For sparse-array semantics, `@` denotes matrix multiplication and `*` denotes elementwise multiplication, making the object's class essential to interpreting an expression. Index-width restrictions also vary by operation, including documented int32 requirements for selected solvers. [22]

Initially, the CuPy backend should use supported CSR operations or a matrix-free kernel. A block-sparse formulation is assigned to a backend that supports the needed representation, such as an appropriately configured PETSc implementation, or to a separately tested custom block kernel. Capability detection should occur at startup, ahead of allocating a large mesh. [17]

Conjugate gradients may suit a symmetric positive-definite system after its preconditioner has been checked for compatibility. General nonsymmetric operators require a different compatible solver, and a pure-Neumann pressure problem requires null-space treatment or a justified pressure reference. The stopping criterion should relate algebraic residual to the accuracy required of the discretized physical solution. Reaching an iteration limit must be reported as an unsuccessful solve unless the acceptance criteria have already been satisfied.

3.11.7 Adaptivity across regime, scale, and complexity

At the chosen mesh scale, physical regime determines the operator and conditioning influences the effort required to solve it. Uncertainty sampling adds forward evaluations whose cost must share the

same resource plan. Adaptive controls respond to these measured demands within approved limits on resolution, precision, and statistical error.

Table 24: Adaptivity across regime, scale, and complexity

Adaptation dimension	Diagnostic	Allowed response	Protected assumption
Flow regime	Mach number, density variation, separation, and unresolved gradients	Select from pre-approved flow models or request review	Governing equations match the intended physical regime
Spatial scale	Error indicators and quantity-of-interest sensitivity	Refine or coarsen using conservative transfer	Mesh changes preserve integral balances
Time scale	Stability estimate, local error, and event timing	Adjust time step and solver tolerances	Fast load transitions receive adequate resolution
Algebraic complexity	Condition estimates, iteration growth, and matrix structure	Change preconditioner or solver backend	The same discrete system is solved to the declared accuracy
Memory pressure	Measured free memory and high-water allocations	Reduce batch size or choose a lower-memory approved method	Physical domain and accuracy target are preserved
Precision	Residual stagnation and comparison against FP64	Increase precision or use checked mixed precision	Error stays within the quantity-specific budget
Tensor rank	Compression error and rank growth	Raise rank, disable compression, or change representation	Tail loads and local peaks retain adequate accuracy
Statistical complexity	Chain mixing and effective sample size	Extend sampling or tune during warmup	Retained samples target the declared distribution

In a one-dimensional advection discretization with a finite CFL stability limit, grid spacing Δx and characteristic velocity magnitude $U_{\max}$ give the CFL time-step requirement. A compressible solver includes the relevant wave speed, and diffusion imposes an additional time-step restriction.

$$\Delta t \leq C_{\mathrm{CFL}} \frac{\Delta x}{U_{\max}}.$$

For the chosen discretization, the CFL coefficient follows the scheme's stability limit, with every time-step change recorded against the requested output times. Preserving this timing prevents an apparent speed gain from concealing omitted physical events.

3.12 Metropolis–Hastings inference and rare-event analysis

Across plausible material and loading inputs, the same deterministic model predicts a distribution of mechanical limits and delivered service. Metropolis–Hastings explores that uncertainty under a declared probability model, separating statistical sampling from the physical evolution solved inside each evaluation. Rare-event methods extend this calculation toward failure conditions that ordinary sampling may encounter too infrequently.

3.12.1 Assign the sampler a statistical task

Around the deterministic solver, Metropolis–Hastings samples uncertain parameters or conditional distributions used in a risk calculation. Its acceptance rule preserves the intended stationary distribution under the required chain conditions, assigning repeated states the residence time needed for correct weighting. A transition along the chain has no automatic interpretation as elapsed physical time. [23]

Given measurements $\boldsymbol{y}$ and uncertain parameters $\boldsymbol{\theta}$, the likelihood $p(\boldsymbol{y} \mid \boldsymbol{\theta})$ quantifies their compatibility under the chosen observation model. Multiplication by the prior $p(\boldsymbol{\theta})$ gives the posterior density up to its normalizing constant, with the proportionality symbol preserving that qualification.

$$\pi(\boldsymbol{\theta} \mid \boldsymbol{y}) \propto p(\boldsymbol{y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta}).$$

At the collector, pressure and power measurements constrain loss coefficients whose uncertainty propagates into predicted yield. For the rotor, damping uncertainty changes dynamic response, with material scatter and storm-profile variation affecting the margin to failure. The observation model must separate measurement noise from discrepancy in the physical equations, preventing an imperfect turbulence or bearing model from acquiring unjustified certainty during fitting.

3.12.2 Acceptance rule and correct GPU batching

Given a proposal density $q(\boldsymbol{\theta}' \mid \boldsymbol{\theta})$, the acceptance probability is defined by the acceptance relation. A rejected proposal retains the current state in the chain, which is necessary for correct sampling.

$$\alpha(\boldsymbol{\theta}, \boldsymbol{\theta}') = \min\left[1, \frac{\pi(\boldsymbol{\theta}' \mid \boldsymbol{y})\, q(\boldsymbol{\theta} \mid \boldsymbol{\theta}')}{\pi(\boldsymbol{\theta} \mid \boldsymbol{y})\, q(\boldsymbol{\theta}' \mid \boldsymbol{\theta})}\right].$$

Evaluating log densities reduces underflow in the acceptance calculation, and separate random streams preserve the intended initialization of independent chains. Independent forward-model evaluations provide parallel GPU work, permitting batches wherever the sampling algorithm allows; likelihood contributions may also be reduced in parallel. Successive states of an ordinary chain retain sequential dependence, limiting acceleration along that axis.

Keeping only accepted proposals would remove the residence time associated with rejection and distort the target distribution. Selecting “important” mesh cells with the same acceptance rule would define a different numerical method whose conservation and bias need separate analysis. For the proposed software, deterministic field evolution retains the complete required physical coupling, with the outer statistical sampler exploring uncertainty in that computed response.

3.12.3 Adaptation of proposals and physical discretizations

For an adaptive Markov chain, persistent changes to the transition rule may compromise convergence to the intended distribution. Diminishing adaptation limits those changes, and containment controls deterioration in the convergence behavior of the kernels visited by the chain. A convergence argument must address the applicable conditions prior to treating an adaptive run as a posterior sample. [24]

During warmup, proposal scales and covariance adapt to improve exploration, with those settings frozen for production sampling. Independent starts and diagnostics must still test whether the resulting chain explores the target adequately. Changes to the physical mesh or surrogate also change the likelihood, requiring a fixed production discretization or a derived correction that preserves the intended posterior.

3.12.4 Statistical diagnostics

Within a Markov chain, successive draws carry correlated information, making raw sample count an incomplete measure of statistical precision. Effective sample size connects that dependence to the uncertainty in a chosen observable, with trace behavior exposing trapping that a count alone would hide. Sampling effort must follow the precision required for the engineering decision.

For a stationary scalar chain with N_s retained draws and lag-k autocorrelation ρ_k, an approximate effective sample size is defined from the autocorrelation sum, subject to reliable estimation of that sum.

$$N_{\text{eff}} \simeq \frac{N_s}{1 + 2\sum_{k=1}^{\infty} \rho_k}.$$

Comparing independent chains tests whether different initial states lead to compatible distributions, and trace inspection exposes trapping or slow movement between regions. Each design-driving observable should carry an effective sample size and Monte Carlo standard error, quantifying finite-sampling uncertainty independently of prior sensitivity or assumed model discrepancy. A high acceptance fraction may coexist with small, poorly exploring steps; adequate inference must instead be demonstrated for outputs such as peak stress or energy shortfall, whose tails may converge more slowly than mean power.

3.12.5 Rare failure probabilities

For limit-state function $g(\boldsymbol{\theta})$, failure corresponds to $g(\boldsymbol{\theta}) \le 0$. The indicator $\mathbb{I}[\cdot]$ equals one inside that failure set and zero outside it. In the failure-probability integral, $\pi(\boldsymbol{\theta})$ denotes the declared uncertainty density for the risk study, which may differ from the conditional posterior used for parameter inference.

$$p_f = \Pr[g(\boldsymbol{\theta}) \le 0] = \int \mathbb{I}[g(\boldsymbol{\theta}) \le 0]\, \pi(\boldsymbol{\theta})\, \mathrm{d}\boldsymbol{\theta}.$$

To estimate small failure probabilities, subset simulation replaces a rarely reached event with a sequence of more frequently sampled conditional events. [25] A campus application might couple an intake blockage during high wind to degraded bearing performance, reducing both recharge opportunity and the safe storage envelope. If export is also unavailable, the joint probability model must preserve that dependence and its effect on the available energy-release paths.

With nested events ending in final failure set F_L, the intermediate levels define successive conditional probabilities. Their product gives the final event probability without assuming independence.

$$F_L \subset F_{L-1} \subset \cdots \subset F_1.$$

$$\Pr(F_L) = \Pr(F_1) \prod_{\ell=2}^{L} \Pr(F_\ell \mid F_{\ell-1}).$$

Within conditional chains, correlated draws reduce precision in the estimated event probabilities, requiring uncertainty intervals that account for the sampling dependence. Threshold selection changes each conditional problem, with physical-model uncertainty additionally shifting the failure boundary. Observing zero failures in a finite campaign leaves a nonzero range of plausible risk, making the estimates suitable for comparative design and experiment planning within the limits of the chosen model.

3.12.6 Optimization and inference serve different objectives

Across candidate designs, optimization seeks an admissible improvement in a declared objective, with posterior sampling quantifying uncertain performance at each candidate. Smooth design problems may favor gradients or adjoint calculations, reserving stochastic evaluation for the uncertainty that affects the decision. Discrete topology and mode choices require methods suited to those variables.

Each optimization task must identify the quantity to improve and the constraints that define an admissible design. A probability model is additionally required wherever the objective or constraints involve uncertain outcomes. The stopping criterion should then state how much numerical or statistical uncertainty is acceptable in that decision. Calling the procedure a generic "Metropolis–Hastings optimizer" would obscure the different roles of sampling and convergence toward a preferred design.

3.13 Software organization and solver alternatives

Across backend changes, a multiphysics implementation must preserve the physical meaning of each field and interface exchange. Separating geometry and state from numerical operators permits solver libraries to change without silently changing units or governing equations. Diagnostics and run records then tie each accepted result to the configuration that produced it.

3.13.1 Proposed source organization

For an initial implementation, a small tested package gives the physical models stable interfaces, with a notebook or command line exposing each scientific case. Controls belong near the top of the notebook and in a reusable configuration file, supporting both interactive inspection and larger runs. PEP 8 formatting and PEP 257 docstrings make those interfaces easier to inspect, with types and unit metadata identifying the quantities accepted across module boundaries.

Table 25: Proposed source organization

Module or directory	Main responsibility	Required input/output interface	Test emphasis
`config.py`	Define controls, approved modes, and resource limits	Typed configuration with units and allowed ranges	Reject inconsistent or unsafe combinations
`units.py`	Convert physical units and define dimensional checks	SI internal representation and labeled external values	RPM/rad s^{-1} and J/MWh conversions
`geometry.py`	Store geometry and material regions	Versioned mesh and component identifiers	Radius/diameter consistency and positive dimensions
`graphs.py`	Build mesh or component connectivity	Compact adjacency/incidence arrays and geometry weights	Connectivity, boundary classification, and orientation
`state.py`	Own physical field buffers and checkpoints	Declared shapes, dtypes, locations, and units	Restart equivalence and field integrity
`physics/flow.py`	Evaluate conservative fluid operators	Face fluxes, pressure coupling, and boundary conditions	Manufactured solutions and mass conservation
`physics/solid.py`	Evaluate structural and rotor models	Loads, displacement, stress, and stability observables	Analytic ring/disk comparisons and mesh convergence
`physics/em.py`	Evaluate machine and bearing models	Field, current, force, torque, and loss maps	Constitutive and energy-balance checks

Module or directory	Main responsibility	Required input/output interface	Test emphasis
`physics/thermal.py`	Evolve thermal states	Heat inputs, boundary transfer, and temperatures	Heat conservation and limiting solutions
`physics/storage.py`	Track kinetic and other energy stores	State of charge, losses, power envelope, and reserve	Closed-form charge/discharge cases
`backends/`	Isolate CPU, CuPy, and optional PETSc adapters	Common operator and solver interfaces	Cross-backend agreement
`adaptivity.py`	Select approved mesh, time-step, and precision changes	Error indicators and resource diagnostics	Conservative transfers and stable switching
`sparse_policy.py`	Plan sparse storage and solver execution	Operator structure, budgets, and capability checks	Memory forecasts and fallback paths
`inference.py`	Execute Metropolis–Hastings and posterior diagnostics	Fixed target, proposal, seed, and retained samples	Known target distributions and rejection bookkeeping
`risk.py`	Evaluate limit states and rare-event methods	Scenario distributions and failure functions	Analytic probability examples and uncertainty intervals
`control_advisor.py`	Produce bounded dispatch recommendations	Approved plant state and confidence information	No direct bypass of independent protection
`io.py`	Record data, manifests, and restart files	Schema version, units, timestamps, and hashes	Round-trip data integrity
`visualization/`	Prepare scientific figures and scene adapters	Labeled arrays and geometry with provenance	Correct units, scales, and data-to-geometry mapping
`tests/` and `benchmarks/`	Maintain correctness and performance campaigns	Reproducible cases with acceptance conditions	Regression, scaling, and resource accounting

3.13.2 Implementation behavior without a full code listing

At startup, configuration checks associate the physical case with a reference discretization and a resource plan, exposing incompatible assumptions ahead of expensive allocation. The backend adapter must confirm support for the required array operations and index widths, including the solver arguments and random-number methods used in production. A small smoke test exercises those exact paths, catching interface failures at a scale that is inexpensive to diagnose.

During repeated field updates, persistent GPU arrays avoid transfers that would otherwise accumulate across every time step. Profiling identifies cell-by-cell Python work suitable for array operations or tested kernels, with host-device exchange reserved for declared checkpoints and reduced outputs. This placement keeps transfer cost visible in the complete benchmark.

For sampling, the random-number adapter should expose only methods covered by the smoke tests. API compatibility is class-specific, and assuming parity between NumPy-like generator interfaces recreates avoidable attribute errors. Missing capabilities should produce a clear diagnostic or a mathematically checked alternative implementation.

After a failed solve, the last accepted state provides a restart point whose residual and resource diagnostics explain the interruption. Broad exception handling must avoid advancing a corrupted state, as successful execution alone supplies no test of physical correctness. Linting and type checks expose implementation defects, with targeted debugging followed by independent numerical acceptance tests.

3.13.3 Alternatives and complementary tools

Across finite-element and sparse workloads, specialized backends offer different capabilities for distributed execution and preconditioning. Visualization introduces separate requirements for mesh exchange and rendering, supporting a modular choice of tools at these interfaces. Any replacement must reproduce the physical quantities and acceptance criteria governing the primary implementation.

Table 26: Alternatives and complementary tools

Option	Suitable role	Strength for this project	Limitation or integration cost	Recommended use
NumPy/SciPy CPU reference	Small physical models and correctness baseline	Simple inspection and broad numerical coverage	Limited acceleration for some large workloads	Implement first and retain in regression tests
Custom CuPy implementation	Batched scenarios, regular kernels, selected sparse systems	Direct Python-level access to GPU arrays	Requires careful solver, memory, and API design	Main experimental acceleration branch
PETSc/TAO backend	Parallel sparse systems and optimization	Developed CPU/GPU solver infrastructure	Configuration and domain decomposition require expertise	Use for difficult or distributed algebraic systems [17]
AMReX-based application	Block-structured adaptive field models	Established refinement and parallel data infrastructure	The physical solver still requires implementation	Use if adaptive CFD exceeds a manageable custom prototype [15]
FEniCSx/DOLFINx with suitable solver backend	Finite-element prototypes and coupled weak forms	Documented finite-element and parallel workflows	End-to-end GPU execution requires a checked configuration	Compare for structural, thermal, and electromagnetic submodels [26]
Quanscient Allsolve	Commercial multiphysics comparison	Vendor-supported modeling environment	License, model support, export, and performance require confirmation	Treat as a benchmark or specialist solver option [27]
Reduced-order or surrogate models	Fast scenario sweeps and advisory calculations	Low latency after a qualified training campaign	Error and extrapolation behavior govern trust	Use inside a declared training envelope
Quantum research path	Study future fluid algorithms	A distinct experimental computing direction	Present project-scale benefit is unestablished	Keep outside the critical path [28]

3.13.4 Scope of commercial and quantum-computing branches

Within a selected commercial solver configuration, available flow models determine whether the collector's turbulence and moving boundaries are representable. Quanscient's public physics description includes laminar CFD, requiring a separate matched demonstration for the turbulent collector duty or any performance comparison. [27] Its quantum research activities provide no demonstrated turbulence closure for the proposed Allsolve calculation. [29]

Ahead of solver selection, request a matched case using the turbulence treatment required for the collector. Moving interfaces must reproduce the intended exchange with the rotor or compliant structure, and the material models must represent the mechanical or electromagnetic regime being tested. Exported fields should preserve the quantities needed for independent comparison. Cloud deployment and nominal core count offer no replacement for measuring the accepted solution's accuracy and cost.

On a superconducting processor, an eight-qubit experiment demonstrated fluid vortex interactions within its reported model and hardware scope. [28] Extending that result to site-specific tornado prediction would require a new accuracy and resource assessment, leaving the campus design independent of a presumed quantum advantage.

Quantum-inspired low-rank methods running on classical GPUs belong to the classical numerical branch and should be evaluated under the same error and performance criteria as other approximations. Naming a tensor representation "quantum-inspired" provides no independent speed or accuracy guarantee.

3.13.5 Reproducibility and distributed execution

For each accepted result, the case identifier links to a recoverable geometry and the material-data version used by its constitutive laws. The solver configuration and random seeds define numerical choices and stochastic initialization, with package versions and hardware details locating the execution environment. Linking that record to the code version makes the calculation reproducible across subsequent model changes.

Changes to geometry or physical assumptions should create distinct cases, preserving earlier outputs for comparison. Calibration measurements must retain separate identities from the independent test data used to assess predictive performance.

Across a partitioned PDE domain, each device owns a defined region and exchanges boundary values with its neighbors, preserving the coupled equations at partition interfaces. Global reductions support convergence checks, introducing communication costs that belong in the solver plan alongside local arithmetic. Independent CuPy processes require this coordination to solve one shared domain. Strong-scaling tests hold total problem size fixed, with weak-scaling tests holding work per device fixed; both must report communication time and load imbalance at the accepted accuracy.

3.14 Safety, development stages, and decision gates

Increasing stored energy increases the consequences of a protective failure, making measured margins from smaller experiments essential to scale-up. Independent shutdown and containment must retain their authority as numerical models or experimental equipment change. Each increase in scale requires tests of the corresponding failure paths and the qualified means of limiting their consequences.

3.14.1 Keep protection independent of numerical prediction

At the supervisory interface, a prediction needs a timestamp and validity envelope, permitting dispatch to reject stale or out-of-regime advice. Local protection must detect overspeed and adverse vibration independently, with bearing faults invoking the qualified response to support loss. The uncertainty associated with a recommendation informs its permitted use without delaying these protective actions.

For abnormal pressure, overheating, or electrical faults, the affected equipment must trip according to its qualified local response. Loss of the GPU process, communications, or weather feed removes the numerical adviser from service without disabling that protection.

Within the proposed control hierarchy, "self-managed" means that approved computational settings adjust automatically and that a supervisory controller chooses among authorized setpoints. Physical limits require a separate, controlled configuration and qualified protection hardware. A user-facing sparse-memory knob has no authority to relax a rotor-speed or flood-response limit.

3.14.2 Risk register for the integrated campus

During a severe event, environmental loading may trigger several failures that share access, cooling, or electrical infrastructure. A risk register must connect each initiating condition to its consequence and protective layer, exposing assumptions that require an integrated test. This treatment captures coupled sequences that isolated component checks leave unresolved.

Table 27: Risk register for the integrated campus

Hazard chain	Early indicators	Proposed consequence-reduction measures	Required study or test
Wind load exceeds collector envelope	Wind, strain, vibration, pressure, and generator state	Derating, bypass, module isolation, and secured access	Coupled load and shutdown testing
Debris blocks an intake	Differential pressure and flow imbalance	Accessible barriers, obstruction alarms, and bypass capacity	Clogging and impact campaign
Rotor overspeed or mechanical damage	Independent speed and vibration monitoring	Local trip, engineered rundown, separation, and containment	Rotor integrity and fault-response qualification
Bearing cooling or force capability is lost	Temperature, lift, gap, and supply diagnostics	Controlled rundown and qualified support-loss response	Dynamic fault tests at representative energy levels
Vacuum degrades rapidly	Pressure rise and rotor-loss increase	Isolation, thermal response limits, and controlled venting	Air-ingress transient analysis
Utility grid disappears during high input	Voltage, current, and breaker status	Collector curtailment, storage headroom, and energy-rated dump paths	Islanding and load-rejection tests
Floodwater enters lower levels	Water level, seepage, and drainage status	Compartmentation, protected power, and site-specific water exclusion	Combined flood and power-loss study
Fire or process gas release	Temperature, gas, and ventilation alarms	Independent process enclosure and emergency access	Process hazard assessment

Hazard chain	Early indicators	Proposed consequence-reduction measures	Required study or test
Forecast or sensor data is corrupted	Cross-sensor disagreement and stale timestamps	Reject the affected advisory result and use conservative state logic	Fault injection and communications tests
Software or GPU fails	Watchdog, heartbeat, and calculation-integrity checks	Independent control and safe advisory timeout	Compute-loss demonstration
Several modules share a common failure	Shared cooling, bus, foundation, or access disruption	Physical and functional separation of critical dependencies	Common-cause analysis
Post-storm hidden damage affects restart	Inspection findings and changed vibration signatures	Staged recommissioning with module-level isolation	Recovery procedure demonstration

Under a shared storm exposure, intake obstruction may reduce generation just as grid damage increases demand on the reserve. If flooding delays repair and disrupts cooling, the capacity available for discharge may fall during the same interval. A joint hazard model must preserve these relationships, as independent probabilities may underestimate positively correlated damaging combinations.

3.14.3 Development stages

At each increase in physical scale, stored energy and environmental loading extend the consequences of failure. Progression requires smaller tests to demonstrate the margins and protective responses needed by the larger configuration. Integrated trials then assess interactions that component tests leave unresolved, defining a bounded pilot envelope.

Table 28: Development stages

Stage	Scope	Deliverable	Gate for progression	Result that redirects the project
0	Service and physics study	Energy ledger, stress-limited sizing, and baseline alternatives	A plausible service is achievable within a qualified physical envelope	Required rotating mass, losses, or encounter rate defeats the chosen service
1	Numerical reference package	Analytic tests, CPU models, and first GPU kernels	Correctness and reproducibility demonstrated	GPU path lacks benefit for the selected workloads
2	Component experiments	Collector, bearing, rotor-loss, material, and machine data	Models predict independent measurements within accepted error	A critical component has inadequate performance or unsafe failure behavior
3	Small integrated module	Protected electrical coupling and storage buffer	Energy balance, fault response, and maintenance access demonstrated	Integration losses or common-cause hazards outweigh benefit
4	Field pilot within a qualified envelope	Measured availability, useful energy, and recovery behavior	Independent review accepts the service and hazard basis	Extreme-wind contribution is too small or too costly
5	Modular campus decision	Site design, quotations, service agreements, and environmental assessment	Reliability-adjusted benefits justify construction	A simpler renewable-plus-storage option performs better
6	Optional advanced branches	HTS upgrade, materials process, or enhanced storm collector	Each branch demonstrates an incremental benefit	Added complexity reduces resilience or lifecycle performance

Controlled tests followed by field exposure inside an approved envelope provide a progression for measuring performance without requiring direct tornado interception. Initial qualification must use hazards and stored-energy levels whose consequences are addressed by the test facility.

3.14.4 Readiness assessment without inflated ratings

At the campus boundary, demonstrated component performance provides inputs to integration without assigning readiness to the complete installation. A motor tested at full power and a cryogenic bearing tested at its rated load establish different portions of the operating envelope. Readiness assessment must connect those portions using tests of the assembled interfaces and protective functions.

For a near-term pilot, a small protected energy module would connect collector measurements to actual storage delivery. Measured loads would constrain the flow model, with loss measurements determining the energy available across the service interval. Matched-accuracy computational results would also give the software a transferable contribution, independent of the eventual feasibility of the mega-rotor branch.

3.15 Economics, sustainability, and collaboration

At a common service boundary, mechanically feasible alternatives may differ in the cost of maintaining reserve across the same outages. Including losses and downtime in delivered-energy accounting exposes those differences in availability. Resource use and implementation risk then enter the comparison wherever they change the service-equivalent cost or environmental burden.

3.15.1 Compare equivalent delivered services

For a fair economic comparison, each candidate must serve the same essential load under the same outage histories and reserve policy. Routine renewable charging supplies the base case, with conventional backup providing a comparison for service continuity independent of storm harvesting. Thermal storage receives credit for the electrical cooling demand it avoids, preserving the distinction between reduced load and generated electricity.

Until measured local yield supports a dependable contribution, storm-derived revenue belongs in a sensitivity range whose assumptions stay separate from the base financing case.

For a consistent delivery boundary, levelized storage cost compares discounted expenditure with discounted electrical energy delivered. Initial capital cost is C_0, and recurring net cost in year t is C_t; annual delivered electricity is $E_{\mathrm{del},t}$. Discount rate r_d applies over assessment horizon Y years in the levelized-cost relation. Salvage value and financing arrangements would require project-specific treatment beyond this reduced comparison. Schmidt and colleagues establish the importance of the selected service in comparing storage economics. [12]

$$\mathrm{LCOS} = \frac{C_0 + \sum_{t=1}^{Y} \dfrac{C_t}{(1+r_d)^t}}{\sum_{t=1}^{Y} \dfrac{E_{\mathrm{del},t}}{(1+r_d)^t}}.$$

Charging electricity must be assigned to the same energy boundary used for delivered service, with auxiliary consumption added only where it has yet to be included. Replacement parts and scheduled maintenance contribute recurring cost as the equipment ages, and end-of-life obligations require an allowance that includes removal and disposal inside the chosen assessment horizon. Reduced unserved

essential load should be reported as a resilience benefit alongside arbitrage revenue, making its assigned value visible without hiding it inside a nominal storage price.

3.15.2 Required economic sensitivities

Across plausible service and cost assumptions, a change in material limits or storage duration may reverse the ranking of candidate systems. Sensitivity analysis identifies the inputs responsible for that reversal, directing early measurement toward uncertainty that changes the investment decision. The comparison must preserve the required service as those inputs vary.

Table 29: Required economic sensitivities

Variable	Why it changes the decision	Required comparison
Usable wind encounter frequency	Determines how much value comes from the storm branch	Ordinary-weather-only case versus measured storm contribution
Net collector coefficient	Links capture geometry to electricity	Pressure losses and auxiliary drives included
Allowable rotor stress	Controls specific energy and mass	Qualified values with material scatter and fatigue reductions
Rotor and auxiliary losses	Determine the value of long holding periods	Speed- and temperature-dependent chronological model
Construction and containment cost	Often dominates a mega-rotor installation	Modular alternatives with equivalent consequences of failure
Converter power rating	Sets charging and discharge rate	Power equipment sized separately from stored energy
Maintenance and storm recovery	Affect both revenue and emergency availability	Correlated downtime and access constraints
Longer-duration technology	changes the preferred division of labor	Same delivered service and reserve requirement
Materials-processing revenue	creates an apparent benefit from unqualified feedstock	Conservative zero-revenue baseline until product acceptance
Computing and engineering effort	Custom software has a development cost	Reuse, maintainability, and specialist solver alternatives

Without current construction quotations and supplier offers, the cost model describes scenarios outside procurement-grade pricing. Local tariffs set the value of electrical exchange, and any grant contribution requires the terms of an active solicitation. Retaining these assumptions in the model permits recalculation as project-specific inputs arrive.

3.15.3 Sustainability and materials recovery

Across construction and service life, steel and concrete account for major material commitments, with excavation adding a burden tied to the selected site. Cooling demand and replacement equipment extend the inventory beyond installation, increasing impact per delivered unit of service if utilization is low. Eventual steel recovery reduces a future burden without removing the initial production and construction commitment.

For recovered debris, compatible fractions may retain material value, although contamination and transportation reduce the benefit of local reuse. Cryogenic electricity and cooling water add recurring resource demands, with process gases requiring an emissions inventory tied to the selected manufacturing

route. Noise and land use affect residents beyond the energy boundary, requiring comparison with a declared alternative over the same service life.

3.15.4 Collaboration roles without assumed commitments

At the concept stage, naming an organization identifies a possible technical contact without establishing participation or equipment suitability. Supplier and funding discussions need a bounded question whose answer informs a design decision. Any commitment or qualified product offering requires confirmation by the relevant organization.

Table 30: Collaboration roles without assumed commitments

Candidate collaborator category	Prospective examples	Useful initial question	Deliverable to request
Weather and atmospheric research	NOAA, NCAR, and university meteorology groups	What local wind and joint-hazard data supports site screening?	Data-access plan and uncertainty assessment
Wind-turbine research and suppliers	Challenergy and flexible-rotor research teams	What generation and survival envelopes have been demonstrated?	Test reports and a defined interface specification
Rotating-machine suppliers	AMSC and conventional industrial machine manufacturers	What reversible duty across the torque-speed envelope is available and warranted?	Application-specific technical offer
Superconducting and magnet laboratories	MagLab and university laboratories	Which bearing or coil experiment would resolve the largest uncertainty?	Instrumented component study
Magnetic-material developers	Niron Magnetics and other material suppliers	Does a candidate material meet the required temperature and demagnetization duty?	Measured material curves and supply qualification
Utility and community partners	Local utility, emergency services, and community institutions	Which loads and recovery functions should receive priority?	Agreed service requirements and interconnection process
Corporate weather and logistics teams	Walmart, FedEx, UPS, Amazon, BP, and Cargill examples	Is there a shared resilience problem suitable for a bounded pilot?	Defined use case, without assumed financial commitment
Materials and recycling specialists	Glass, mineral-processing, recycling, and waste experts	Which captured fractions have a safe and economic destination?	Assay protocol and product/disposal acceptance criteria
Public research funding bodies	DOE, ARPA-E, NSF, and state energy offices	Which current solicitation matches a tested research question?	A funding review based on the active solicitation

For residents affected by the site, the proposed service benefit must be assessed alongside exclusion zones and access during an emergency. Community participation must precede siting and risk decisions, with engagement involving Tribal lands or interests following the relevant community's governance and consultation processes. Long-term maintenance and emergency responsibilities must identify who carries those obligations after construction.

3.15.5 Publication and intellectual-property planning

Across development, dated design records preserve each contribution and the calculations supporting it. A software license governs released code, with measurement access addressed by a separate data-sharing policy. Potentially protectable inventions and third-party material require qualified institutional or legal review prior to disclosure, including assessment of defensive publication or provisional filing as project-specific options.

3.16 Branched architecture map and terminology

Between coupled campus functions, an environmental input affects more than generation, as the same weather may interrupt access and reduce cooling availability. Physical dependencies connect these disturbances to stored energy and ultimately to community service. Consistent terminology identifies each technology and proposed branch without implying performance beyond its tested scope.

3.16.1 Architecture map

From environmental input to community service, conversion and storage create dependencies that extend beyond a simple component inventory. Computation informs bounded decisions, with independent protection limiting the consequences of adverse equipment states. Materials processing draws from the available surplus under a separate process envelope.

Service objective
- Essential electrical loads during outages
- Summer cooling-peak reduction
- Recovery services and inspected restart

Energy input
- Ordinary renewable generation
  - Site-qualified wind
  - Solar or other available sources
- Storm-tolerant research capture
  - Qualified generation envelope
  - Separate survival envelope
  - Debris-aware intake and bypass
  - Site encounter probability and net yield

Protected energy routing
- Collector-local generator and converter
- Sectionalized electrical buses
- Grid and islanding interface
- Energy-rated curtailment and dump paths

Storage portfolio

- Fast flywheel buffer
  - Stress-limited material and geometry
  - Qualified bearings and machine duty
  - Vacuum and auxiliary-loss budget
  - Independent containment and rundown
- Separately sized electrical energy reserve
- Thermal storage
  - Cooling-peak service
  - Optional process-heat service

Civil and environmental systems

- Site-specific wind and flood design
- Rotor foundation separated from compliant skin
- Feeders to off-site community loads
- Materials, water, noise, and lifecycle assessment

Optional debris recovery

- Hazard screening and assay
- Compatible recycling or mineral recovery
- Glass research after feedstock qualification
- Silicon reduction and purification as a separate study

Numerical architecture

- Adjacency and conservative physical operators
- GPU tensor-shaped field arrays
- Hybrid domain solvers
- Automatic resource planning
  - Matrix-free versus budgeted CSR
  - Mesh, time-step, and precision controls
  - CPU or distributed-solver fallback
- Metropolis–Hastings inference
  - Fixed production target
  - Independent-chain batching
  - Tail-risk and model-discrepancy checks
- Scientific archive and visualization adapter

Independent assurance

- Hardware protection outside numerical autonomy
- Analytic and experimental comparisons
- Measured performance at matched accuracy
- Community-defined service and risk limits
- Stage gates with simpler alternatives retained

3.16.2 Direct and indirect connections

At the boundary between subsystems, one design variable may alter several energy and hazard paths. Greater cryogenic demand consumes part of the reserve that supports superconducting equipment, linking machine choice to service duration. Weather similarly couples generation to site access, affecting the time needed to restore damaged equipment.

Table 31: Direct and indirect connections

Connection	Physical or mathematical relationship	Consequence for the project
Radius, RPM, and material stress	Rim speed combines radius and angular speed; ring stress scales with its square	Large radius limits acceptable RPM at fixed material allowable
Magnetic support and seasonal economics	Contact-free support still requires a complete thermal and electrical loss budget	Bearing choice changes the preferred storage duration
Intake filtration and energy capture	Pressure drop dissipates available fluid work	More protection may reduce net energy yield
Structural flexibility and precision machinery	Load alleviation changes displacement and alignment	Compliant exterior structures should be isolated from precision supports
Mesh adjacency and sparse algebra	Local physical couplings create sparse discrete operators	Sparse structure reduces memory without removing physical interactions
Tensor compression and rare-event risk	Low-rank approximation may smooth localized extremes	Compression criteria must include design-driving tail quantities
MH sampling and adaptive numerical models	Changing the likelihood model changes the statistical target	Production sampling needs a fixed or correctly corrected target
Summer cooling and thermal storage	Electrical peak demand is shifted by storing cooling	The original service may be met with less electrical storage
Semiconductor materials and storm debris	Purity requirements impose separation and refinement steps	Abundant mineral mass alone offers little information about useful silicon yield
Grid resilience and storm exposure	The same weather hazard also damages the collection and delivery system	Joint availability is more important than isolated nameplate output

3.16.3 Terminology and proposed names

For proposed project names, the associated device or method requires demonstrated performance under its assigned duty. Naming alone assigns neither trademark ownership nor manufacturer endorsement, leaving those questions to separate confirmation.

Table 32: Terminology and proposed names

Term	Construction or meaning	Intended use
Typhoon-bine	Proposed blend of “typhoon” and “turbine,” retained as an informal project coinage	Informal name for the storm-tolerant turbine research branch
Maglev-kinetics	Proposed combination of magnetic levitation and kinetic storage	Communication label for the bearing and rotor branch
StormStore	Proposed blend of “storm” and “store”	Campus-level project nickname, subject to naming review
Adjacency Tensor Sampling	Proposed descriptive name for the numerical architecture	Separates connectivity, field representation, and inference responsibilities

Term	Construction or meaning	Intended use
Metropolis–Hastings	Method name honoring its historical algorithmic lineage; the general acceptance construction is cited here via Hastings	Statistical sampling method, distinct from physical time integration [23]
Tensegrity	Established structural term used here for prestressed tension-and-compression assemblies	Refers to a class requiring geometry-specific stiffness analysis [9]
Magnus-effect collector	Collector using aerodynamic force associated with rotating bodies in a flow	Identifies the physical principle used by the cited manufacturer [5]

3.16.4 Acronym glossary

Across atmospheric and electromechanical models, abbreviations need a consistent expansion that identifies their physical or computational role. Keeping storage quantities separate from numerical representations prevents similar terminology from implying interchangeable functions.

Table 33: Acronym glossary

Acronym	Meaning	Role in the proposed campus
AC	Alternating current	Grid and machine electrical quantities
AMR	Adaptive mesh refinement	Spatial resolution management
CFD	Computational fluid dynamics	Collector and local wind-flow modeling
CFL	Courant–Friedrichs–Lewy condition	Time-step stability guidance
COP	Coefficient of performance	Cooling and refrigeration energy accounting
CPU	Central processing unit	Reference and fallback computation
CSR	Compressed sparse row	Candidate sparse numerical representation
DC	Direct current	A possible internal electrical bus
DOE	United States Department of Energy	Potential research-funding category
EF	Enhanced Fujita	Tornado damage-based intensity classification
FEM	Finite element method	Structural and selected field models
FESS	Flywheel energy storage system	Kinetic storage subsystem
FP32 / FP64	32-bit / 64-bit floating-point arithmetic	Numerical precision choices
GPU	Graphics processing unit	Parallel numerical execution
HPC	High-performance computing	Larger simulation campaigns
HTS	High-temperature superconducting	Optional bearings and machine windings
LCOS	Levelized cost of storage	Service-aware economic comparison
LES	Large-eddy simulation	Candidate turbulence treatment
MCMC	Markov chain Monte Carlo	Statistical sampling
MH	Metropolis–Hastings	Acceptance-based sampling algorithm

Acronym	Meaning	Role in the proposed campus
MPI	Message Passing Interface	Distributed-memory communication
MVDC	Medium-voltage direct current	A possible power-routing architecture
PDE	Partial differential equation	Distributed physical model
PETSc	Portable, Extensible Toolkit for Scientific Computation	Solver backend
QPU	Quantum processing unit	Exploratory computing branch
RANS	Reynolds-averaged Navier–Stokes	Candidate engineering turbulence treatment
RPM	Revolutions per minute	Rotor speed input
SpMV	Sparse matrix-vector multiplication	Core sparse operation
TAO	Toolkit for Advanced Optimization	Optimization capabilities associated with PETSc
USD	Universal Scene Description	Graphics-scene interchange option
VRAM	GPU device memory, commonly called video random-access memory	Compute-memory planning

4 Verification

In the coupled calculations, agreement with analytic limits and conservation balances tests whether the equations have been implemented consistently. Convergence at design-driving quantities then assesses discretization error, with independent measurements testing the physical model within declared uncertainty. Backend performance must be compared at this same accepted accuracy, preserving both numerical and physical requirements.

4.1 Establish correctness prior to acceleration benchmarking

For the storage calculation, recovery of the analytic speed-energy relation tests the inventory that charging and discharge modify. Agreement with ring and disk stress limits checks the mechanical interpretation of the same geometry, linking permissible speed to stored energy. Small problems with known solutions then test dimensional consistency and boundary treatment prior to larger coupled calculations.

Table 34: Correctness checks prior to acceleration benchmarking

Test family	Example case	Quantity checked	Proposed acceptance approach
Dimensional checks	RPM to rad/s, tonnes to kg, joules to GWh	Units and magnitude	Exact conversion tests within floating-point tolerance
Storage accounting	Constant-power charge followed by discharge	Stored energy and delivered energy	Agreement with the analytic energy balance
Elasticity	Thin ring and bore-free disk	Stress and displacement	Convergence toward the appropriate analytic solution
Diffusion	Manufactured Poisson or heat-equation solution	Field error and observed convergence order	Error decreases consistently under refinement

Test family	Example case	Quantity checked	Proposed acceptance approach
Fluid dynamics	Decaying Taylor–Green vortex in an appropriate laminar test regime	Velocity, pressure, and energy decay	Agreement with the chosen analytic or reference solution
Mesh adaptation	Refined/coarsened conservative transport case	Mass and interface flux	Integral conservation and bounded interpolation error
Rotor dynamics	Simple rotor with known support properties	Modal frequencies and orbit response	Agreement with an independent reduced model
Electromagnetic coupling	Simple field and force configuration	Force, torque, and energy	Agreement between independent force/energy calculations
Thermal coupling	Lumped heating and cooling limit	Temperature history	Agreement with a closed-form or trusted reference model
MCMC	Known Gaussian and multimodal targets	Distribution, correlations, and effective sample size	Correct target recovery with adequate exploration
Rare-event estimation	A probability model with a known tail	Failure probability and uncertainty interval	Repeated-run agreement with the known answer
Restart and fallback	Stop, checkpoint, and resume across supported backends	Physical state and observables	Agreement within the declared numerical tolerance
End-to-end protection interface	Invalid sensor data and failed compute process	Advisory output and command boundaries	Safe rejection of invalid recommendations

Across successive meshes, the design-driving observable may converge at a different rate from the algebraic residual. Collector power requires its own convergence history, with peak pressure and integrated support force assessed against the decisions they control. In a rotor, global displacement may settle ahead of a local stress concentration, requiring separate stress convergence prior to assigning an allowable speed.

4.2 Physical experiments and scale similarity

In a scaled collector experiment, geometric resemblance leaves the balance of flow forces undetermined. Reynolds number compares inertia with viscosity, and blockage ratio measures the influence of the test surroundings. Preserving the relationships controlling the selected response is essential to extrapolating that response to the full-size collector.

Within a swirling flow, local loading follows the spatial profile, and reduced frequency relates field variation to the structural response scale. The structure's mass and stiffness ratios then govern motion, yet a one-tenth geometric copy cannot generally preserve these fluid relationships together with gravity and material failure. Each experiment must declare its scaling target, preventing geometric resemblance from standing in for dynamic similarity.

For flexible turbine blades, gravo-aeroelastic scaling connects selected response ratios to field measurements at reduced size. [6] A campus collector requires its own scaling target and an accounting of effects excluded from similarity, confining any extrapolation to the response that the experiment represents.

Table 35: Physical experiments and scale similarity

Experiment	What it should establish	Deliberate boundary of interpretation
Controlled intake-flow test	Pressure drop, net captured power, and bypass response	A laboratory vortex represents a declared flow case
Debris-barrier test	Response to a specified projectile and clogging spectrum	Survival applies to the tested impact classes and conditions
Small rotor loss test	Loss versus speed, pressure, and bearing state	Seasonal estimates require auxiliary power as well as spin decay
Bearing force and stiffness test	Load capacity, damping, and thermal sensitivity	Static levitation alone leaves dynamic stability unresolved
Machine duty-cycle test	Motoring/generating efficiency and thermal response	Nameplate power alone leaves transient duty unresolved
Material and joint tests	Strength scatter, fatigue, fracture, and residual stress	Coupon strength must be translated into a qualified assembly allowable
Coupled module test	Electrical faults, rundown, thermal interaction, and state transitions	A single module leaves common-cause campus failures unresolved
Field pilot inside its approved wind envelope	Availability, maintenance, noise, and useful net energy	Extrapolation to untested tornado exposure requires a separate justification

4.3 Performance benchmarks at matched accuracy

Under a common physical case, measured speedup S compares complete workflows reaching the required error tolerance. Setup and data exchange belong in elapsed time alongside the solve, with required output included at the declared delivery boundary. This common accounting prevents a fast isolated kernel from acquiring the status of application-level acceleration.

$$S = \frac{T_{\mathrm{CPU}}}{T_{\mathrm{GPU}}}.$$

During the first solve, compilation and assembly add costs whose treatment must be identified if repeated runs amortize them. Preconditioner setup may dominate a fast matrix-vector kernel, with convergence determining how often that kernel executes. Transfer timing and device synchronization capture delays hidden by asynchronous launches, and required file output completes the workflow boundary. Separate measurements of these contributions explain the achieved time to an accepted solution.

For p devices at fixed total problem size, T_1 denotes one-device elapsed time and T_p denotes elapsed time using p devices. Their ratio defines strong-scaling efficiency, which is evaluated by the displayed relation; weak-scaling tests answer a different question by holding workload per device approximately constant as total problem size grows.

$$\eta_{\mathrm{strong}}(p) = \frac{T_1}{pT_p}.$$

At the execution boundary, measuring electricity consumed to reach the accepted answer captures a cost that elapsed time alone leaves unresolved. Host and accelerator demand must share that boundary, preserving the accounting if work shifts between devices. Peak arithmetic throughput supplies no substitute for this complete measurement.

4.4 Required figures and comparison plots

For each design-driving quantity, a figure should expose the physical limit and the uncertainty affecting its interpretation. Analytic curves supply reference behavior, with measured or simulated values showing departures under the stated assumptions. Performance belongs beside these quantities only if its error and accounting boundaries support the comparison.

Table 36: Required figures and comparison plots

Figure	Content and comparison	Axes or visual encoding	Data requirement and caveat
1	Campus energy-flow diagram with protection boundaries	Directed power/energy paths and independently isolated zones	Proposed topology; distinguish mechanical, electrical, and thermal paths
2	Ring hoop stress versus RPM for 15 m and 30 m radii	RPM on horizontal axis; stress in GPa on vertical axis	Analytic curves with clearly labeled illustrative allowable-stress levels
3	Gross, usable, and delivered energy for the three stress scenarios	Scenario on horizontal axis; energy in GWh on vertical axis	Calculated table values; include speed-range and efficiency assumptions
4	Collector incident and estimated electrical power versus wind speed	Wind speed in m/s; power in MW	Separate hypothetical 100 m collector from manufacturer-rated small modules
5	Generation envelope versus survival envelope	Wind speed against allowed mode or output	Use the exact product and revision; avoid extrapolating a power curve past cut-out
6	Seventy-day storage and served-demand history	Time in days; stored energy and unmet demand in separate figures	Chronological charging and loss scenarios; enforce capacity and reserve bounds
7	Screen pressure loss versus net collector power	Flow rate or obstruction fraction; loss and useful power	Test data or qualified CFD, with capture-area definition
8	Structural load and displacement histories	Time; force, displacement, or stress in separate figures	Matched turbulent inflow histories and uncertainty intervals
9	CPU, CuPy CSR, and matrix-free runtime versus error	Quantity-of-interest error; elapsed time	Measured end-to-end benchmarks at matched physical conditions
10	Peak memory versus scalar unknown count	Unknown count; peak allocated and reserved memory	Include operator, vectors, preconditioners, and transient assembly
11	Strong and weak scaling	Device count; elapsed time or efficiency in separate figures	Measured partition, communication, and load-balance statistics
12	Mesh and time-step convergence	Resolution or time step; peak load and net energy	At least a credible refinement sequence with consistent boundaries
13	Posterior and rare-event diagnostics	Parameter distributions, chain traces, and failure-probability uncertainty	Distinguish parameter uncertainty from Monte Carlo error

Figure	Content and comparison	Axes or visual encoding	Data requirement and caveat
14	Lifecycle service-cost comparison	Delivered service and cost under shared assumptions	Current project quotations; show uncertainty and exclude speculative revenue
15	Embodied and recurring environmental burdens	Material/process category and lifecycle totals	Traceable inventories and consistent system boundaries

On each quantitative figure, a separate plotting canvas gives scales and uncertainty intervals enough room for interpretation. Numerical axes need units, with legends and direct labels placed clear of extrema or interval boundaries. Matplotlib's `rcParams` should set figure and saved-image resolution to 250 DPI, retaining in-memory previews and vector export where line work benefits from resolution-independent reproduction.

Across plotted series, line style and marker shape preserve identification in black and white, with direct labels reducing repeated reference to a legend if they stay clear of the data. Every interval must identify whether it represents statistical uncertainty or a range of design assumptions. Performance comparisons require measured results at the accepted accuracy, excluding invented timing curves or decorative speedup factors.

4.5 Scientific data and Blender scene exchange

Within a scientific archive, field values require units and an association with the mesh entities on which they were calculated. Timestamps and a coordinate convention align outputs across the common geometry, preserving their physical ordering. Visualization exports must carry the identifiers needed to recover that underlying numerical state.

Numerical arrays may use HDF5 or Zarr, and mesh-oriented exchange may use VTK or XDMF if the selected schema preserves field-to-mesh associations. An immutable analysis result should precede simplified presentation geometry, preventing visualization changes from overwriting the scientific record.

Blender documents interchange formats such as Alembic for geometry-cache workflows and USD for scene exchange, with STL available for triangle-surface geometry. [30] Selection should follow the fields and animation information that the export specification needs to carry, followed by an import comparison against the scientific archive. A graphics scene serves the presentation task without replacing the full numerical record.

For a given case, a versioned parameter table supplies geometry to the proposed graphics adapter, with selected physical fields mapped to scene attributes. Retaining the case identifier links those attributes to the full analysis archive, which an STL surface alone lacks. CSV-driven Geometry Nodes changes require an ingestion script or supported importer; the attribute spreadsheet displays the resulting data without supplying that ingestion path.

During animation, the relationship between physical time and display time must accompany the depicted motion. Slowing a rotor for visibility requires a caption giving its simulated speed and playback scale, with interpolation or motion blur identified as presentation effects. The archived numerical state supplies the basis for assessing structural and numerical accuracy independently of playback smoothness.

4.6 Courtyard screening and computed response

For a storm-facing collection structure, openings alter both airflow and the load paths carrying aerodynamic forces into the foundation. A roofless courtyard calculation examines these precursor responses using a conditional statistical screen followed by a connected elastic model, keeping energy conversion and effects on the surrounding tornado outside the calculated outputs. The retained results describe a research-scale geometric concept within the broader campus proposal. [31]

4.6.1 Geometry and conditional Monte Carlo response

Around the selected courtyard, four tapered reinforced-concrete monoliths contain three ground-level passages on each side, leaving twelve routes for airflow and debris. Candidate selection compares material volume with modeled limit-state exceedance, providing a bounded structural screen under an assumed EF5 wind distribution. The approximately 73 m by 70 m footprint belongs to this collector concept and excludes the eight 30 m radius storage rotors, whose separated campus siting requires a much larger area.

For the statistical calculation, wind speed follows a triangular distribution from 201 to 320 mph with a mode of 250 mph, accompanied by uncertain pressure and debris parameters. Screening covers 4,096 candidates at 65,536 trials each, followed by ten million trials for the selected geometry. With 6,249 modeled failures, the final conditional survival estimate is 99.937510%, exceeding the chosen 99.9% target under these assumptions; uncertainty in structural idealization and the hazard distribution lies outside a sampling-only confidence interval.

Table 37: Selected courtyard geometry and conditional statistical results

Quantity	Recorded value	Interpretation
Outer plan	73.18 m by 69.79 m	Collector concept; storage rotors excluded.
Height	10.92 m	Selected tapered shell height.
Monolith depth	10.48 m at base; 7.79 m at top	Tapered geometric envelope.
Shell wall thickness	1.55 m	Assumed reinforced-concrete section.
Courtyard plan	52.22 m by 48.82 m	Open central region.
Passage dimensions	7.32 m north/south; 6.98 m east/west; 4.37 m height	Three openings per side.
Footprint	5,107.1 m^2	Outer plan area, distinct from concrete volume.
Concrete-volume proxy	7,829 m^3	Geometric quantity used in candidate comparison.
Final trial count	10,000,000	Conditional hazard draws for the selected design.
Conditional shell survival	99.937510%	Fraction passing all included limit-state screens.
Conditional shell failure	0.062490%	6,249 failures within the assumed model.
Mean utilization	0.3028	Mean of the maximum modeled utilization per trial.
99th / 99.9th percentile	0.7019 / 0.9479	Upper utilization quantiles in the retained run.
Maximum sampled utilization	2.0904	At least one modeled limit exceeded unity.
Recorded final throughput	57.2 million trials/s	NVIDIA L4 screening measurement; excludes a matched CPU comparison.

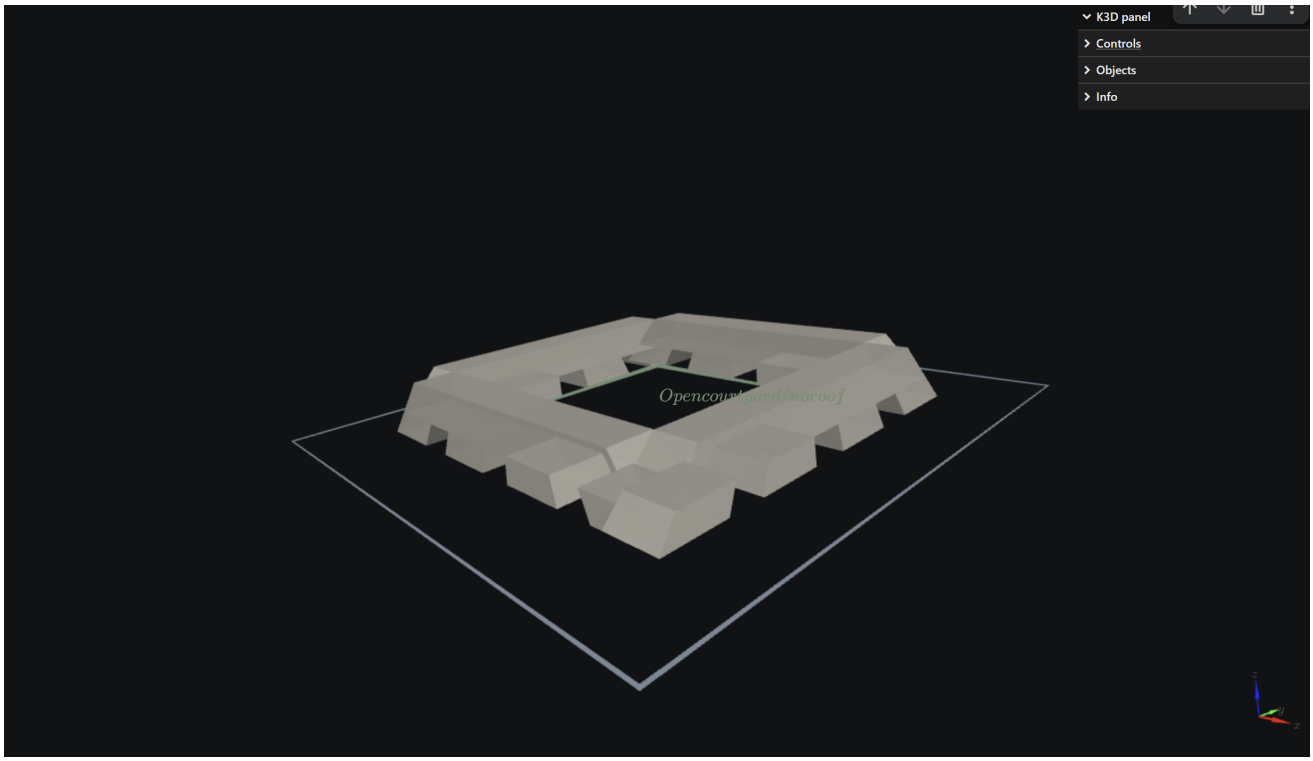


Figure 1: Roofless courtyard concept with four tapered monoliths and twelve ground-level passages. The outer outline represents a 100 m by 100 m reference block; open circulation paths also admit wind and debris, consistent with an industrial collection structure whose intended uses exclude habitation and storm refuge.

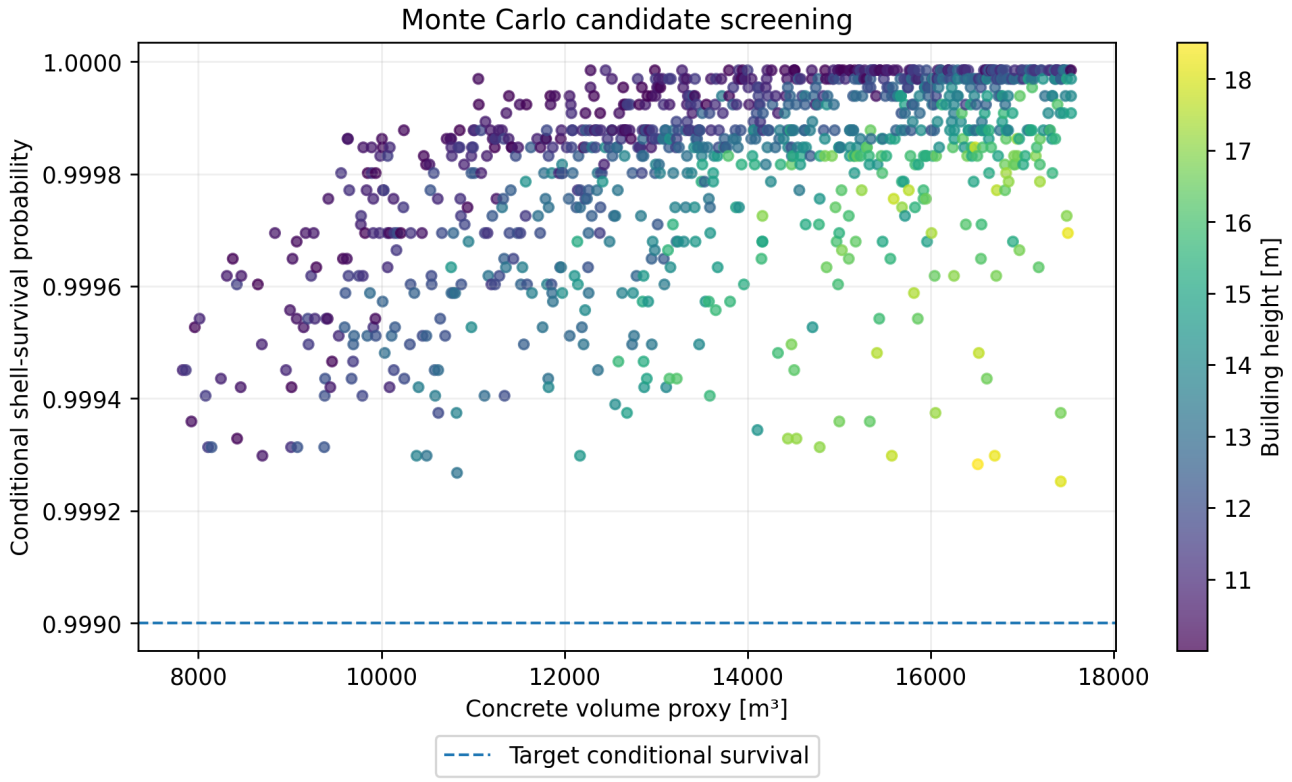


Figure 2: Displayed subset of up to 1,000 candidates from a 4,096-geometry screen, using 65,536 trials per candidate. Concrete-volume proxy is compared with shell-survival probability under the assumed hazard distribution; color denotes height, and the dashed line marks the 99.9% model target. These probabilities describe the reduced structural checks, with site occurrence and energy recovery outside their scope.

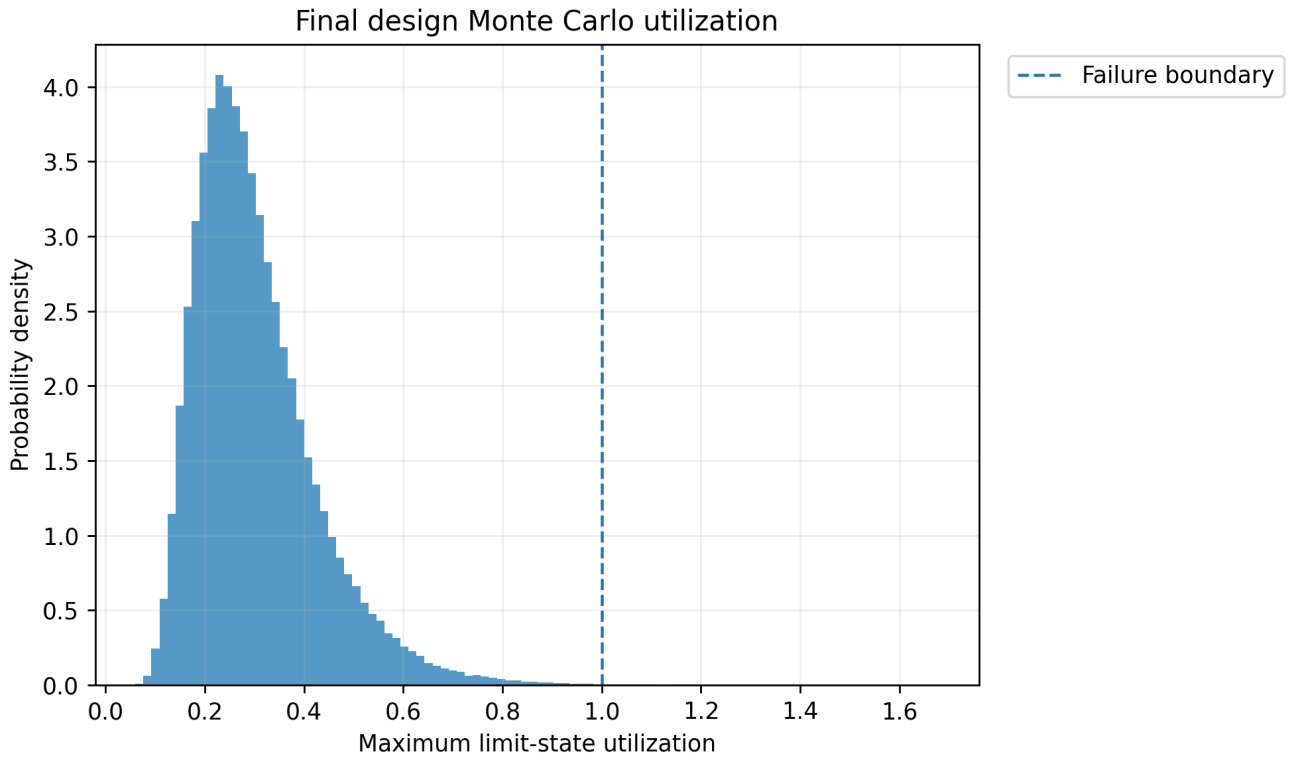


Figure 3: Maximum limit-state utilization for a regularly spaced 200,000-trial subsample of the final conditional wind and debris calculation. Values above unity cross a modeled failure threshold; the final ten-million-trial calculation gives 99.937510% survival and 0.062490% failure within this screening model. The histogram emphasizes the main distribution, with the recorded maximum of 2.0904 extending beyond its displayed horizontal range.

4.6.2 Connected shell and foundation demand

Within the connected shell, tetrahedral elasticity transfers directional wind loads and gravity into a spring-supported foundation interface. Compression contact resolves the distribution of calculated bearing demand, with reinforcement and connection adequacy requiring capacity information beyond this elastic response. The imposed 320 mph scenario uses illustrative loading coefficients and excludes the reduced-Reynolds-number pressure field from its structural inputs.

The finer model contains 20,936 nodes and 88,320 tetrahedra, representing 62,808 displacement degrees of freedom and approximately 7,818.25 m^3 of continuum material. Across the four solved cardinal directions, the largest recorded displacement is 6.881 mm and peak principal tension is 1.516 MPa, both associated with the 90-degree case. Eight additional directions were screened by loading measures without completed structural solutions, limiting the calculated envelope to the four solved cases.

Table 38: Elastic response and interface demand for the selected loading cases

Quantity	Recorded value	Physical scope
Largest displacement	6.881 mm	Maximum across solved cardinal directions.
Largest principal tension	1.516 MPa	Elastic stress screen, with mesh convergence unresolved.
Total compressive reaction	187.127 MN	Sum over the modeled foundation contacts.
Peak footing pressure	323.55 kPa	Demand without supplied geotechnical capacity.
Peak distributed connection shear	36.68 kPa	Demand without supplied connection capacity.
Distributed anchorage tension	0 kPa	Result of this load case; other combinations unresolved.
Maximum base settlement	6.4711 mm	Displacement under the assumed foundation springs.
Maximum base uplift	0 mm	Result of the selected contact solution.
Global friction demand ratio	Unavailable	A missing friction input prevents its assessment.

Zero anchorage tension follows the particular contact and load solution, leaving anchorage design subject to other governing combinations. Site-specific soil properties and verified connection capacities are absent, preventing a demand-to-capacity assessment at those interfaces. The reinforced-concrete strip interaction curve likewise uses unfactored assumptions and supplies a mechanics comparison separate from complete shell design.

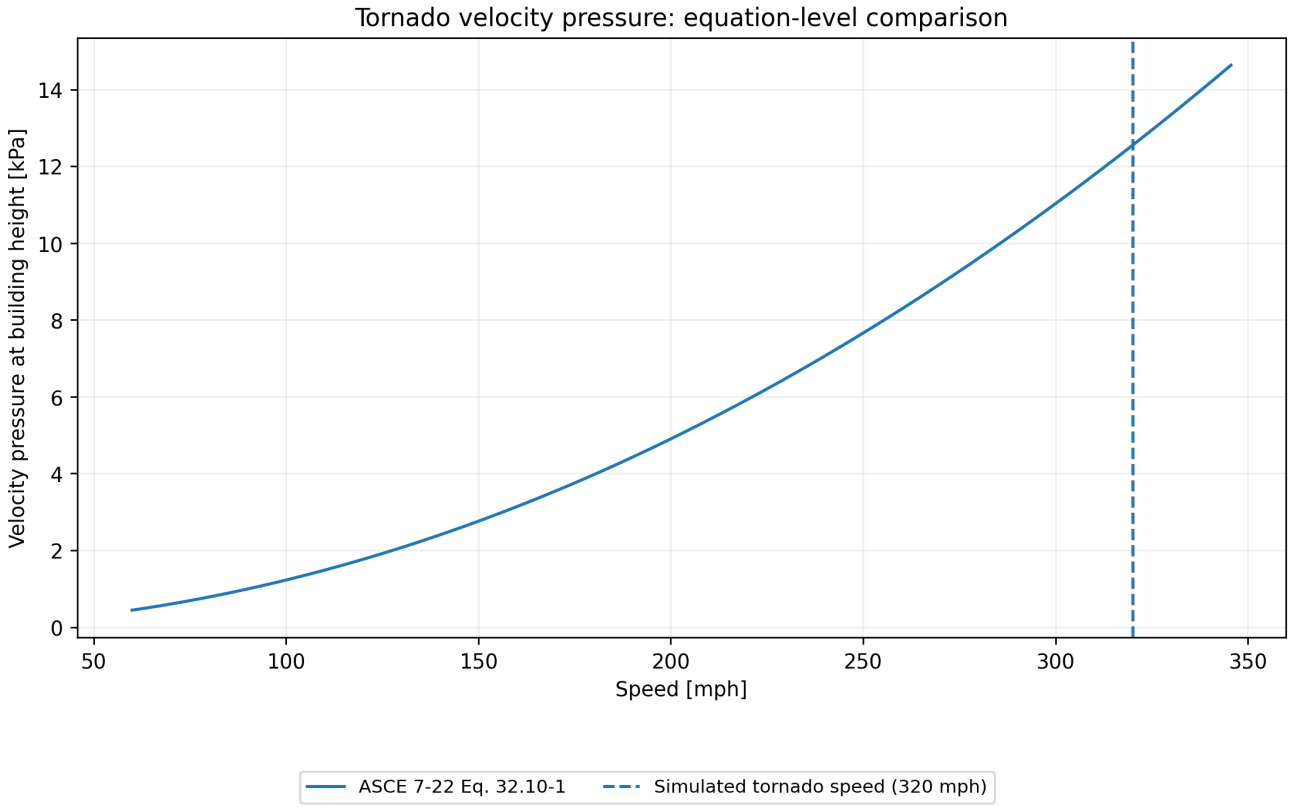


Figure 4: Velocity-pressure comparison over the selected speed range, with a vertical marker at the 320 mph structural scenario. The curve evaluates the implemented pressure relation under its chosen coefficients, providing an equation-level reference without completing the site-specific load procedure or resolving surface pressures around the courtyard.

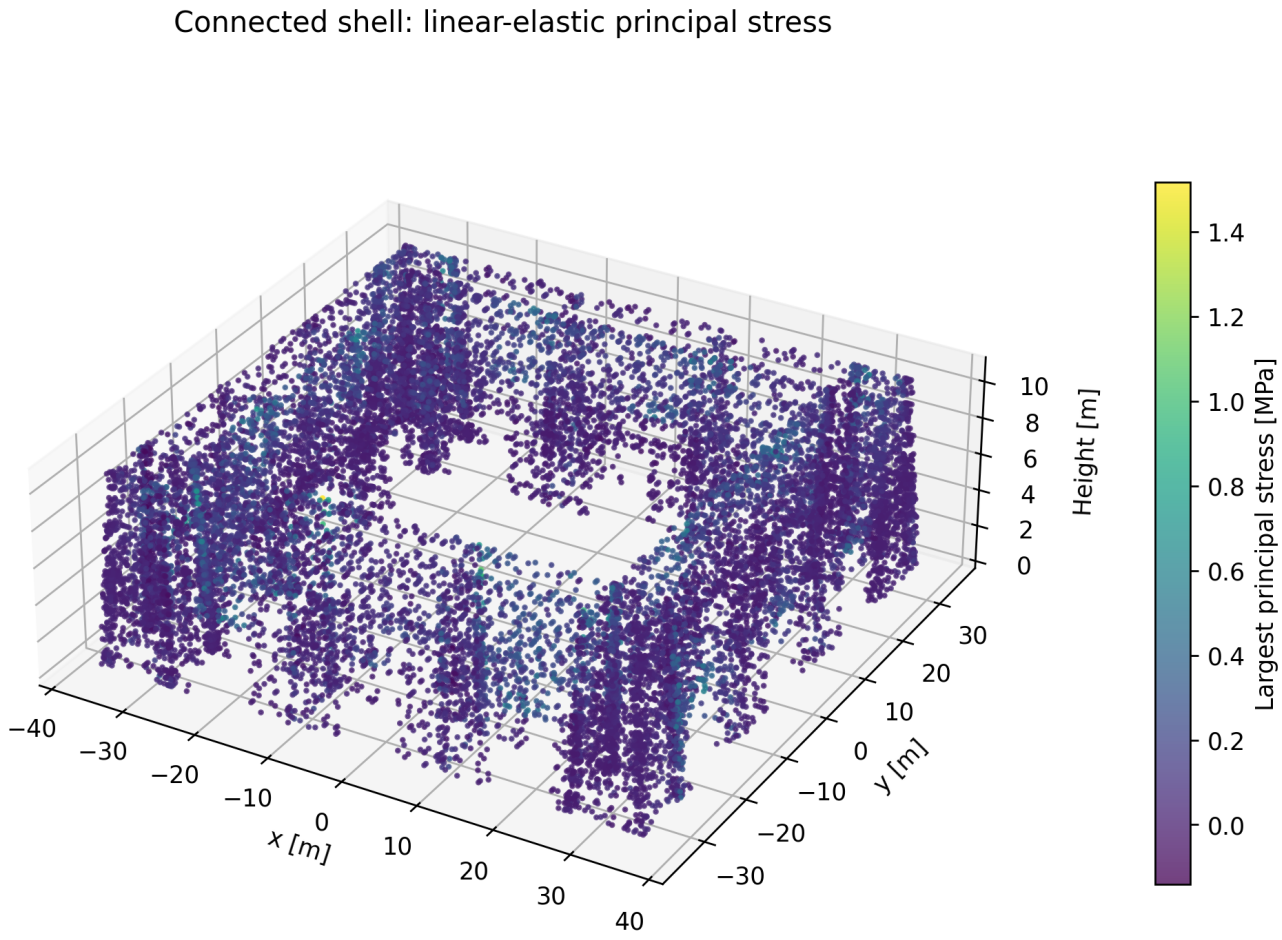


Figure 5: Largest principal stress in the connected linear-elastic shell for the displayed loading case. The tetrahedral model represents the passages and transverse webs, exposing localized stress concentrations around the connected geometry. Across the solved directional cases, maximum principal tension reaches 1.516 MPa, subject to unresolved mesh convergence and the assumed material response.

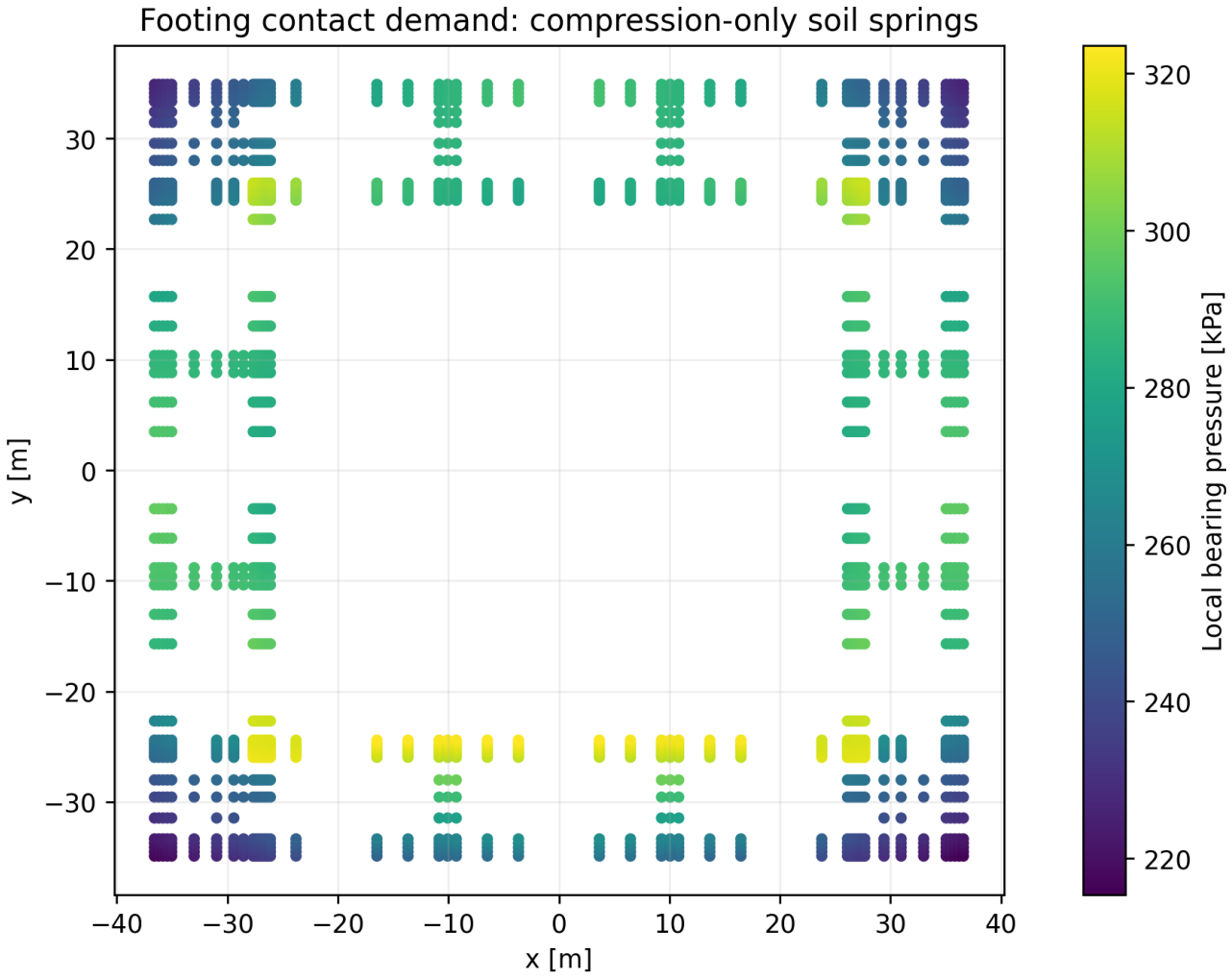


Figure 6: Compression contact pressure at the foundation interface under the modeled elastic loading. The peak calculated bearing demand is 323.55 kPa; geotechnical capacity and a site-specific soil model are required to convert this demand into an adequacy assessment.

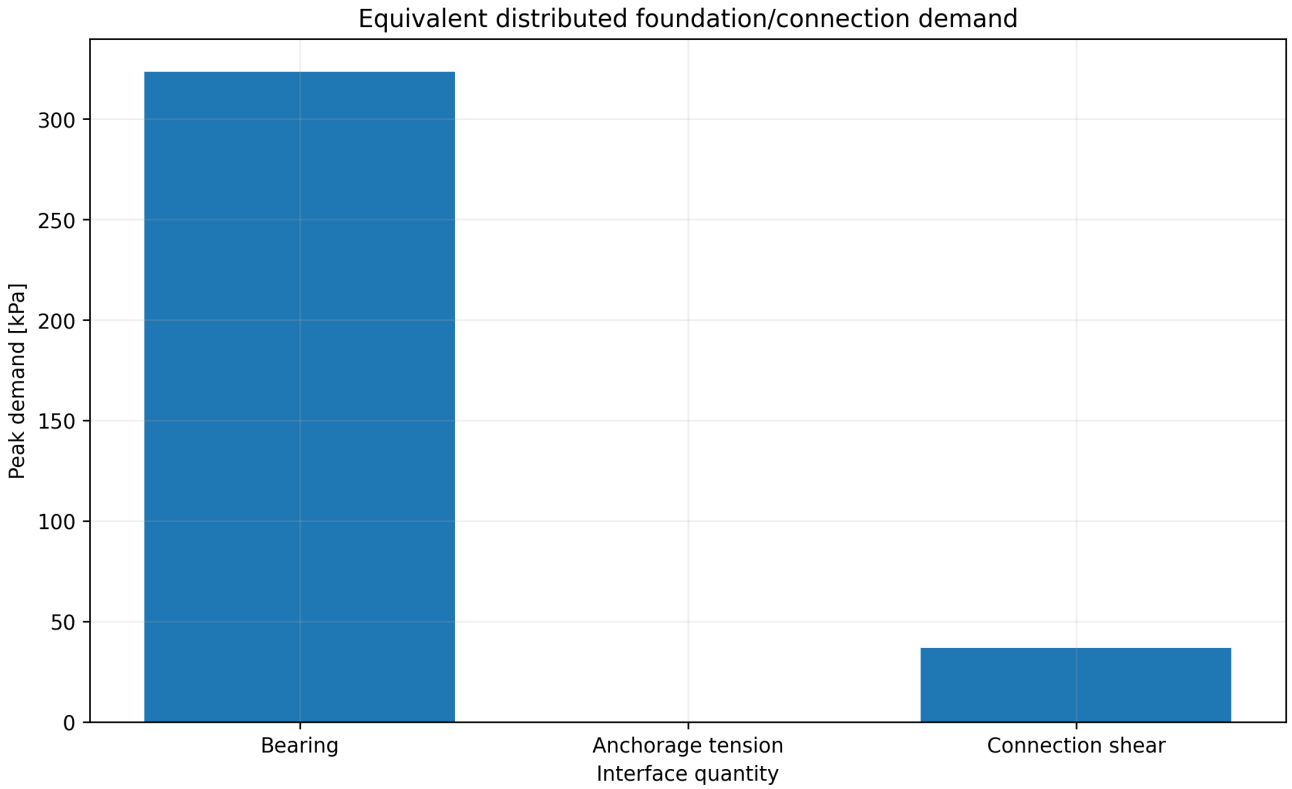


Figure 7: Equivalent distributed interface demands for the modeled case, including 323.55 kPa peak bearing pressure and 36.68 kPa peak connection shear. Zero calculated anchorage tension applies to this load and contact solution; other load combinations and connection capacities require separate assessment.

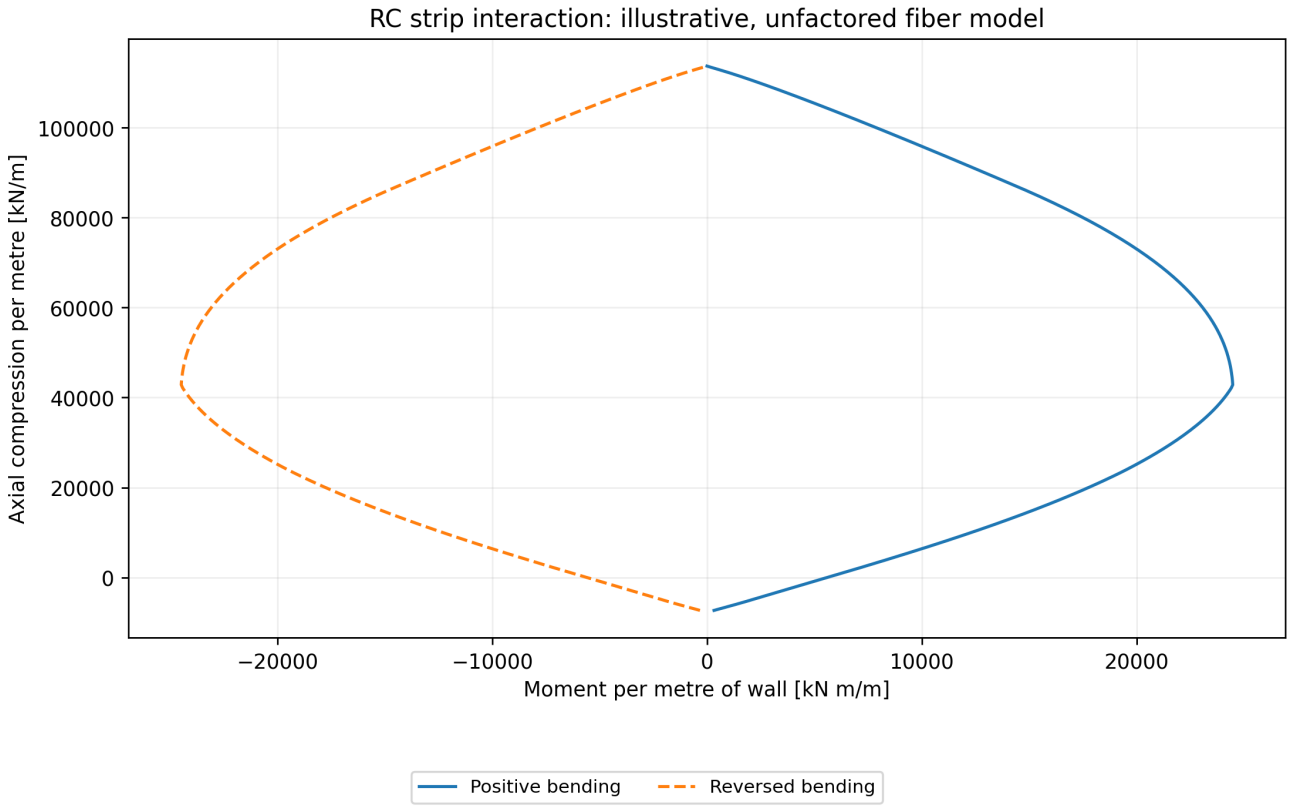


Figure 8: Illustrative axial-force–bending-moment interaction of a reinforced-concrete strip using an unfactored fiber model. Positive and reversed bending trace the two response branches under the assumed reinforcement and material laws, excluding a qualified design resistance for the complete shell.

4.6.3 Support loss and numerical error

Removing a pier-base support redirects force into neighboring portions of the connected shell, increasing deformation and tensile stress. Static elastic trials expose sensitivity to this redistribution, with every tested removal crossing the assumed cracking screen. Dynamic removal and nonlinear material response require separate calculations to determine the subsequent load path.

Table 39: Independent static support-removal cases

Removed support	Displacement (mm)	Displacement factor	Peak tension (MPa)	Tension factor
West pier 2	12.201	1.773	5.070	3.344
East pier 2	12.184	1.771	4.899	3.232
West pier 3	11.904	1.730	4.293	2.831
South pier 3	12.713	1.847	4.187	2.761

Across the two tested meshes, load-displacement work changes from approximately 1.051395 to 1.052913 MJ, a relative change of 0.144%. The corresponding 99.5th-percentile tensile stress changes by 4.42%, illustrating greater sensitivity in a local mechanical observable. An element cap stopped additional refinement, leaving mesh convergence unresolved despite a converged algebraic equilibrium calculation.

For the retained numerical checks, relative force and moment balance errors are approximately 9.89×10^{-10} and 2.35×10^{-9}, with relative work balance error near 3.27×10^{-13}. These checks assess the implementation at the chosen discretization, independent of its continuum error or the adequacy of elastic material response after cracking. Metropolis–Hastings spatial sampling supplies an additional diagnostic of importance-based exploration, with its accepted proposal fraction carrying no structural reliability interpretation.

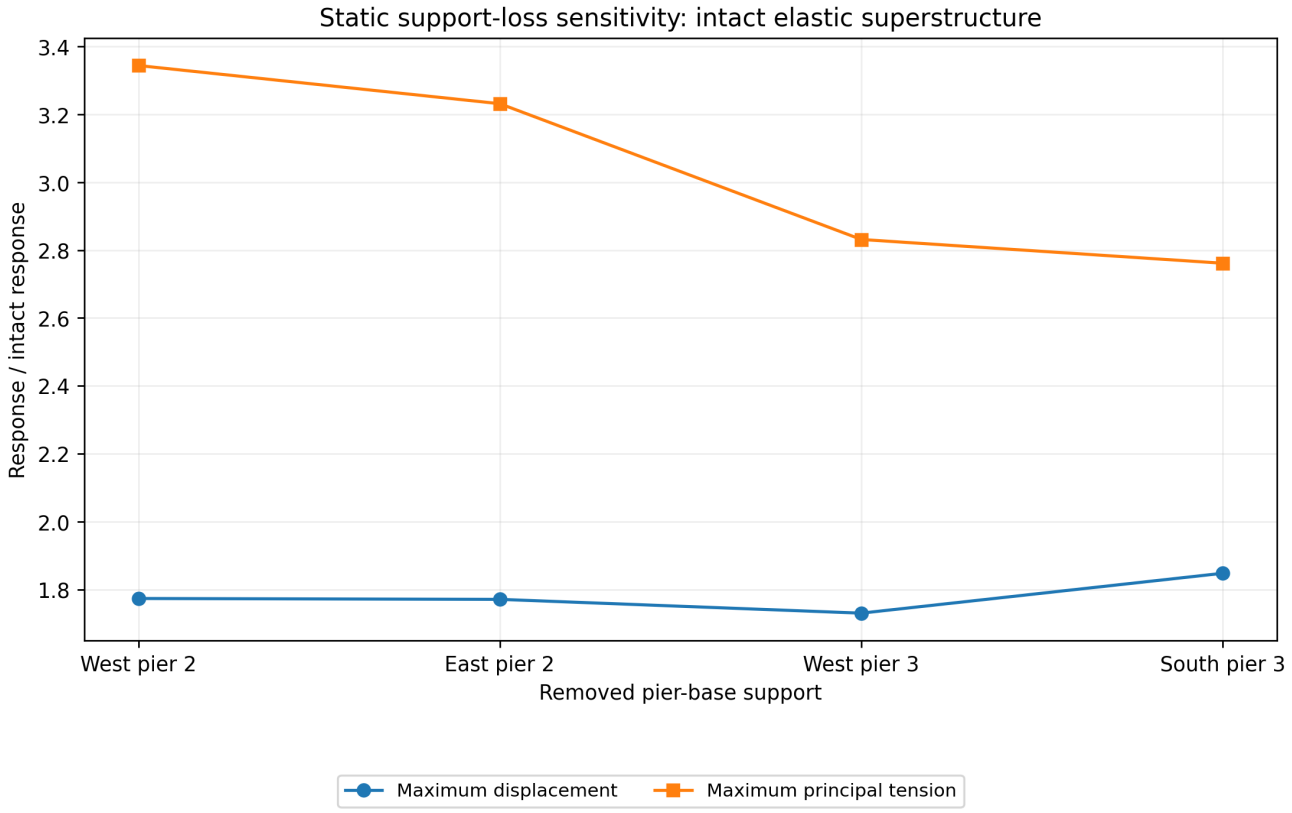


Figure 9: Static sensitivity to independent removal of selected pier-base supports, with the elastic superstructure otherwise retained. Maximum displacement increases by factors of 1.73–1.85 and peak principal tension by factors of 2.76–3.34; every tested removal exceeds the assumed cracking screen. These results motivate nonlinear redistribution and dynamic member-loss analysis.

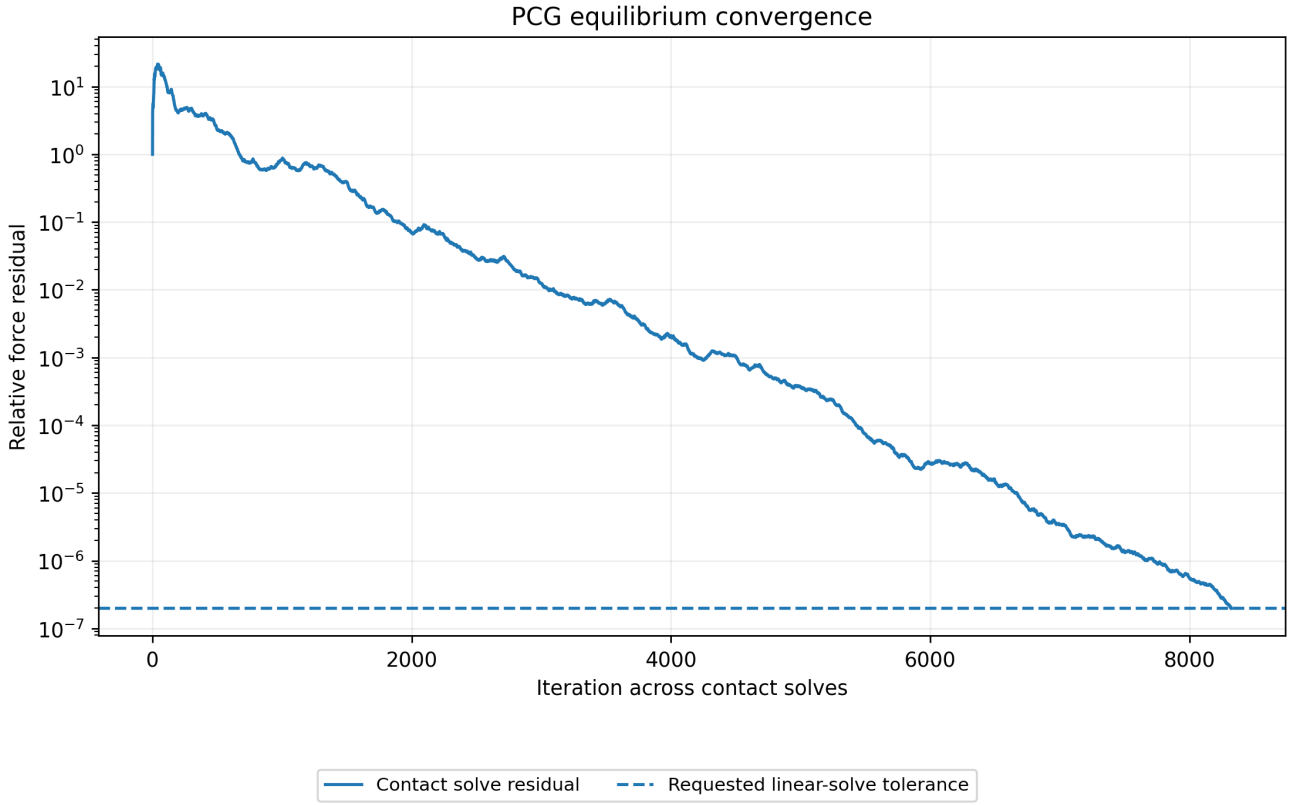


Figure 10: Preconditioned conjugate-gradient convergence for the displayed contact-equilibrium solution, with the dashed line marking the requested linear-solve tolerance. Decay of the relative force residual tests the algebraic solution at a fixed discretization, leaving mesh error and physical-model accuracy to independent checks.

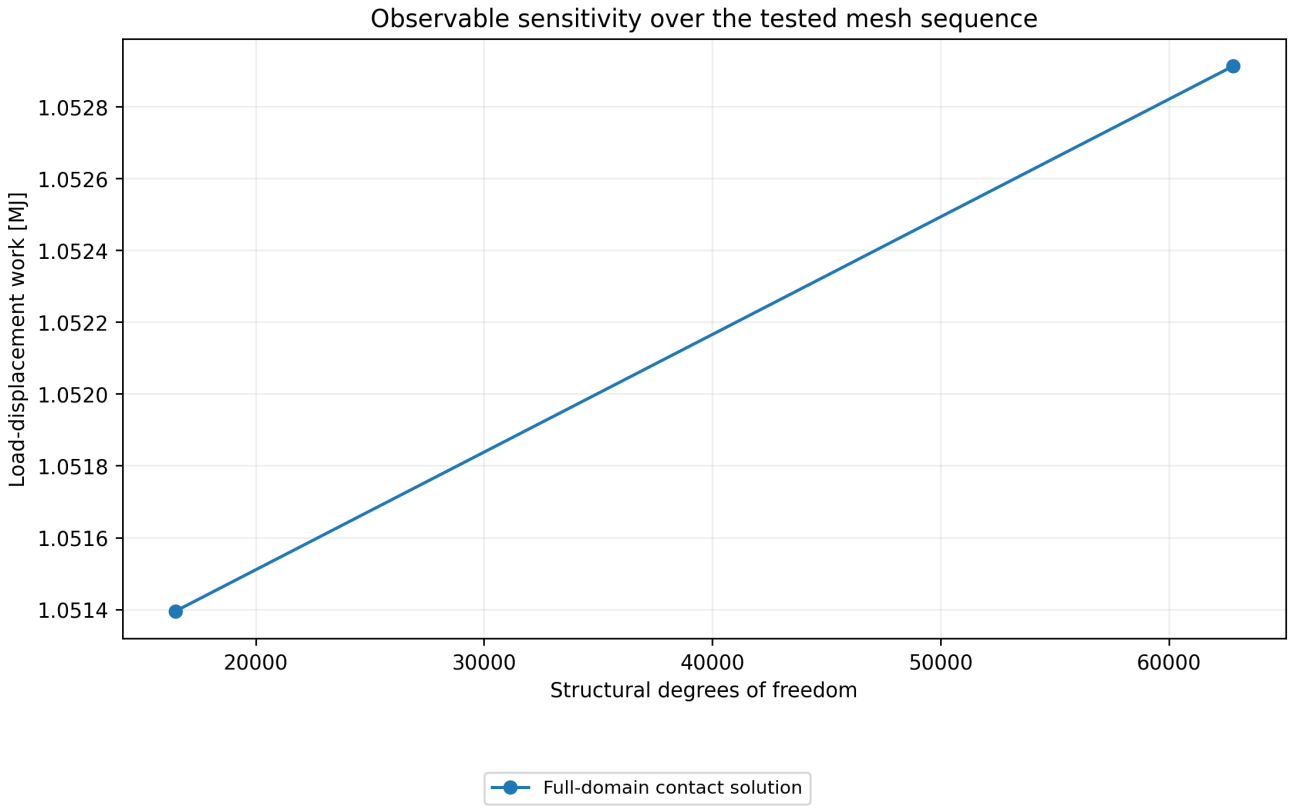


Figure 11: Load-displacement work for meshes containing 16,488 and 62,808 structural degrees of freedom. The work changes by approximately 0.144%, accompanied by a 4.42% change in the 99.5th-percentile tensile stress. The element cap prevented further refinement, leaving continuum convergence unresolved.

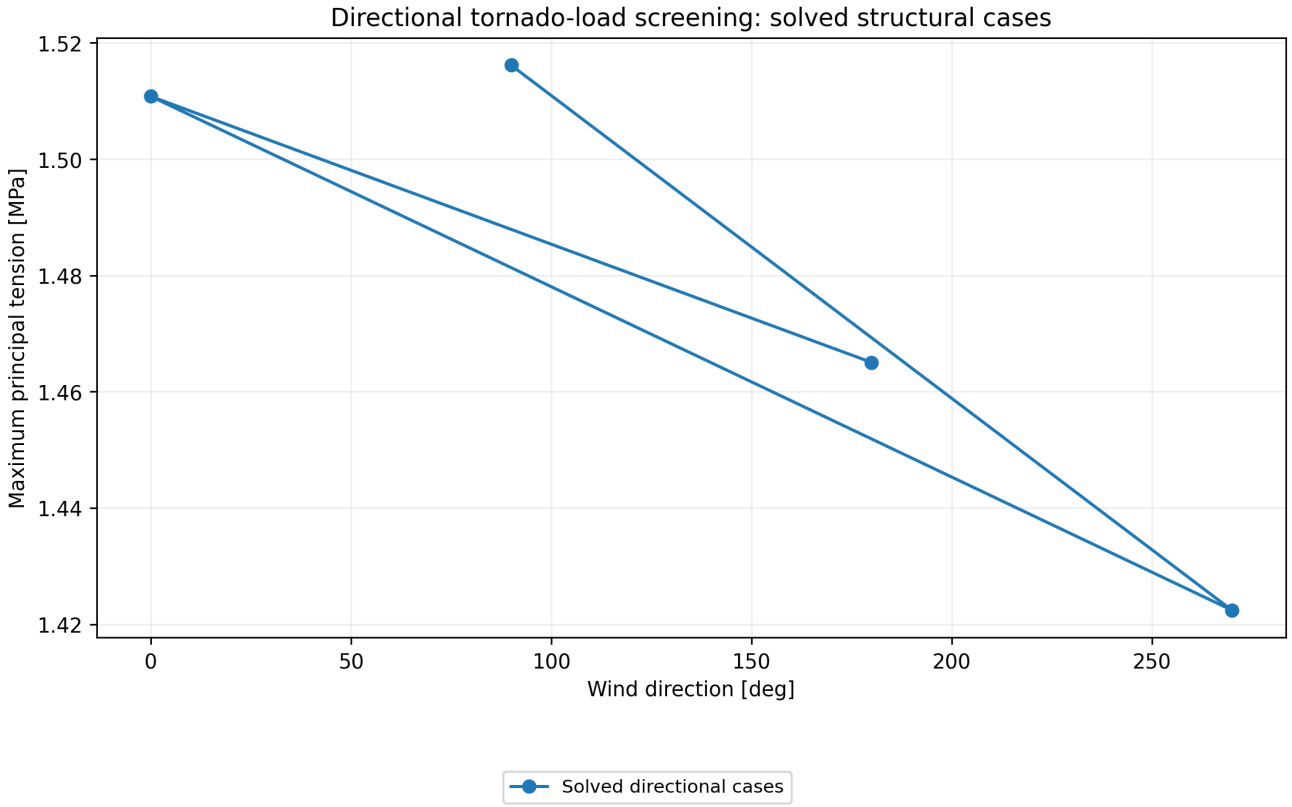


Figure 12: Maximum principal tension for the four solved cardinal wind directions under the 320 mph scenario. The 90-degree case gives the largest recorded value, 1.516 MPa; eight additional candidate directions were ranked without full structural solutions. Connecting segments follow stored case order and provide no continuous angular envelope.

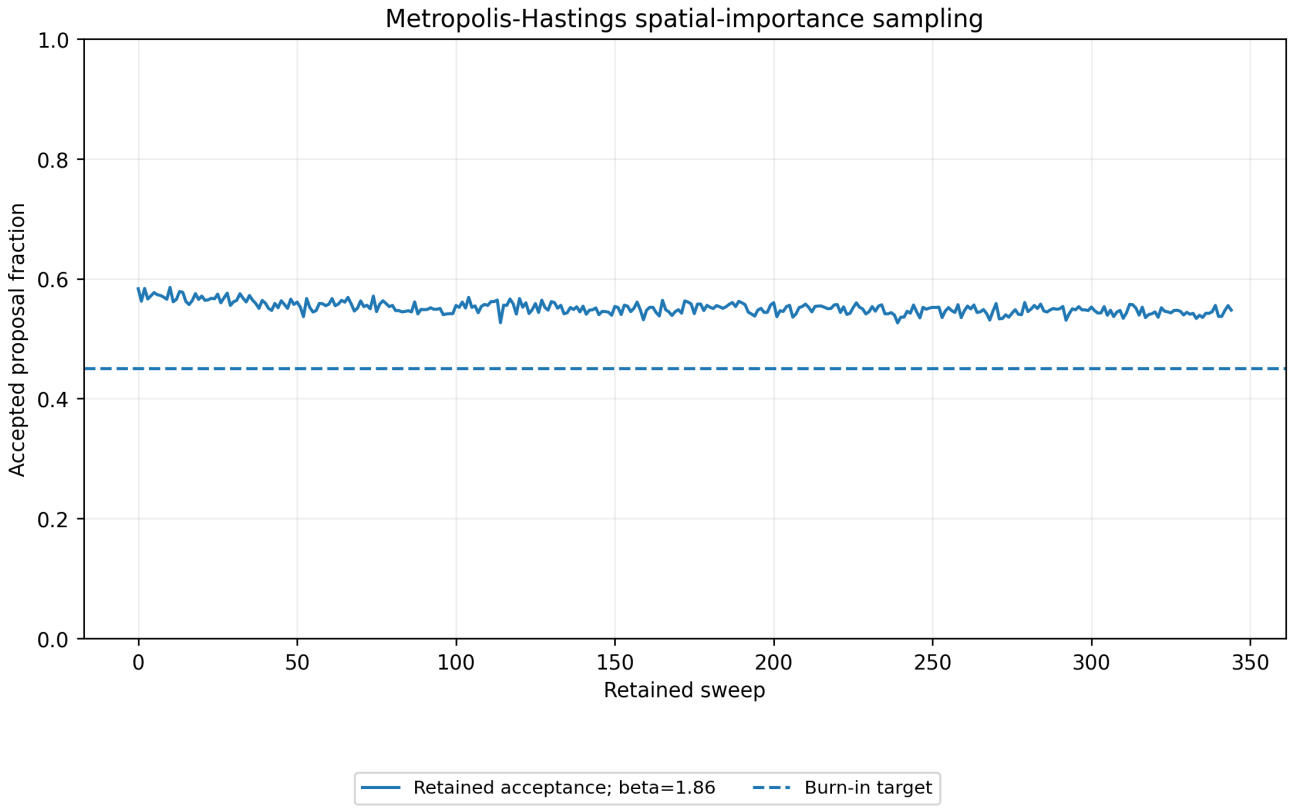


Figure 13: Accepted proposal fraction during retained Metropolis–Hastings sweeps for spatial-importance sampling. The dashed line marks the warmup acceptance target, with the displayed trace documenting sampler behavior without representing shell-survival probability or convergence of the physical mesh.

4.6.4 Debris energy and idealized arrest

For an intercepted projectile, kinetic energy fixes an incident demand that a barrier must redirect or dissipate across a finite contact interval. A 15 lb timber reference permits a transparent comparison between impact speed and idealized arrest force, with the material failure process excluded from this reduced calculation. Impact tolerance requires a qualified assembly response under the applicable test conditions.

At the selected 67 mph and 100 mph reference speeds, projectile energy is approximately 3.05 kJ and 6.80 kJ, respectively. Assuming complete arrest with zero rebound and a triangular force pulse relates peak force to momentum change divided by contact duration, exposing the increase in demand associated with a shorter impact. Dissipating this energy in a barrier supplies an impact-management function without establishing electrical recovery or the intercepted fraction of a tornado's debris field.

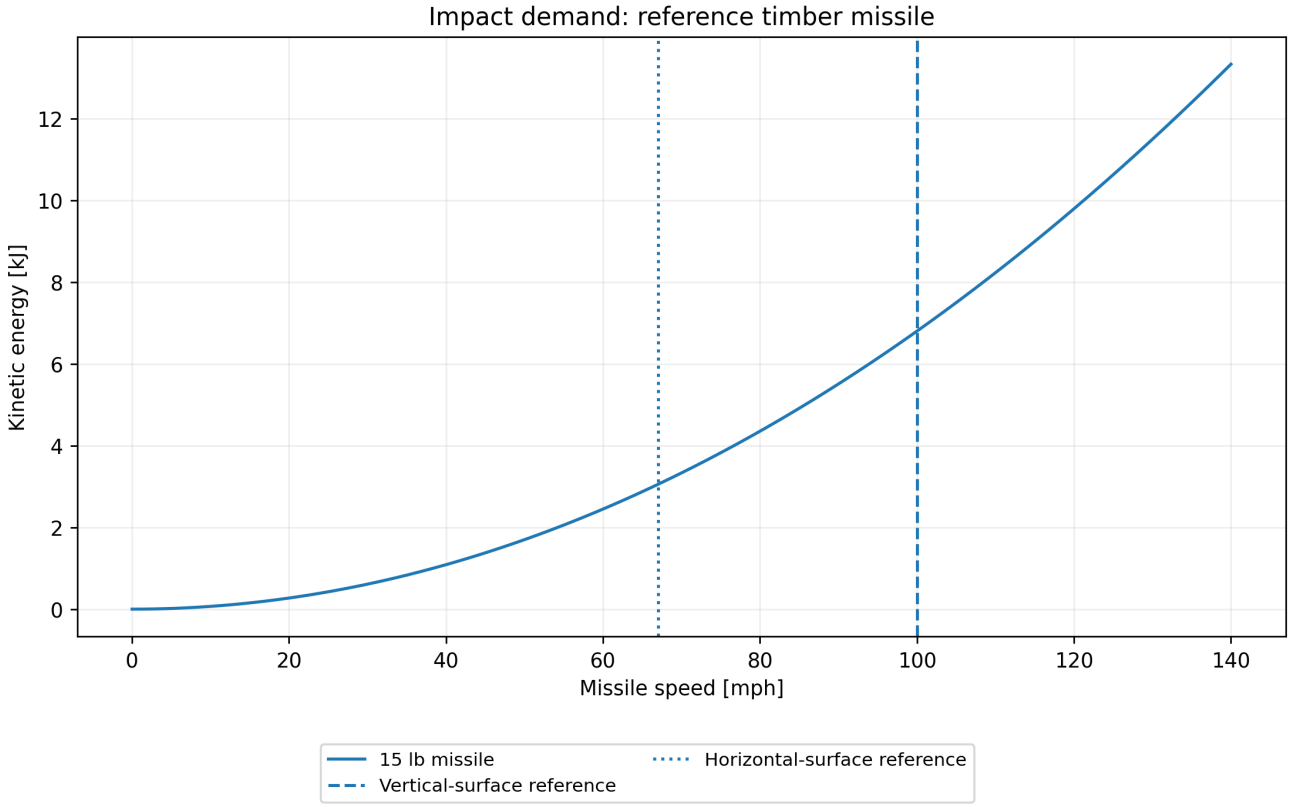


Figure 14: Kinetic energy of the assumed 15 lb timber projectile as a function of impact speed. The selected 67 mph and 100 mph reference markers correspond to approximately 3.05 kJ and 6.80 kJ, respectively, quantifying incident demand without establishing barrier penetration resistance or recoverable electrical energy.

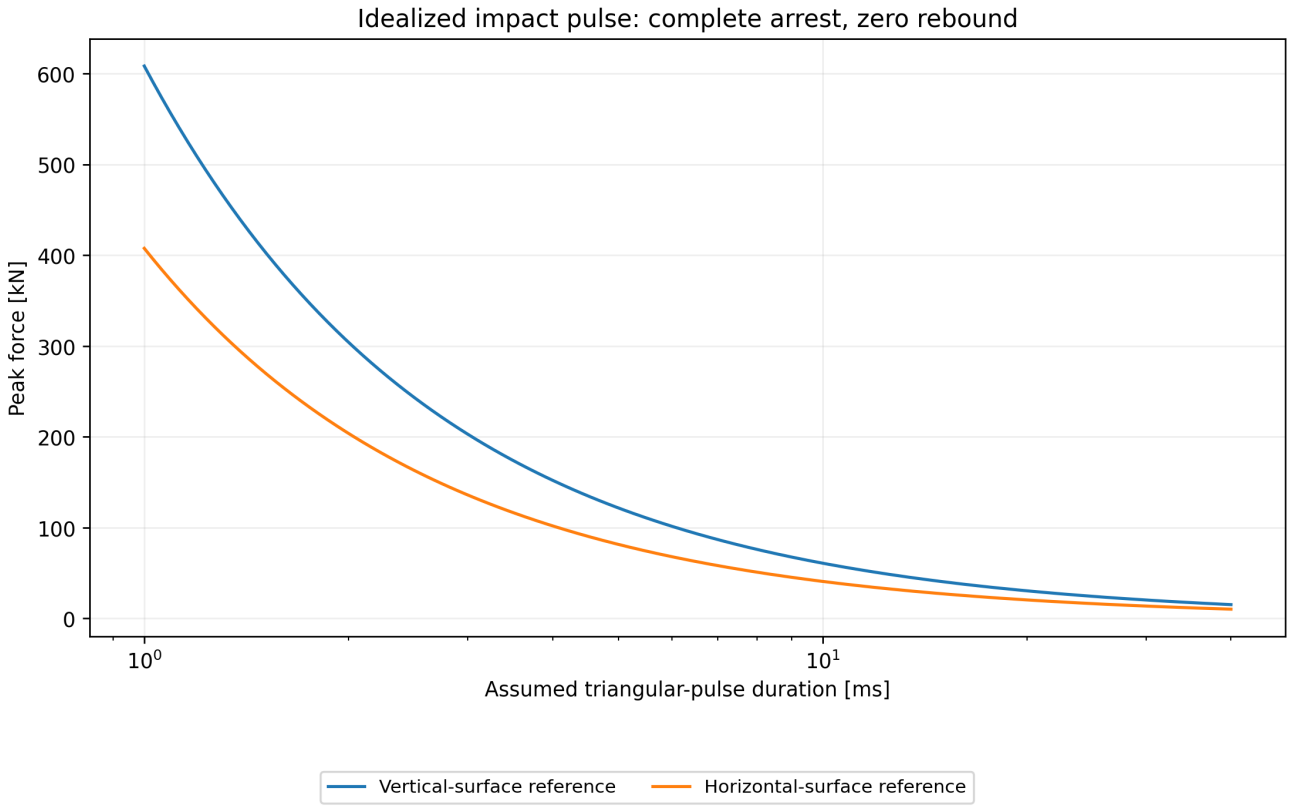


Figure 15: Peak force for complete projectile arrest with zero rebound and an assumed triangular force pulse. For fixed momentum change, shortening the contact duration increases peak force in inverse proportion; the curves describe idealized impact demand without resolving fracture, penetration, or assembly response.

4.6.5 Reduced-Reynolds-number flow and energy-capture limits

Around the open passages, a two-dimensional lattice-flow experiment examines how an imposed stream accelerates and forms a wake. Its Reynolds number of 150 differs from the approximately 6.656×10^8 reference-air value associated with the physical scenario, preventing direct dynamic scaling to a tornado. Pressure and velocity images support qualitative diagnostics within this reduced model.

On a 640 by 420 grid, the flow calculation reached 12,000 lattice steps and exhausted its adaptive step budget with continuing transient behavior. The resulting pressure field was excluded from structural design inputs, preserving the separation between illustrative flow behavior and the imposed elastic load cases. The recorded 181.7 s workflow duration includes the structural and flow diagnostics as implemented, with no matched CPU run available for a speedup estimate.

The retained computations report geometry and mechanical response alongside numerical diagnostics, leaving turbine power and net accepted energy uncalculated. A three-dimensional flow model with physically matched inflow and an energy-conversion boundary must quantify recovered work, followed by experiments measuring local load changes around the installation. Until those results are available, absorbing a usable portion of storm energy is the project's engineering objective, with the magnitude of any reduction in surrounding impacts still to be determined.

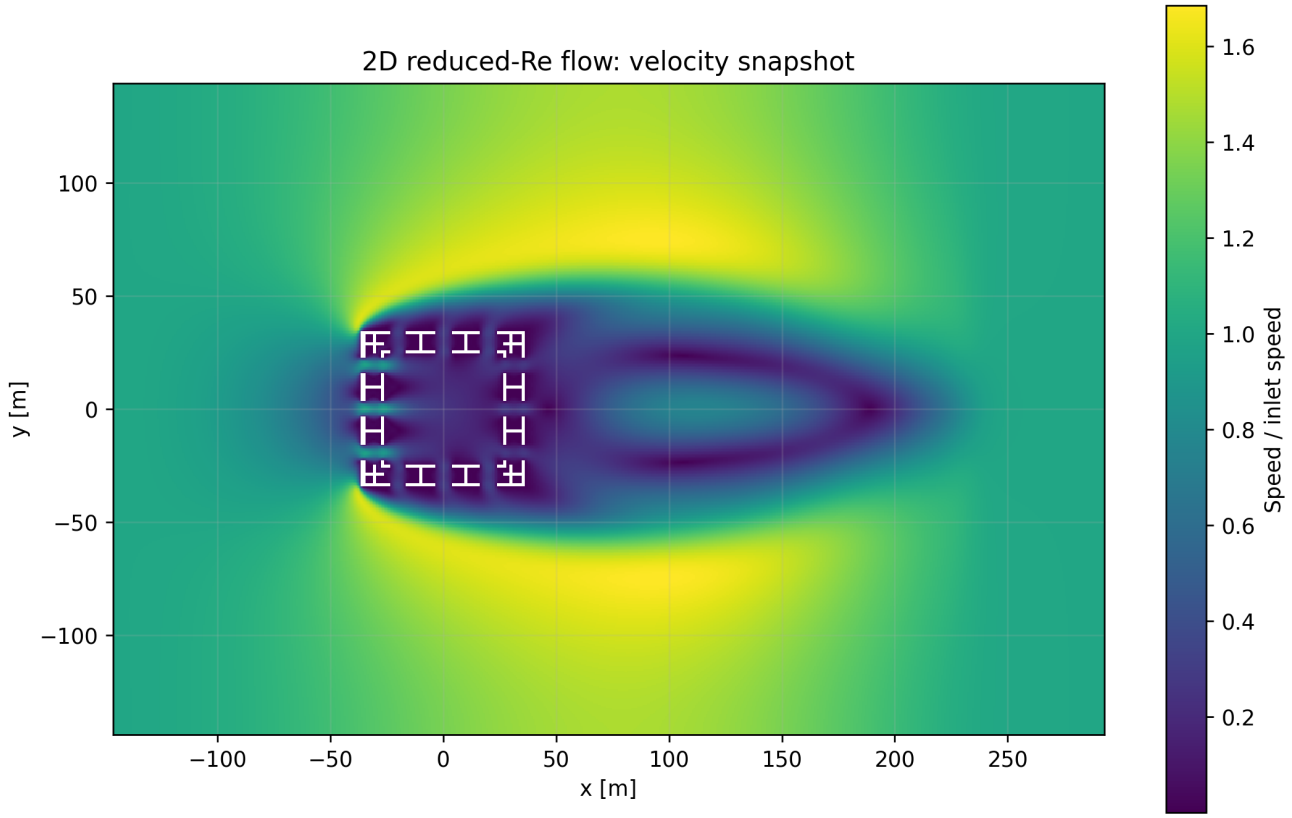


Figure 16: Transient two-dimensional velocity field at Reynolds number 150, normalized by inlet speed. Acceleration near the passages and a downstream low-speed region reflect the reduced-Reynolds-number experiment, whose reference air Reynolds number is approximately 6.656×10^8. The image supplies qualitative flow diagnostics without a full-scale tornado-energy capture estimate.

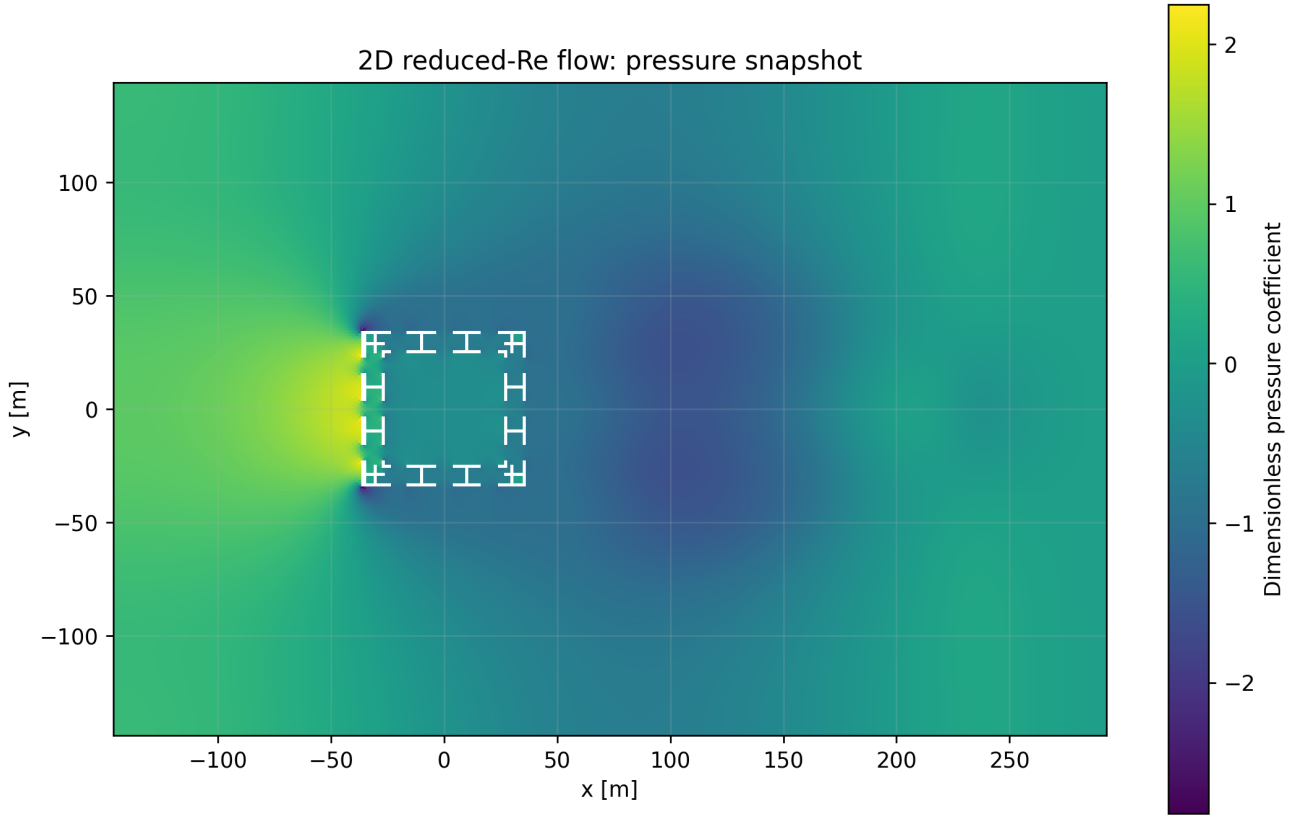


Figure 17: Dimensionless pressure snapshot from the same transient two-dimensional experiment at Reynolds number 150. Spatial pressure differences illustrate the modeled passage and wake response; these values were excluded from structural design inputs and require a physically matched three-dimensional flow calculation for atmospheric application.

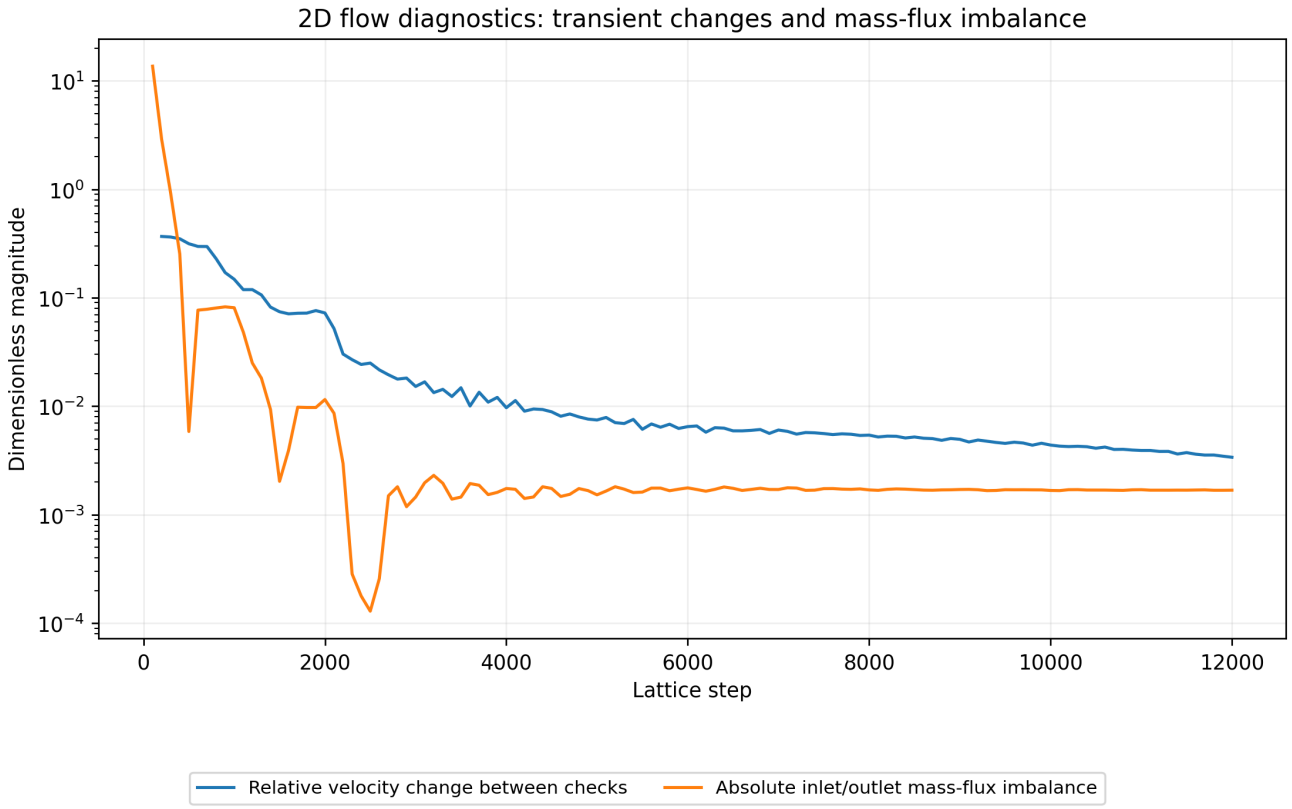


Figure 18: Changes in velocity and inlet-to-outlet mass flux over 12,000 lattice steps. The run reached its adaptive step limit with continuing transient behavior, leaving a steady-flow interpretation unsupported. The diagnostic histories qualify the accompanying snapshots without measuring turbine work or net recovered energy.

5 Conclusion

Extreme-weather harvesting contributes to resilience if the accepted energy reaches essential loads within the plant's qualified mechanical and electrical limits. For the conceptual eight-rotor bank, the 15.85 GWh gross kinetic inventory accompanies an idealized hoop stress of 8.96 GPa, making material strength the controlling constraint on permissible speed. Applying the illustrative 500 MPa limit reduces estimated electrical delivery to approximately 0.597 GWh under the stated assumptions, well below the 8.4 GWh service requirement and supporting separate sizing of the extended reserve. Modular flywheels retain a candidate role in rapid buffering, subject to measured losses and qualified containment. An adaptive GPU implementation supports these engineering comparisons only if complete benchmarks preserve the required accuracy, with independent experiments constraining the physical models. A bounded pilot links those calculations to measured service delivery without relying on hazardous storm interception or an assumed computing advantage. Motivated by EF5 destruction, the architecture pursues locally recoverable storm energy as industrial infrastructure, with living accommodation and storm refuge excluded from its intended function.

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